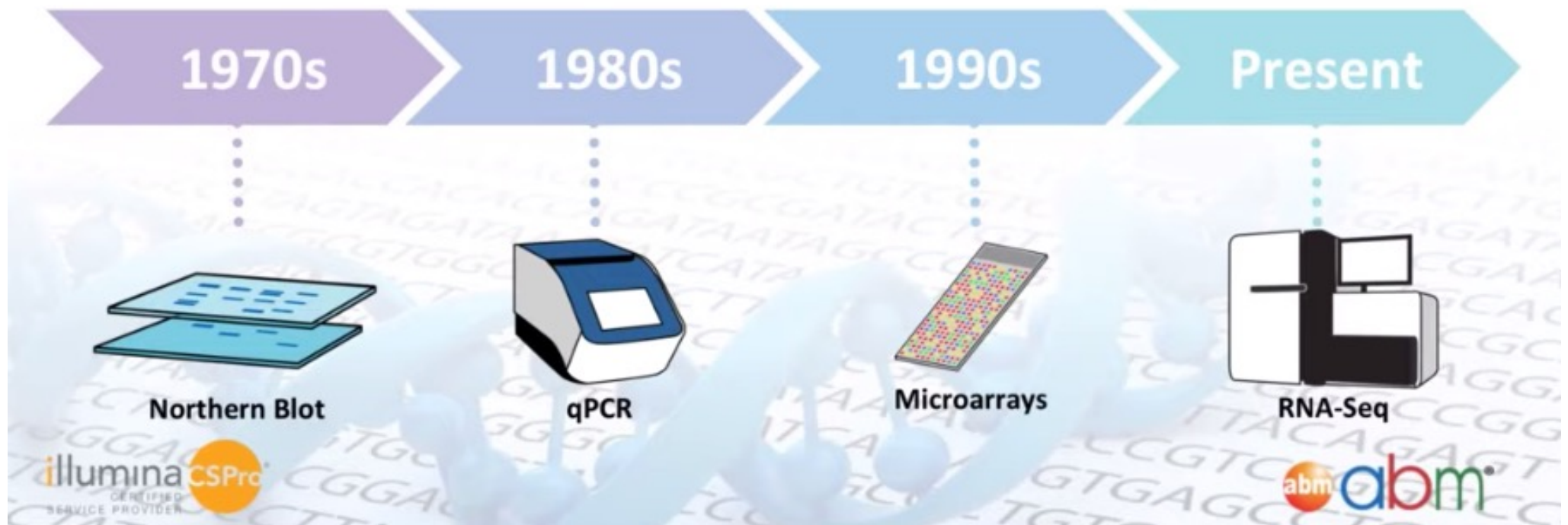


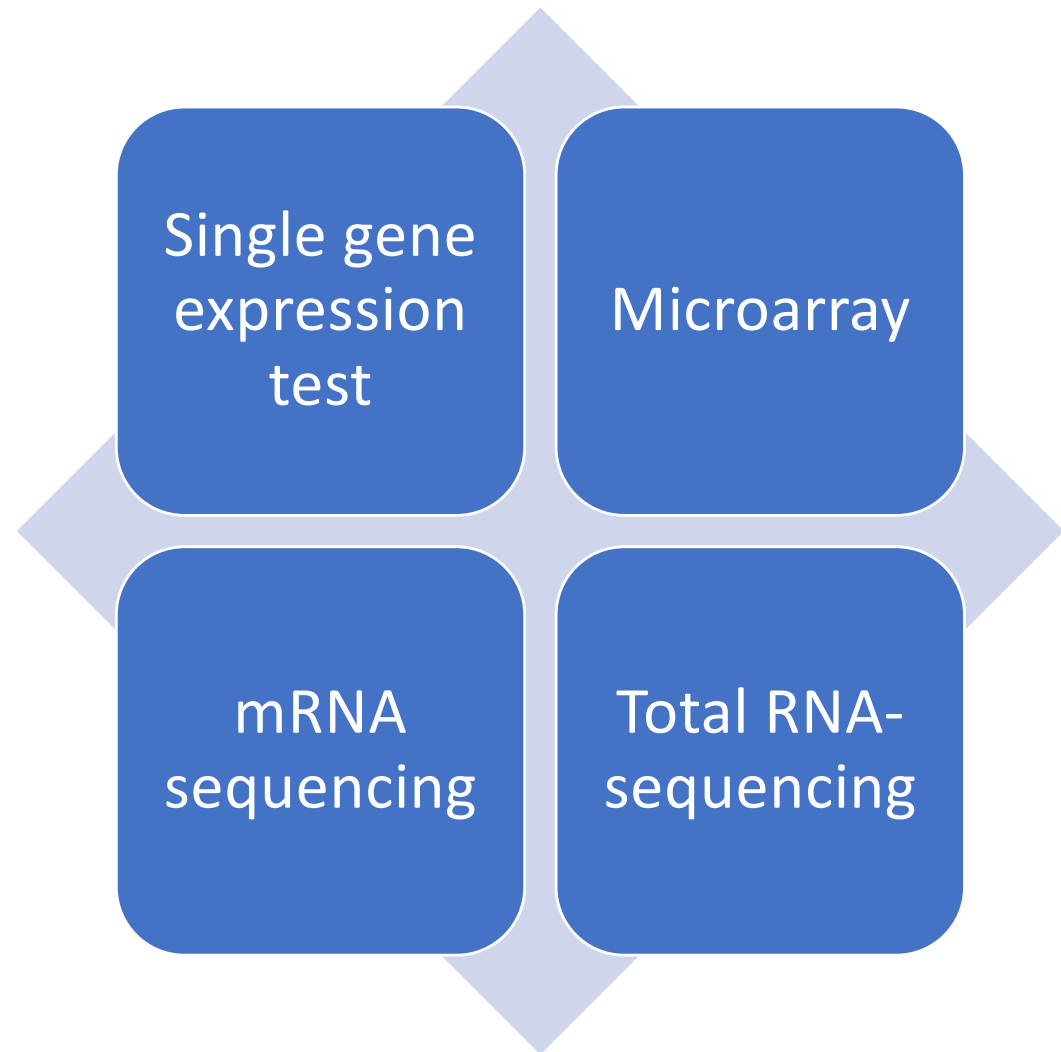
Technique for Transcriptome

Chapter19

Studying the role of genes in development and disease

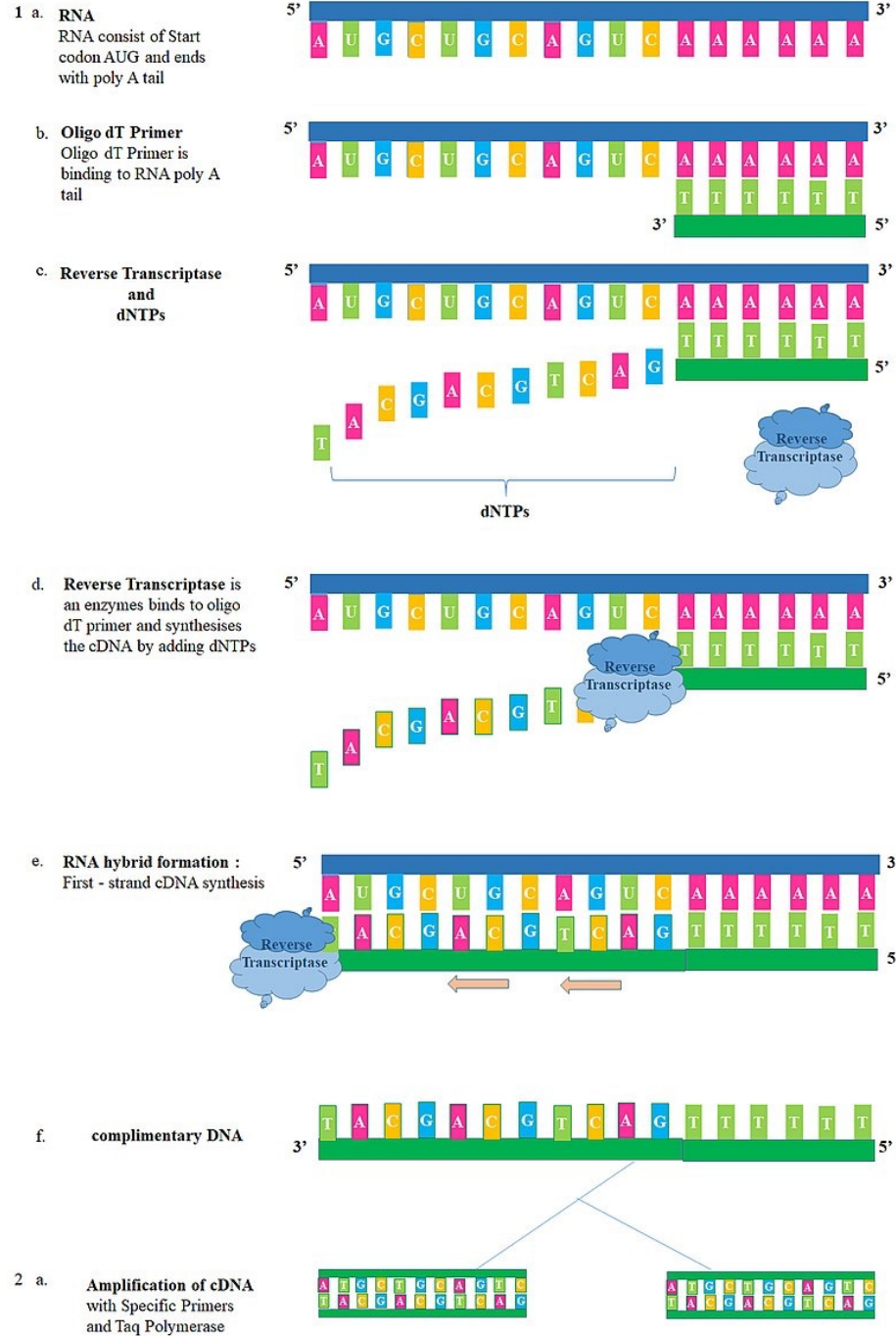


Main methods

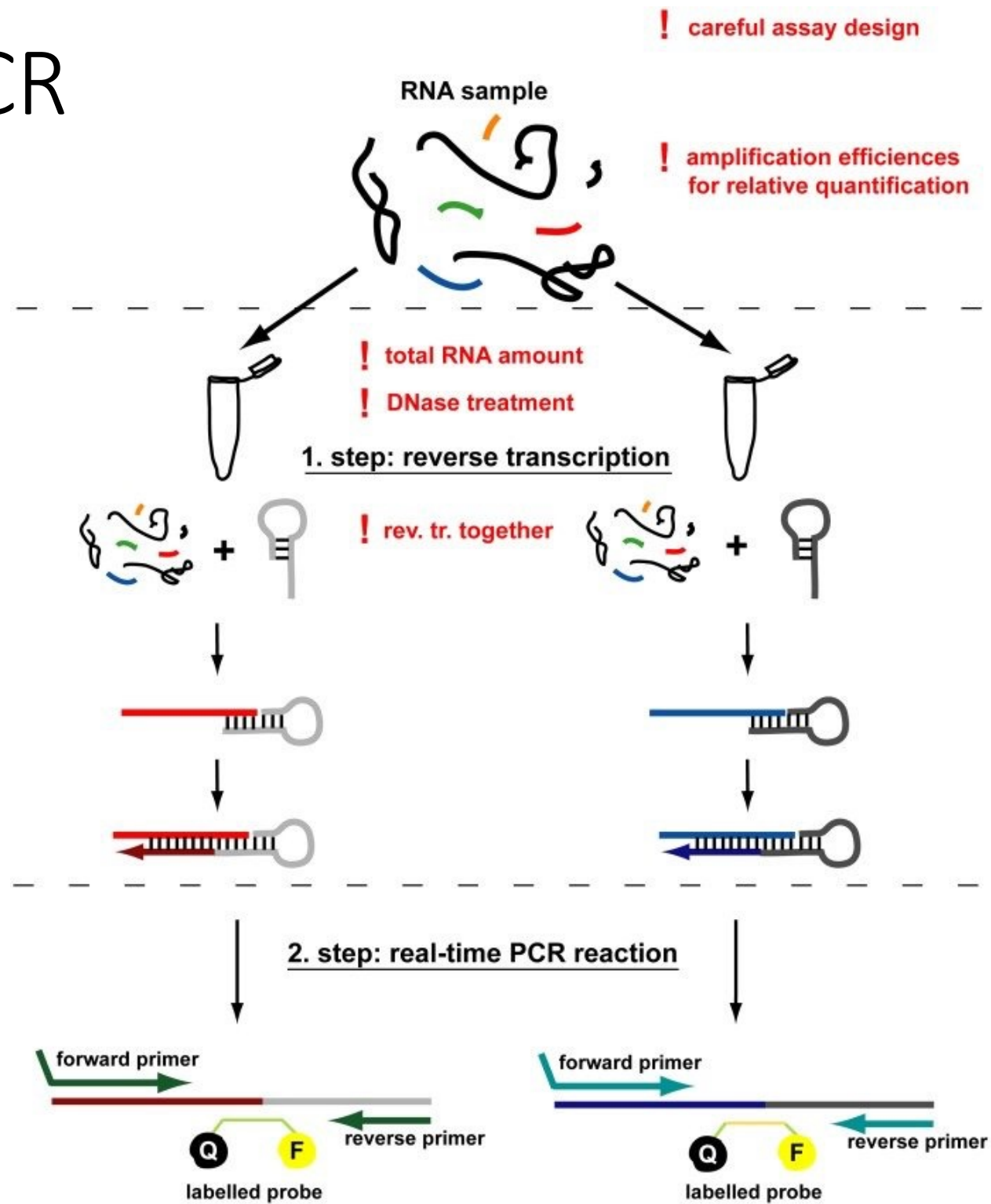


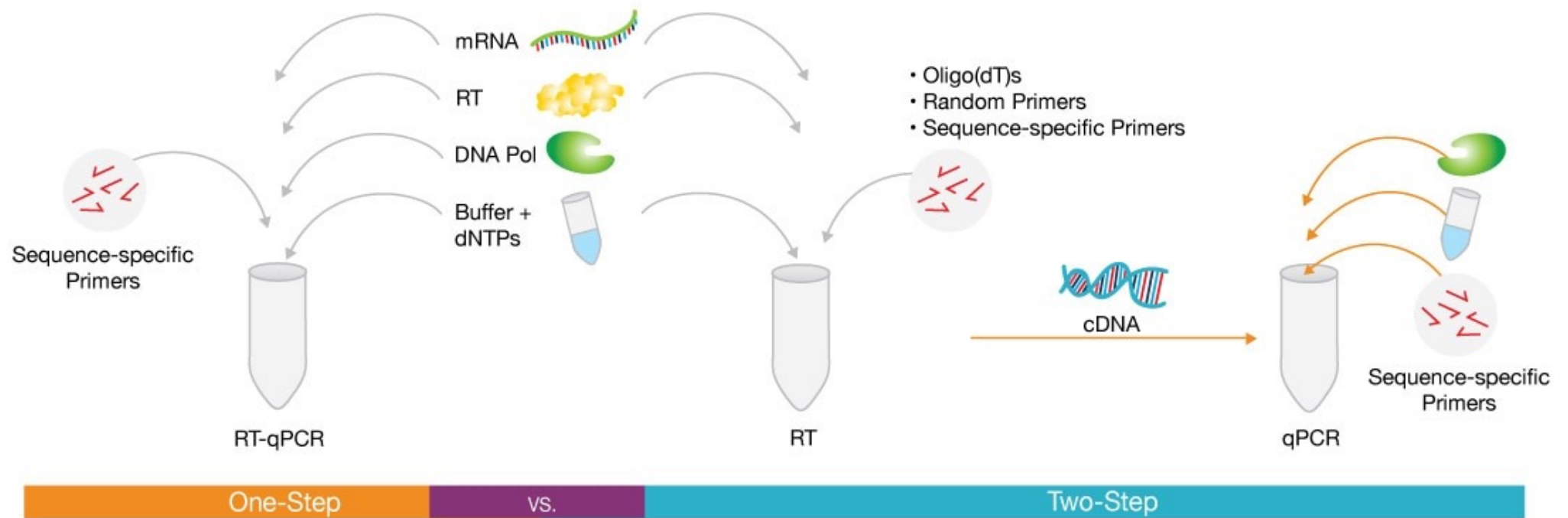
RT-PCR

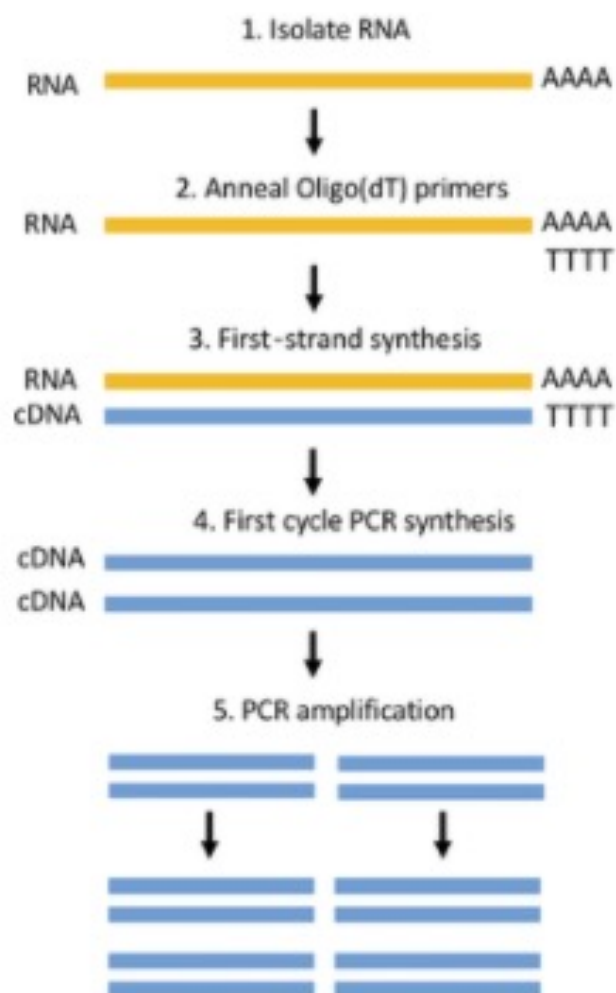
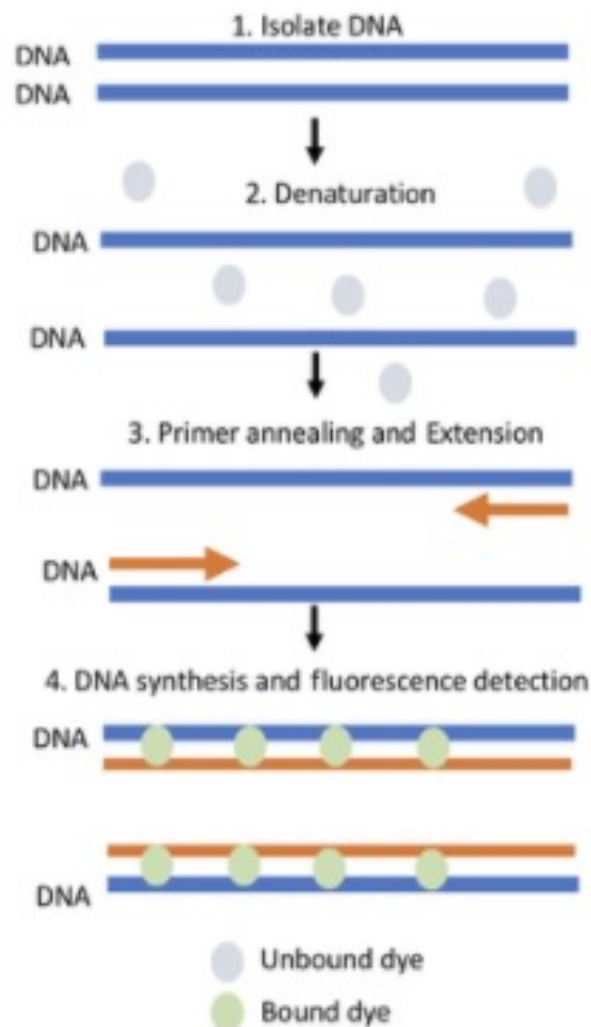
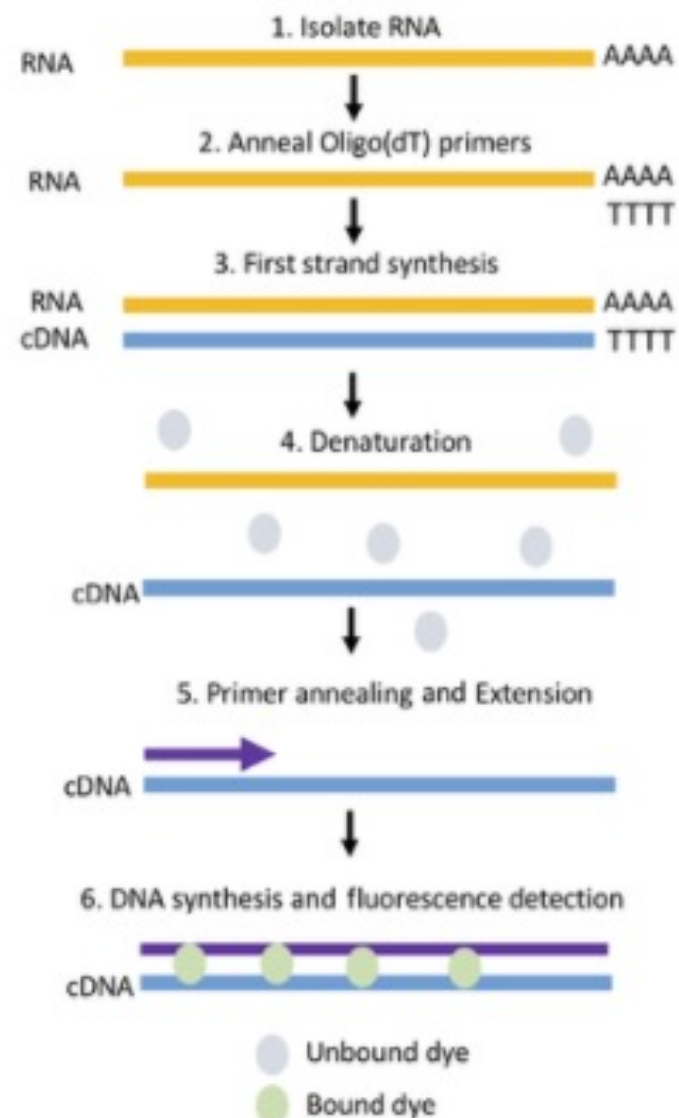
In RT-PCR, The RNA population is converted to cDNA by reverse transcription (RT), and then the cDNA is amplified by the polymerase chain reaction. The cDNA amplification step provides opportunities to further study the original RNA species, even when they are limited in amount or expressed in low abundance. Common applications of RT-PCR include detection of expressed genes, examination of transcript variants, and generation of cDNA templates for cloning and sequencing.



RT-qPCR





A**Reverse transcription-PCR****B****Quantitative real time – PCR (qPCR)****C****Reverse transcription quantitative real-time PCR (RT-qPCR)**

Example

Tests for COVID-19 virus detection

Diagnostic test

Rapid Antigen Test



Real time reverse transcription polymerase chain reaction (RT-PCR)

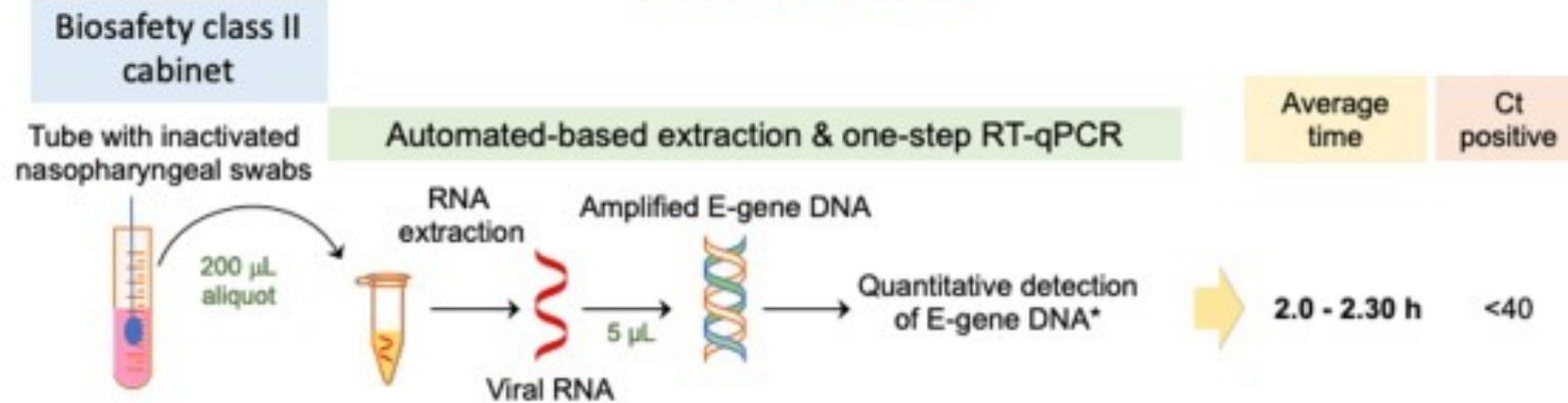


Surveillance test

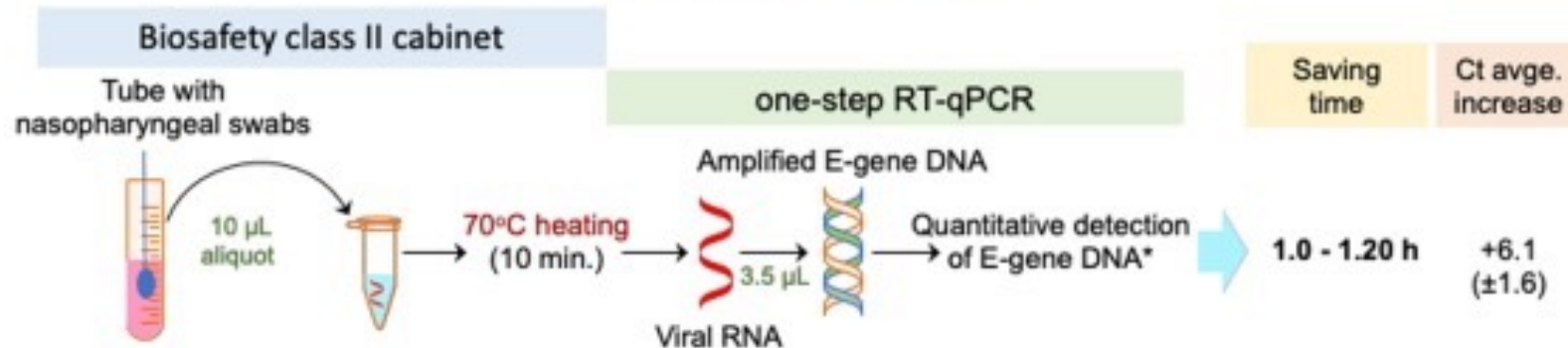
Antibody Test



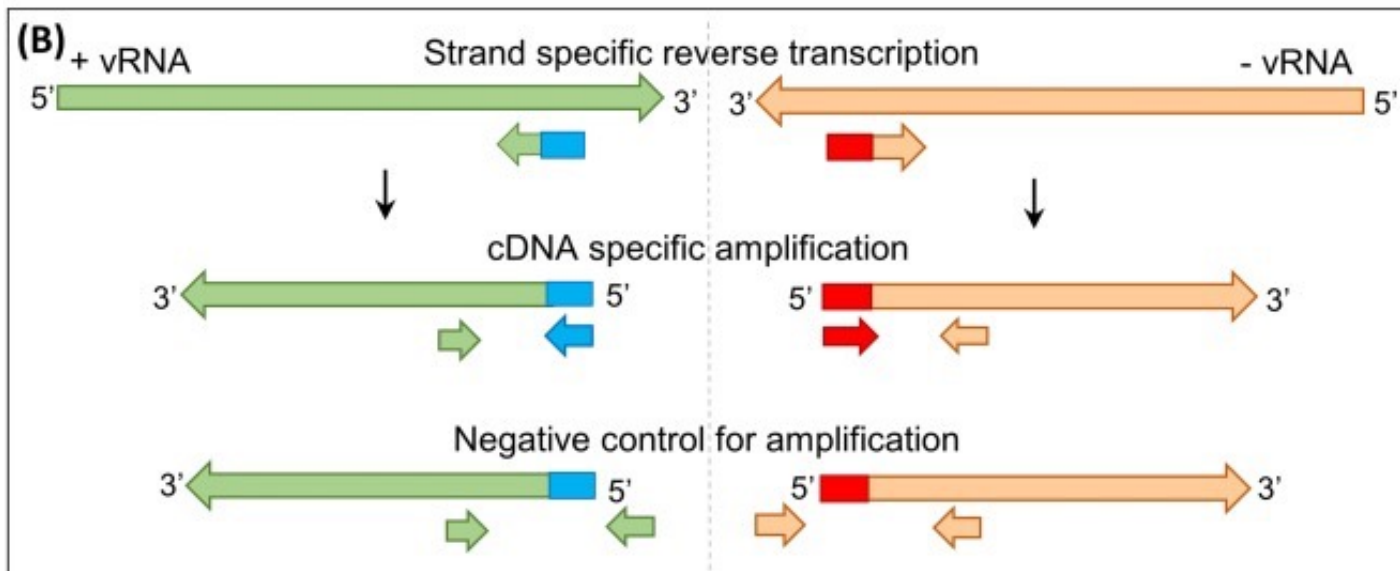
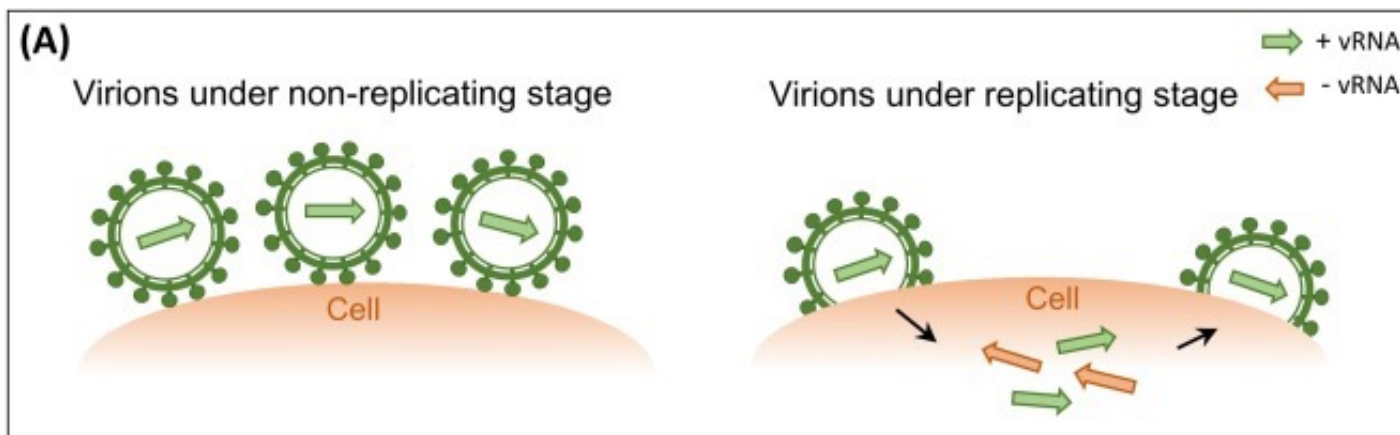
a) Standard procedure



b) Alternative fast protocol



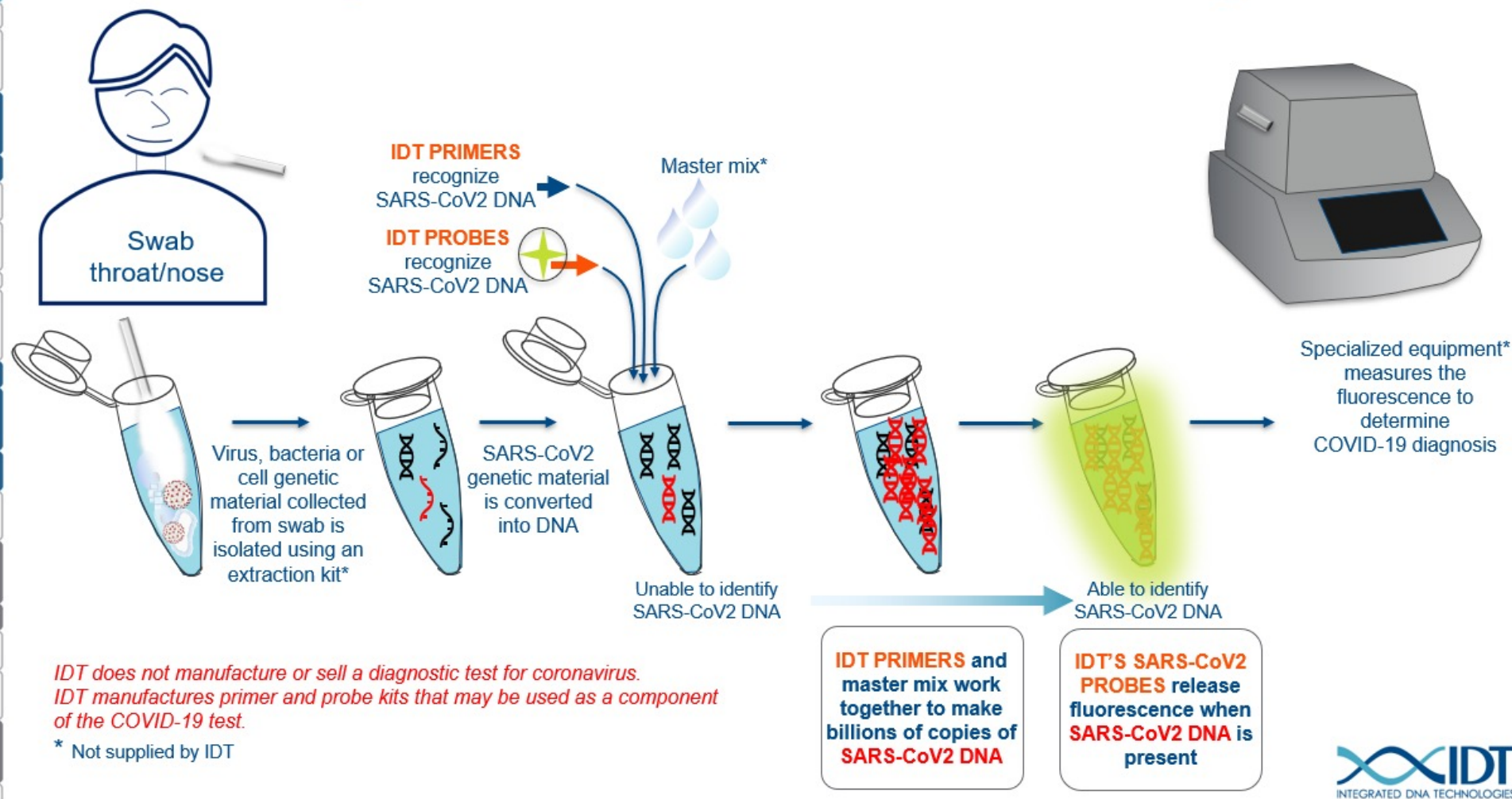
*Simultaneous detection of *ACTB* as internal control

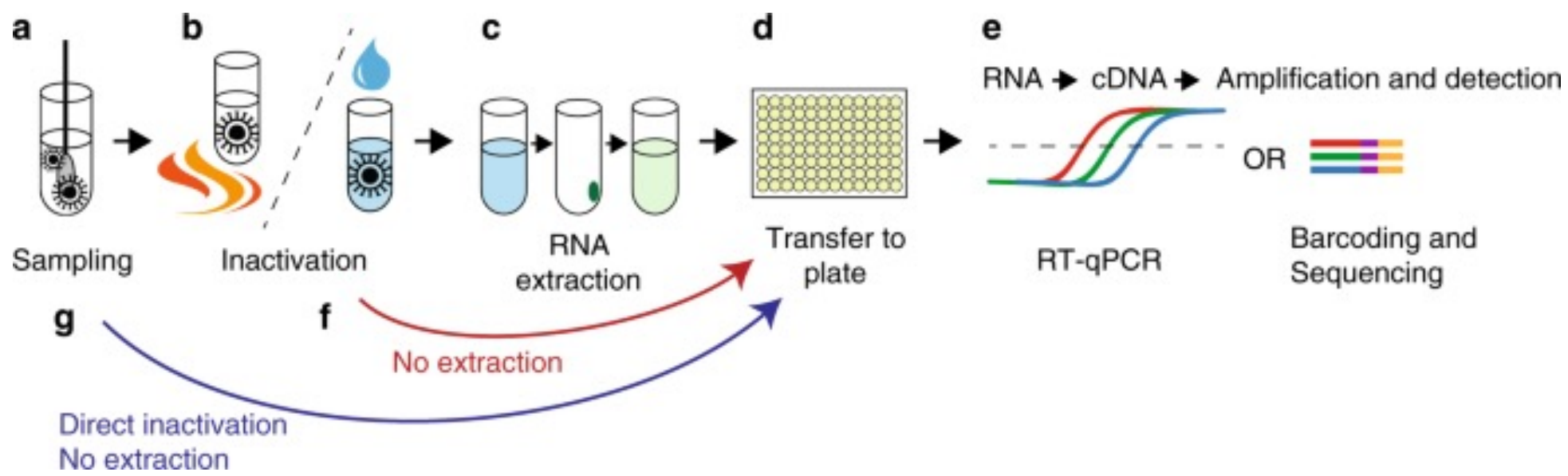


(C)

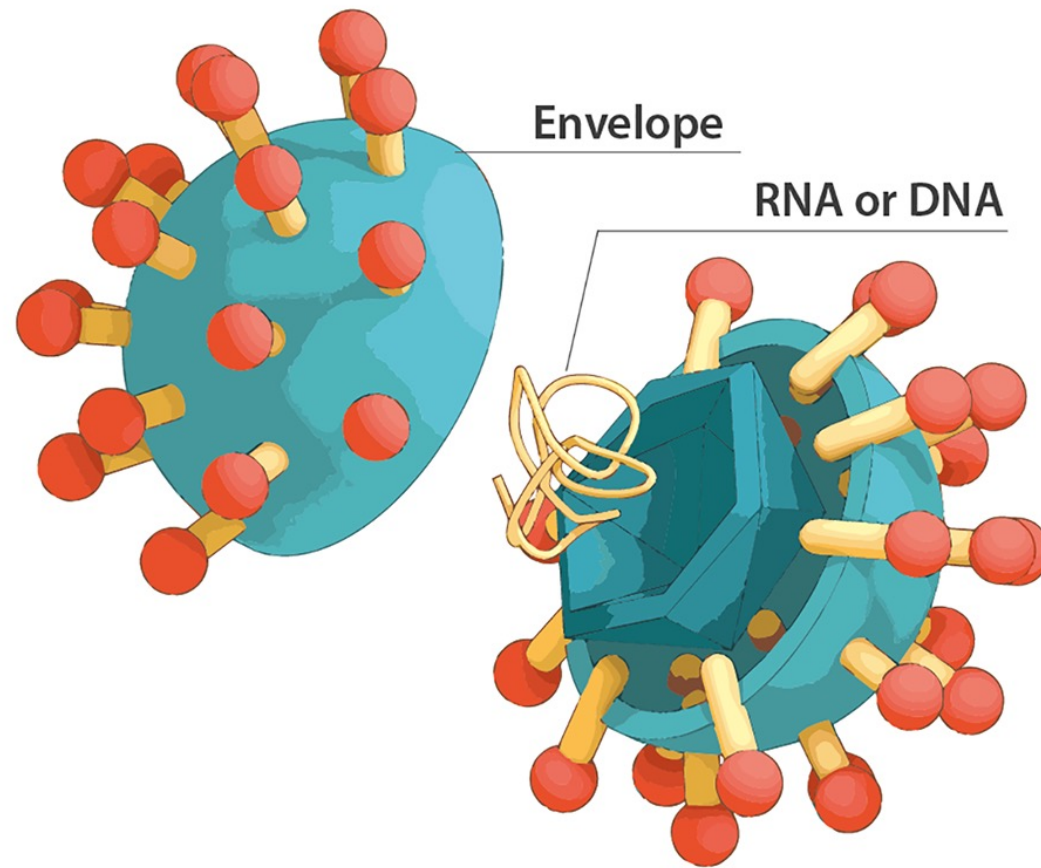
RNA extracted from	+ vRNA (Mean ± SD)	Negative Control for + vRNA	- vRNA (Mean ± SD)	Negative Control for - vRNA
SARS-CoV-2-infected cells	17.9 ± 0.2	NA	32 ± 0.9	NA
Medium from washing virus-attached salmon	24.1 ± 0.2	NA	NA	NA
Mock infected cells	NA	NA	NA	NA
Plasmids control	17.6 ± 0.3	NA	16.6 ± 0.4	NA

Using IDT Primers and Probes for COVID-19 Testing





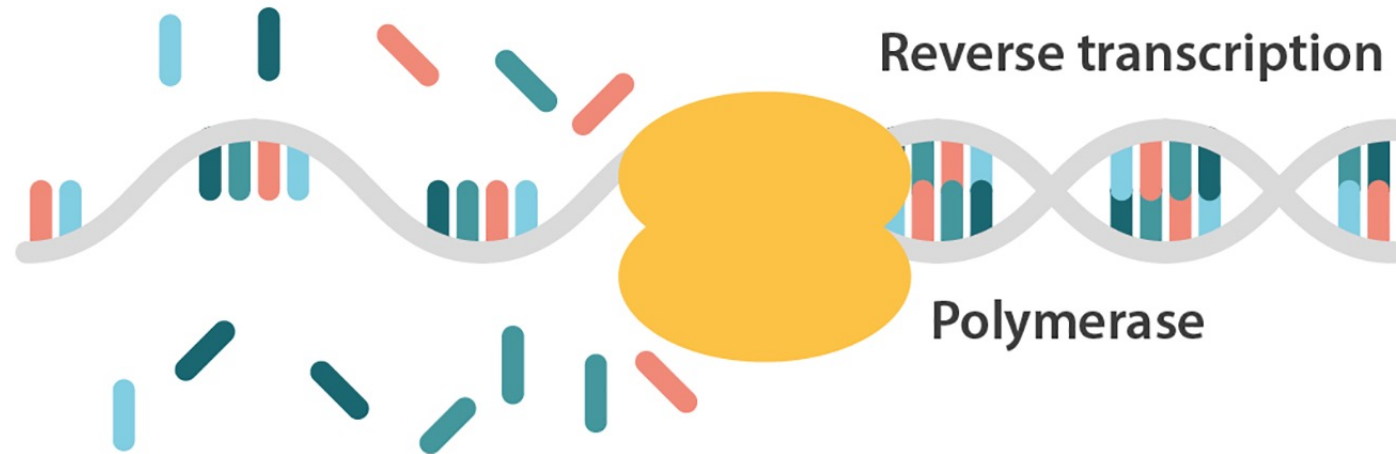
How is the COVID-19 virus detected using real time RT–PCR?



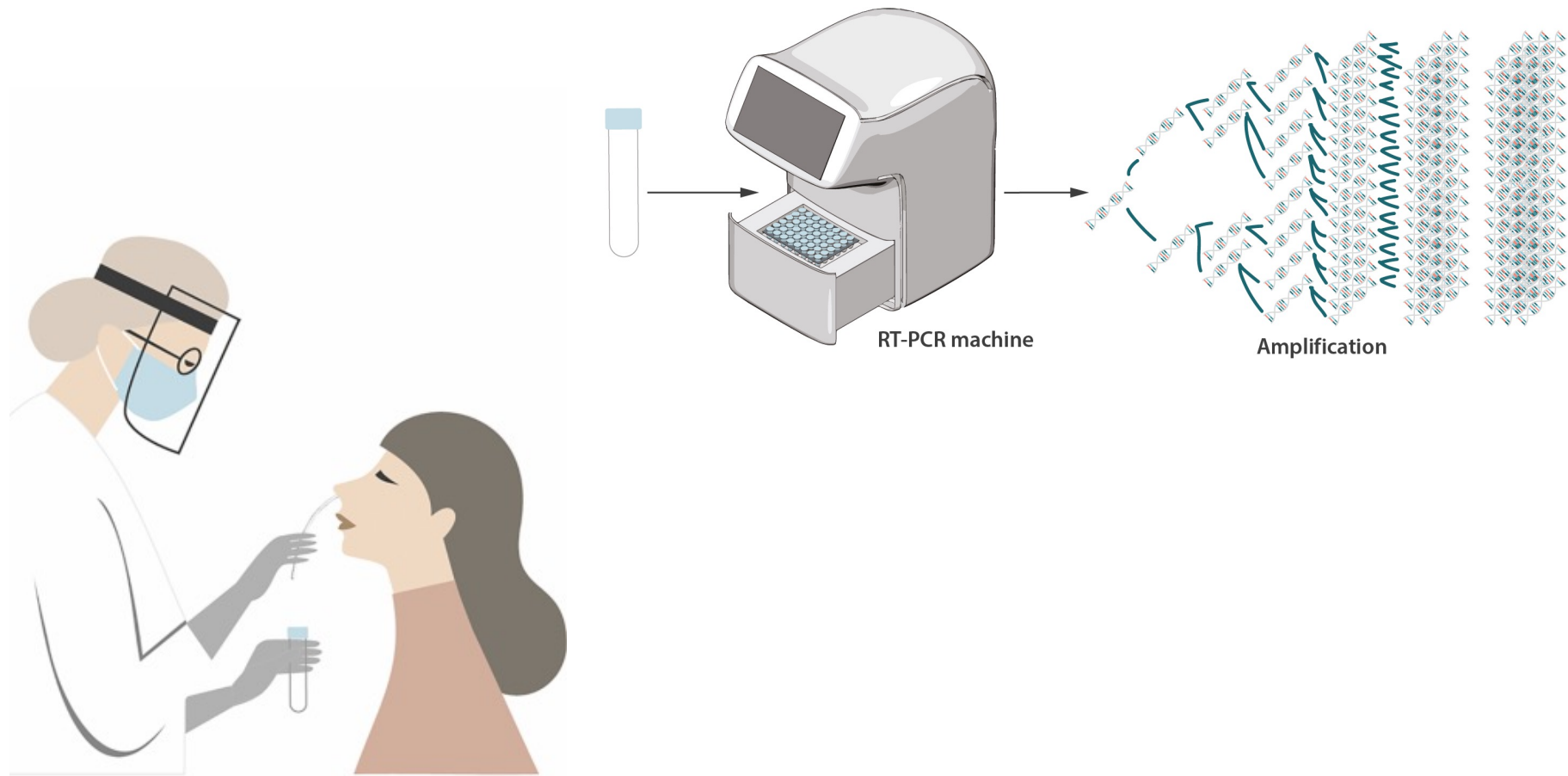
How is the COVID-19 virus detected using real time RT-PCR?



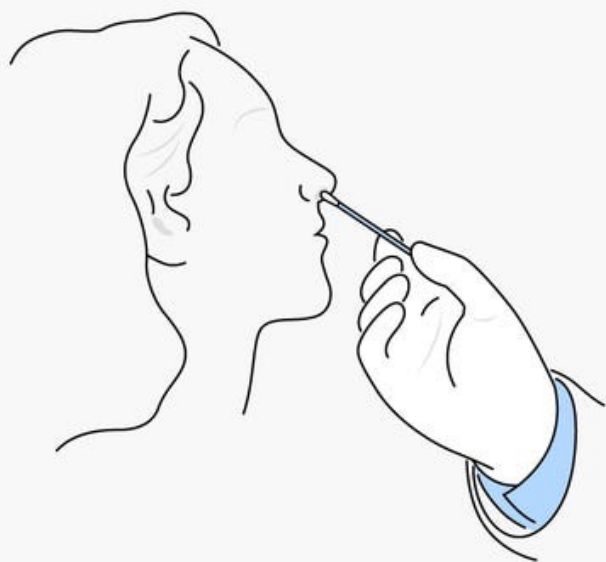
How is the COVID-19 virus detected using real time RT–PCR?



How is the COVID-19 virus detected using real time RT–PCR?



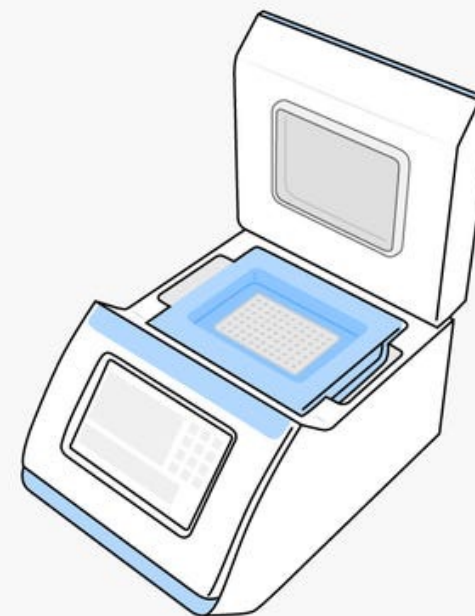
Swab

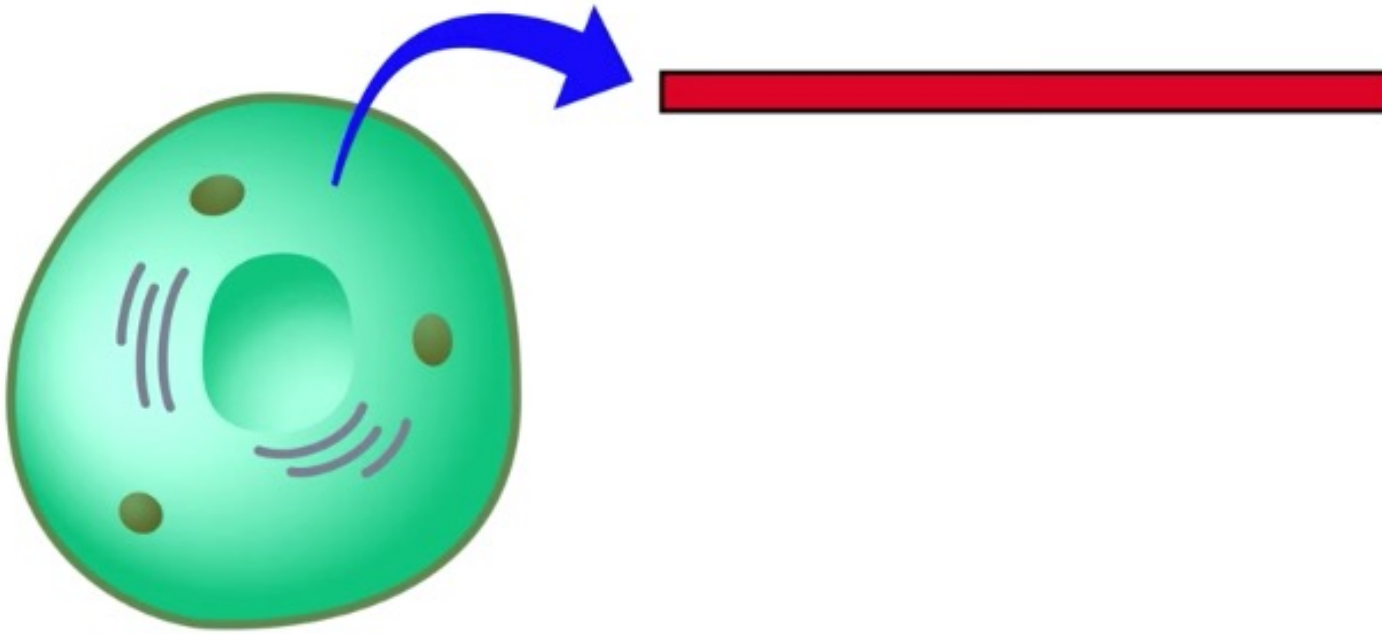


Lysis buffer

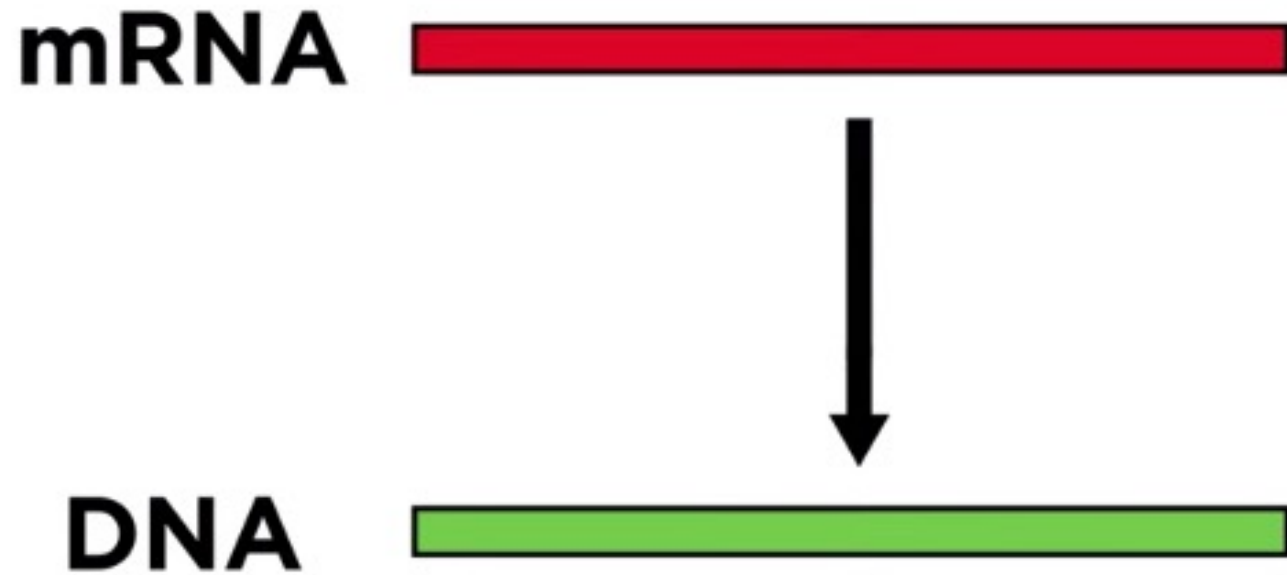


RT-PCR machine

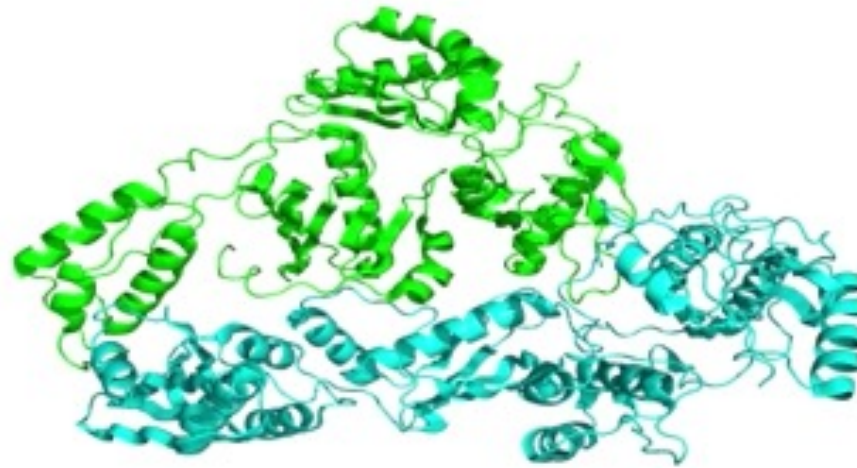
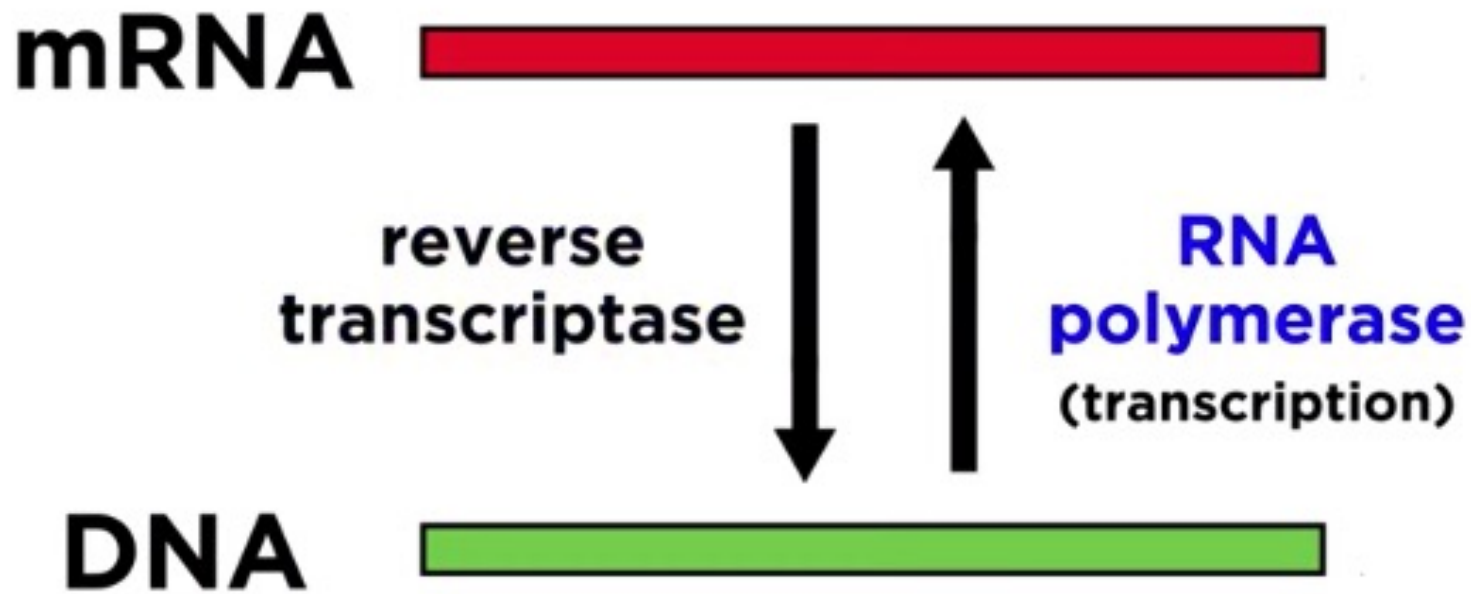




- 1) isolate the **mRNAs** from a cell
- 2) perform the **reverse transcriptase polymerase chain reaction (RT-PCR)**



**we are reverse engineering
the **DNA template** that would
have generated the mRNA**



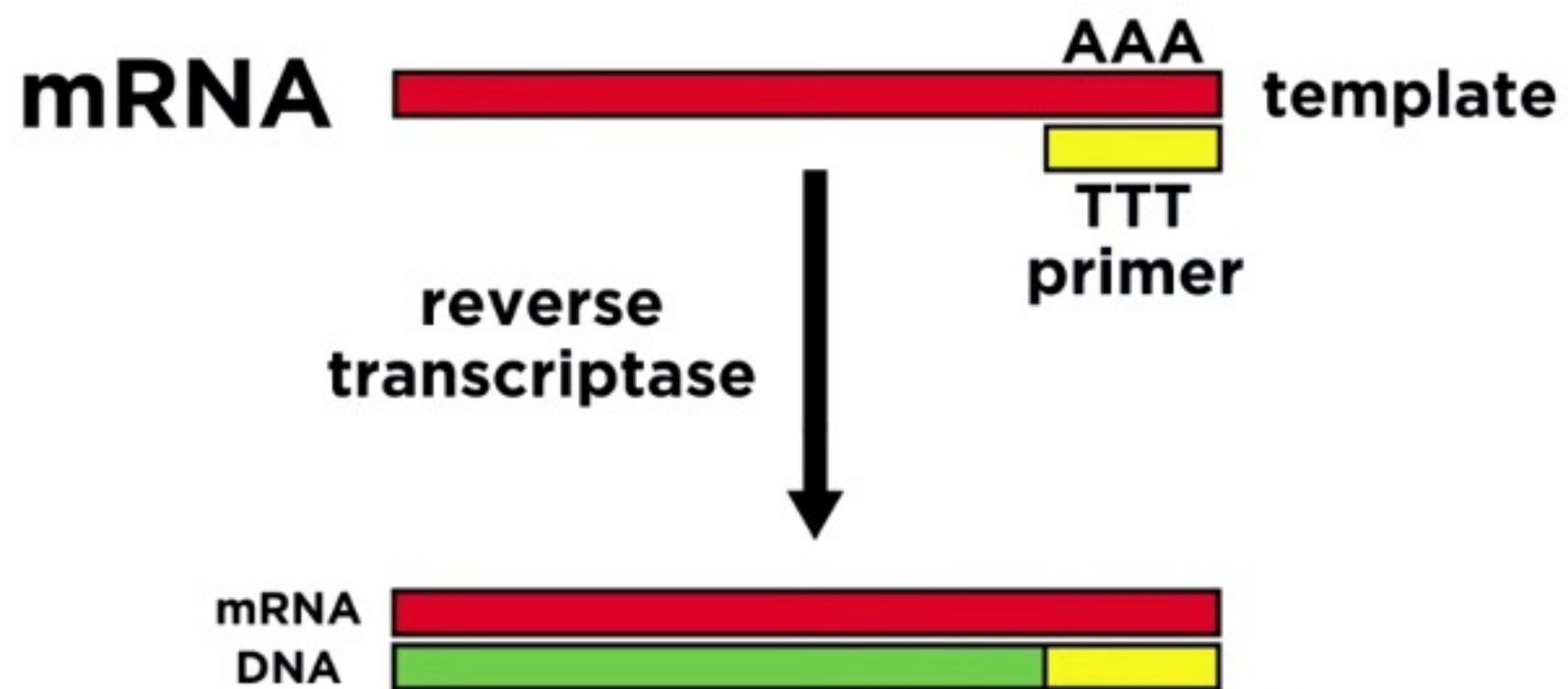
reverse transcriptase

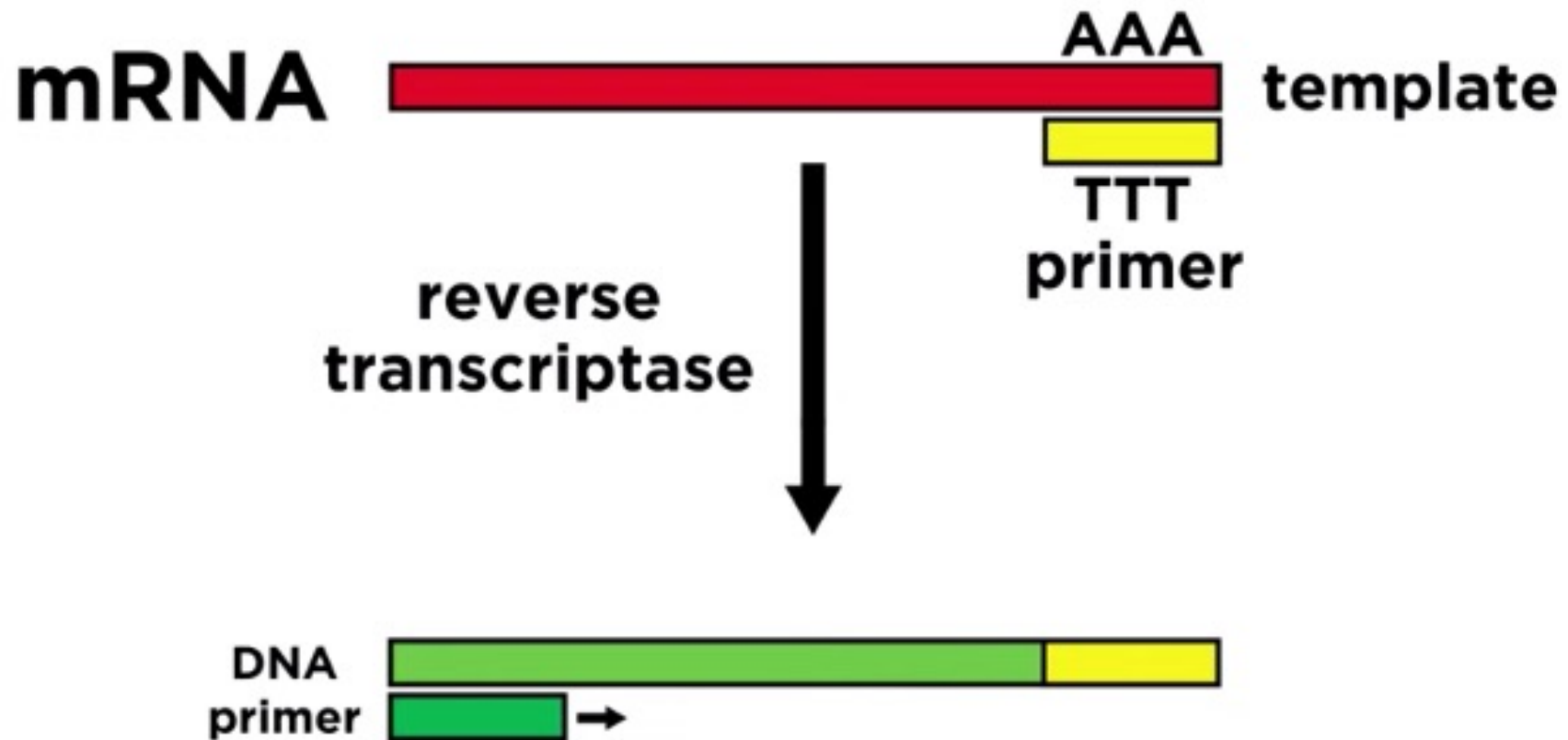
mRNA  **template**

**reverse
transcriptase**

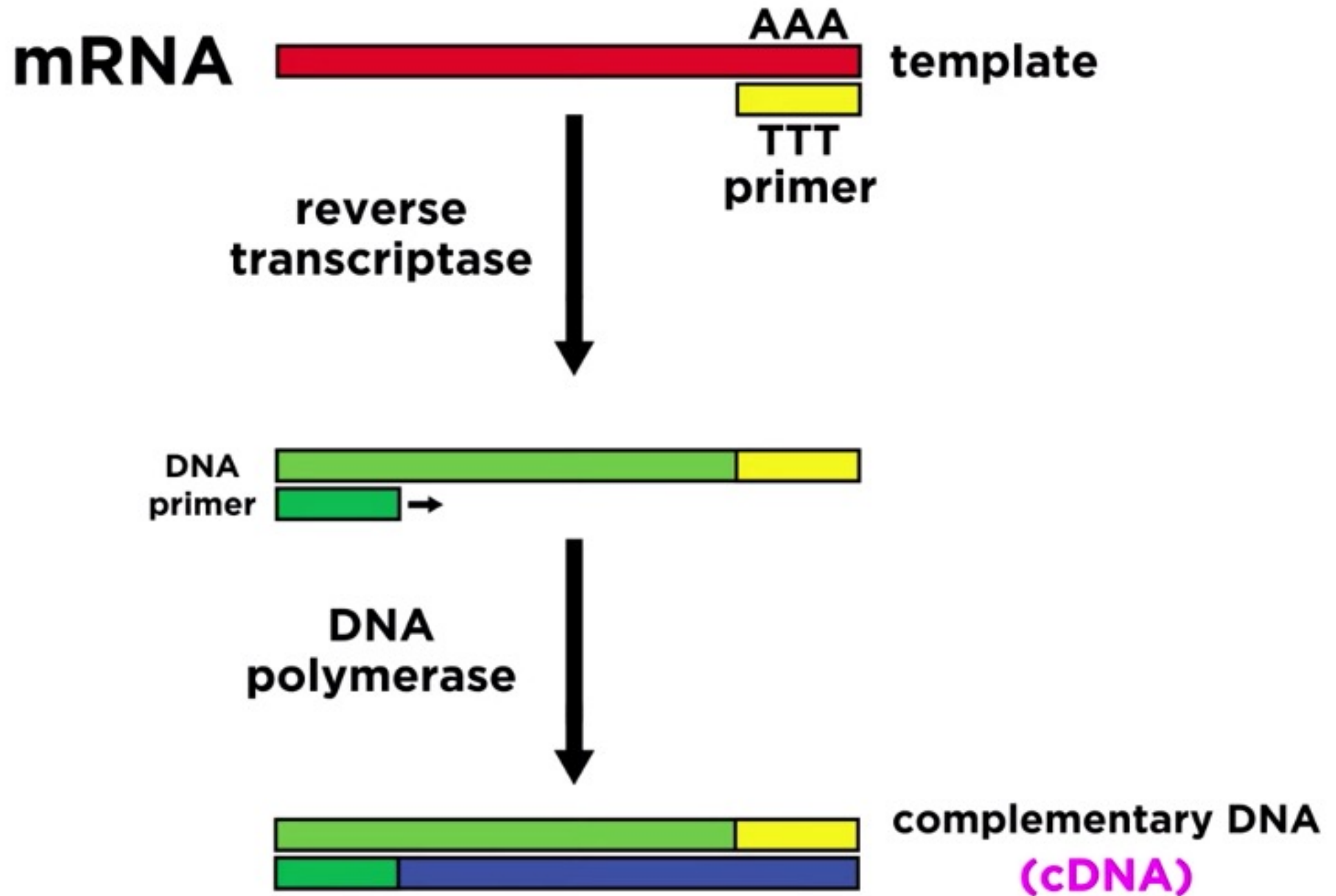


DNA  **single-stranded
complement**





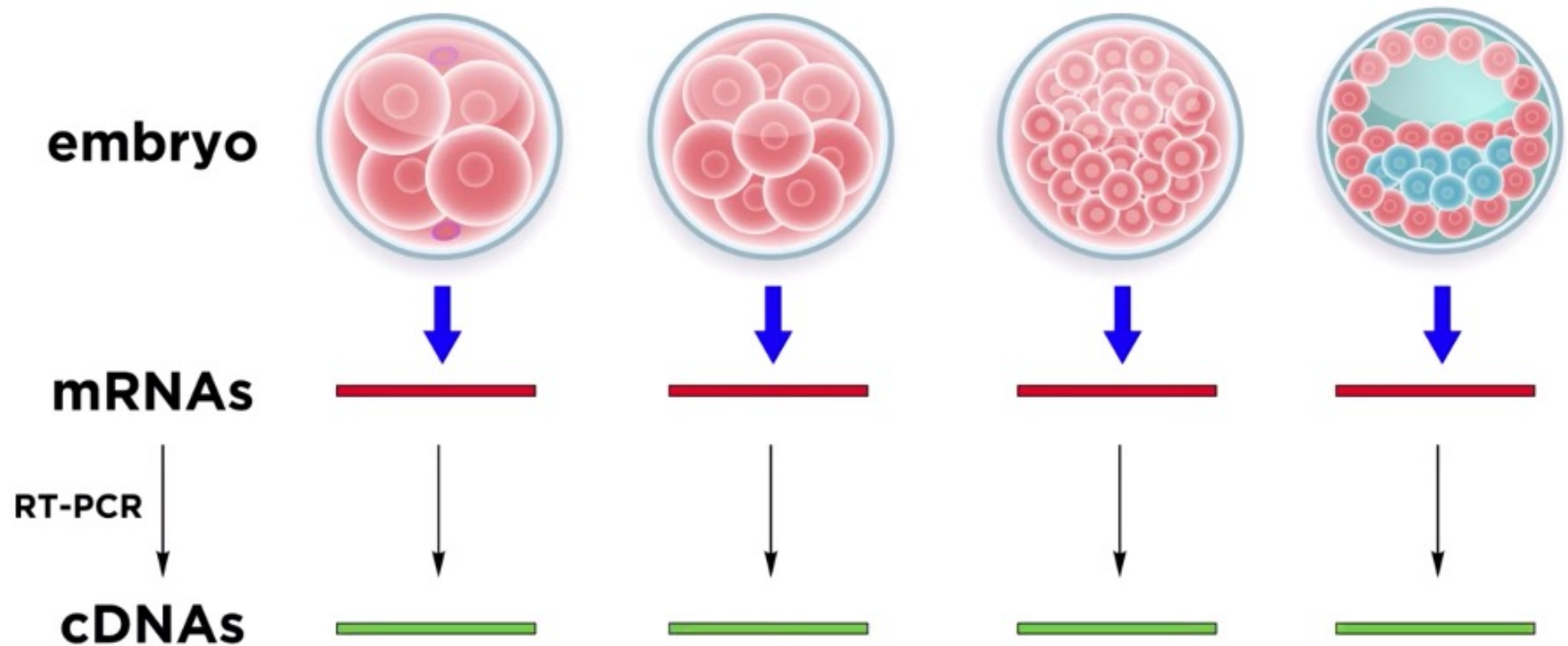
DNA polymerase begins to synthesize the other complementary DNA strand



When is a **gene being expressed** during embryonic development?



**we could isolate mRNAs from
different **stages of development****



gene of interest



corresponding cDNA



the **primer** used for
amplification is
specific to this gene

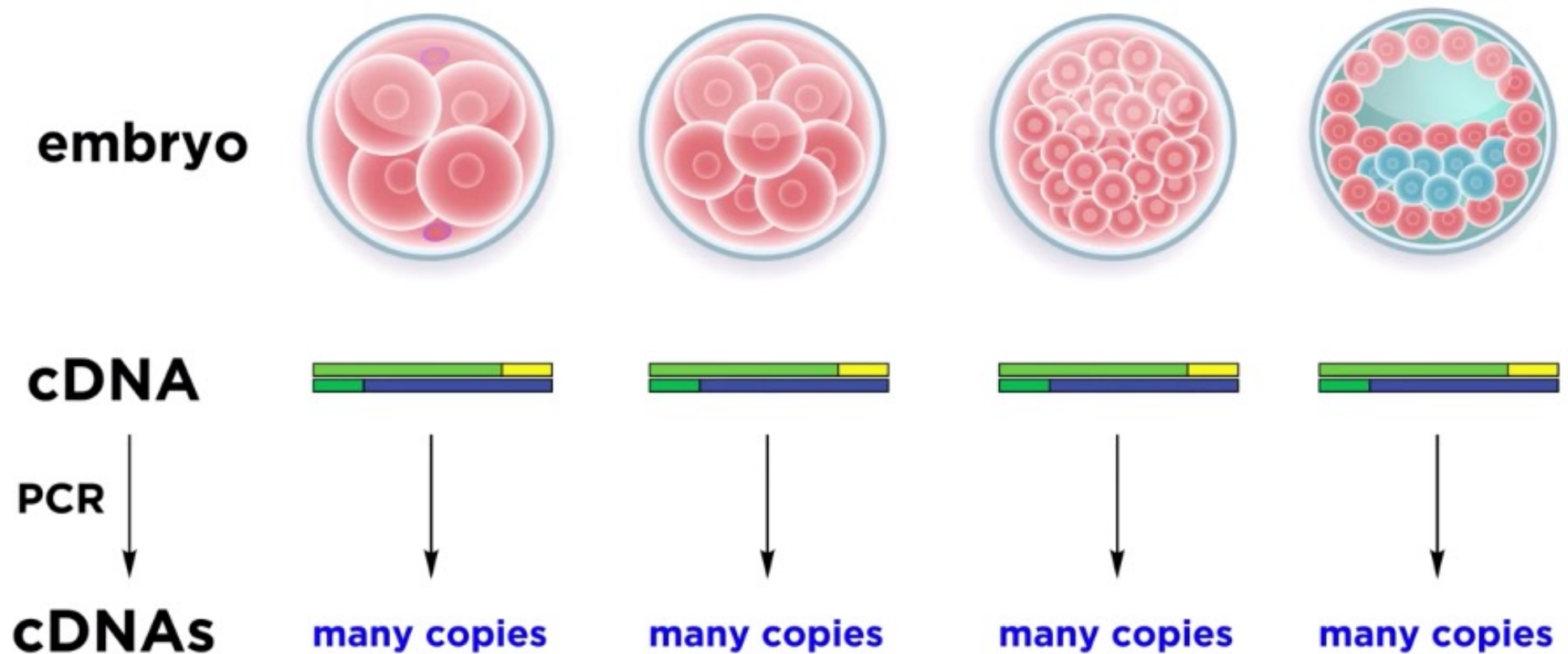


PCR



many **copies** of
this cDNA molecule

we can **amplify the cDNA** that represents the gene of interest for all the stages we are testing



Length of
DNA
fragment
(base pairs)

1,000 bp

900 bp

800 bp

700 bp

600 bp

500 bp

400 bp

300 bp

200 bp

100 bp

Stage 1

Stage 2

Stage 3

Stage 4

Stage 5

Stage 6



this is how we know when
a gene is being **expressed**

Gel Electrophoresis

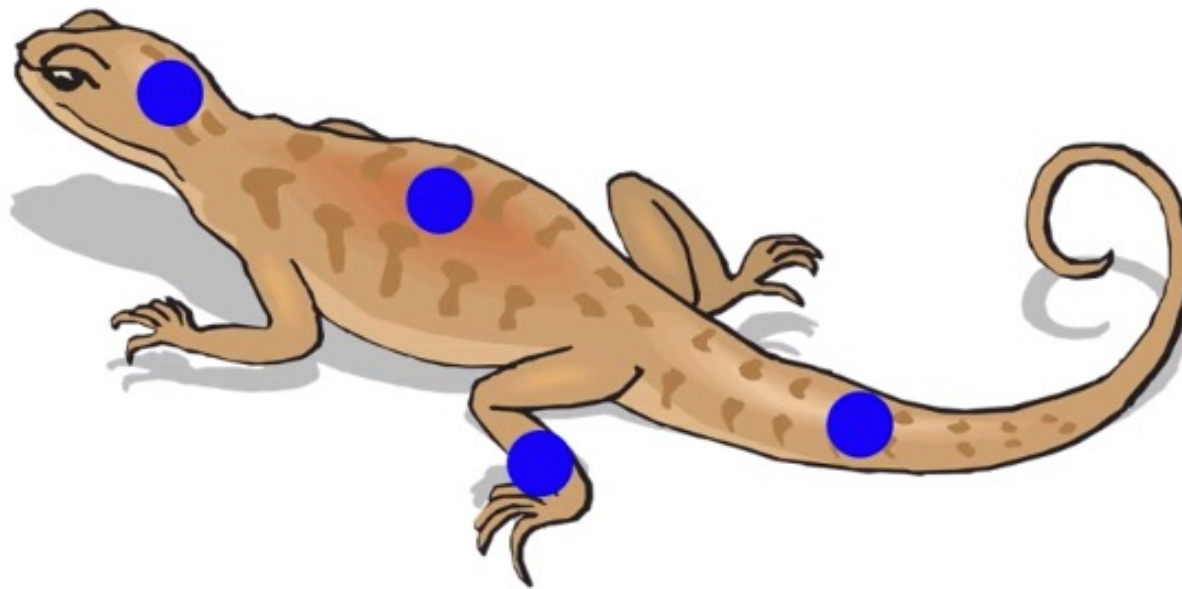


cDNA that
represents the
gene of interest



mRNA is produced
during that **stage**
of development

we can determine which **tissues** in an organism are expressing a particular gene



mRNA → cDNA → gene of interest



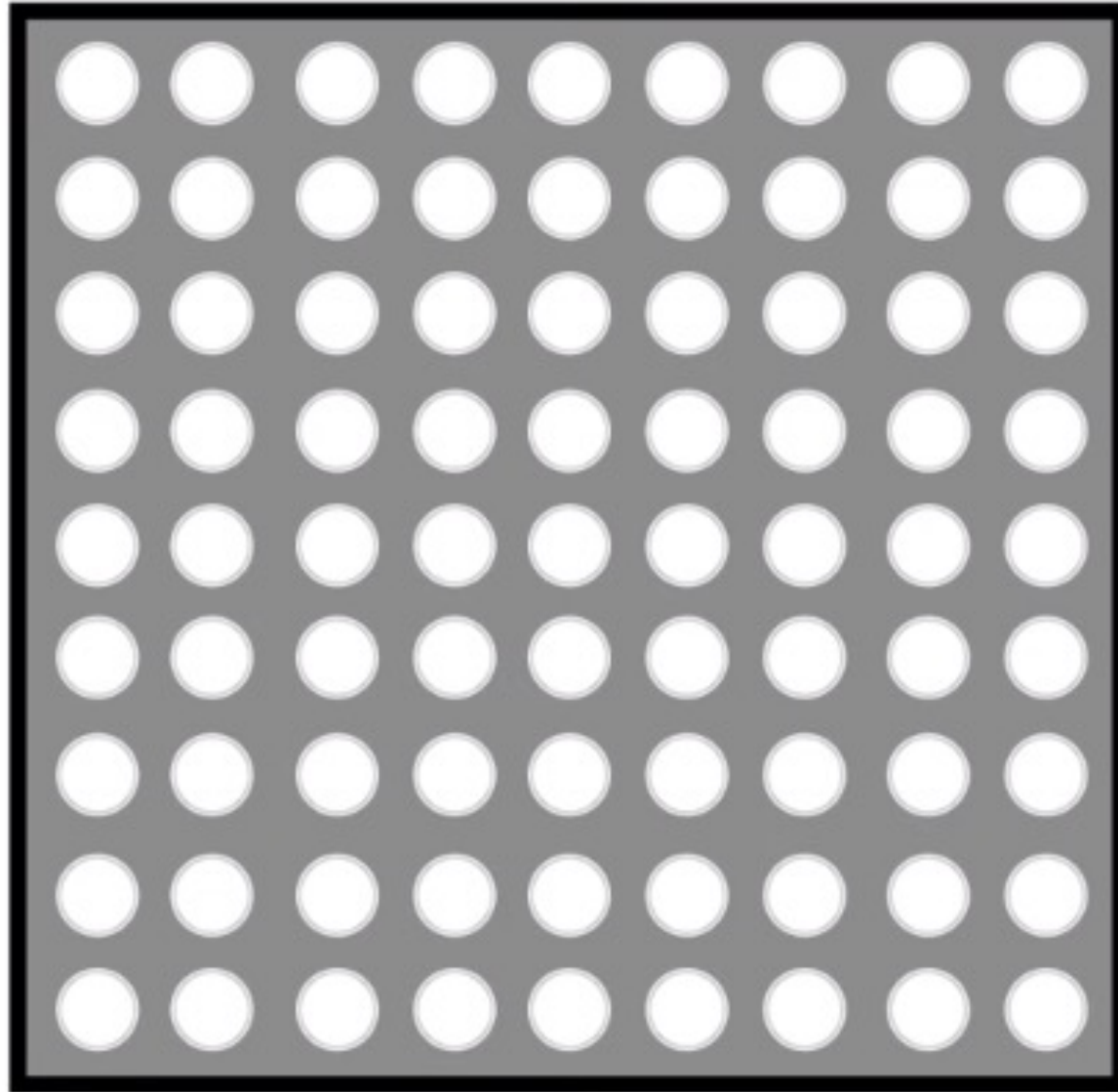
Microarray

we want to understand
genome-wide expression

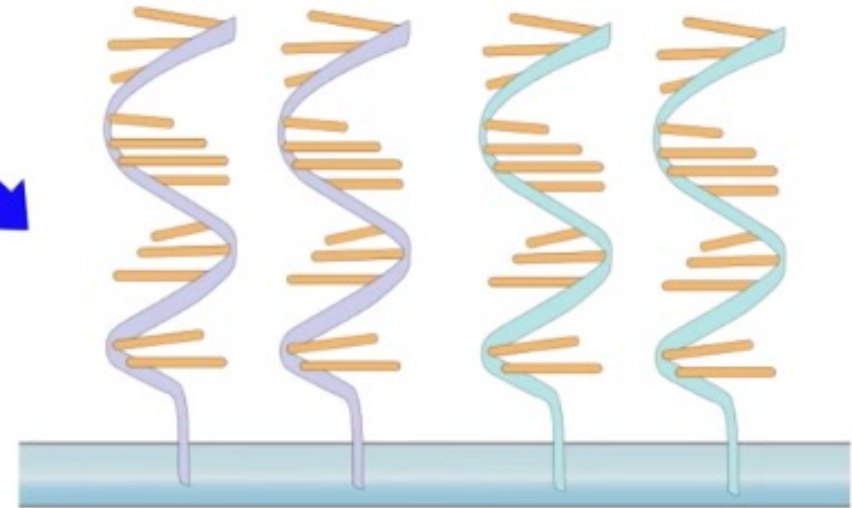
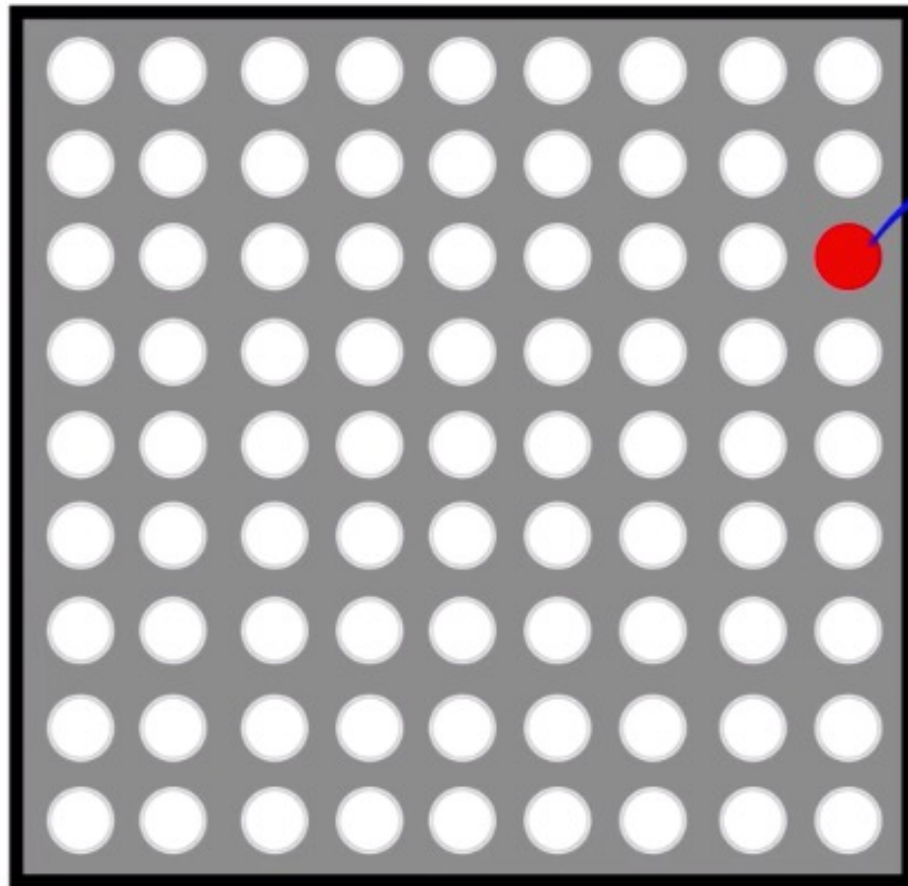


How do all the different genes act together to
produce and maintain this functioning organism?

DNA Microarray Assay

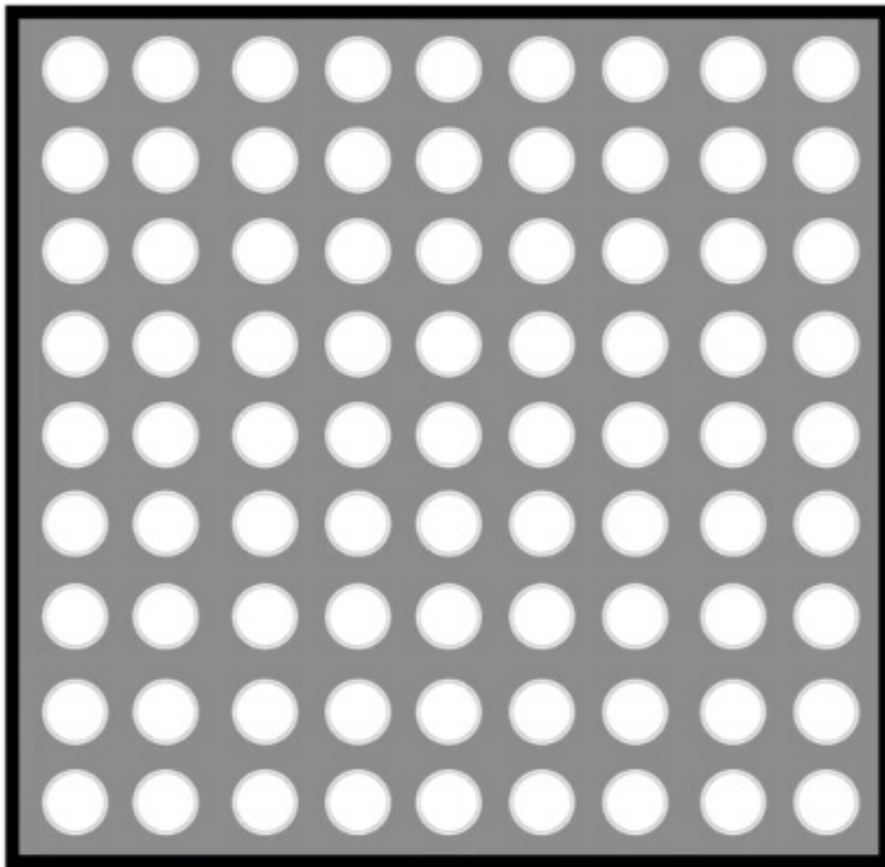


DNA Microarray Assay

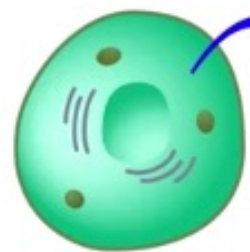


- single-stranded **DNA probes** tethered to a solid support
- these represent a particular **gene**

DNA Microarray Assay



this is a grid containing
many or even all of the
genes of an organism



cell of
interest

mRNA



AAA

template



TTT
primer

reverse
transcriptase



DNA
primer



DNA
polymerase

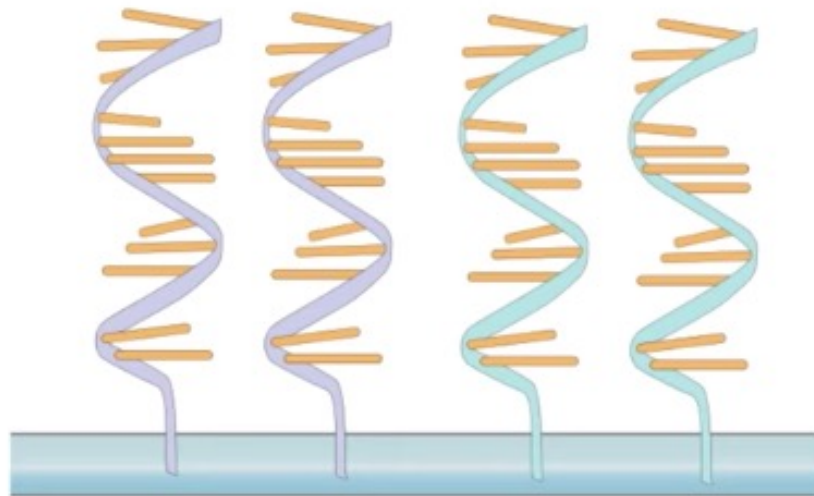
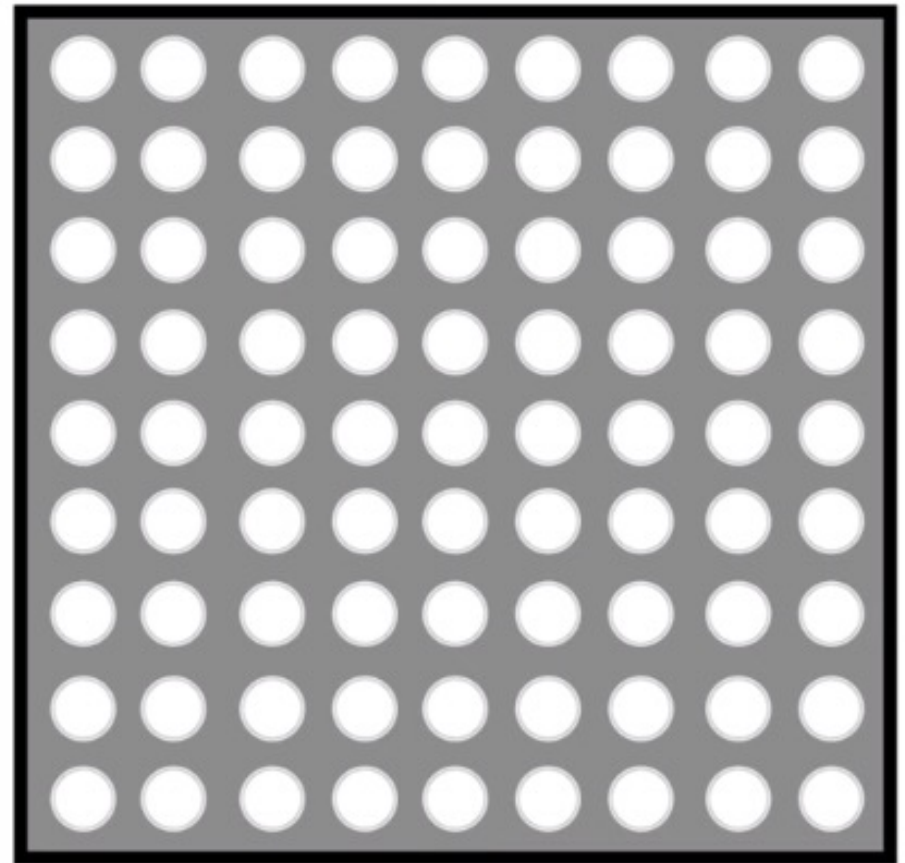


complementary DNA
(cDNA)

the cDNAs are labeled
with **fluorescent tags**



and introduced
to the **array**



hybridization with **probes**

DNA



template for transcription

transcription



mRNA



reverse
transcription



cDNA



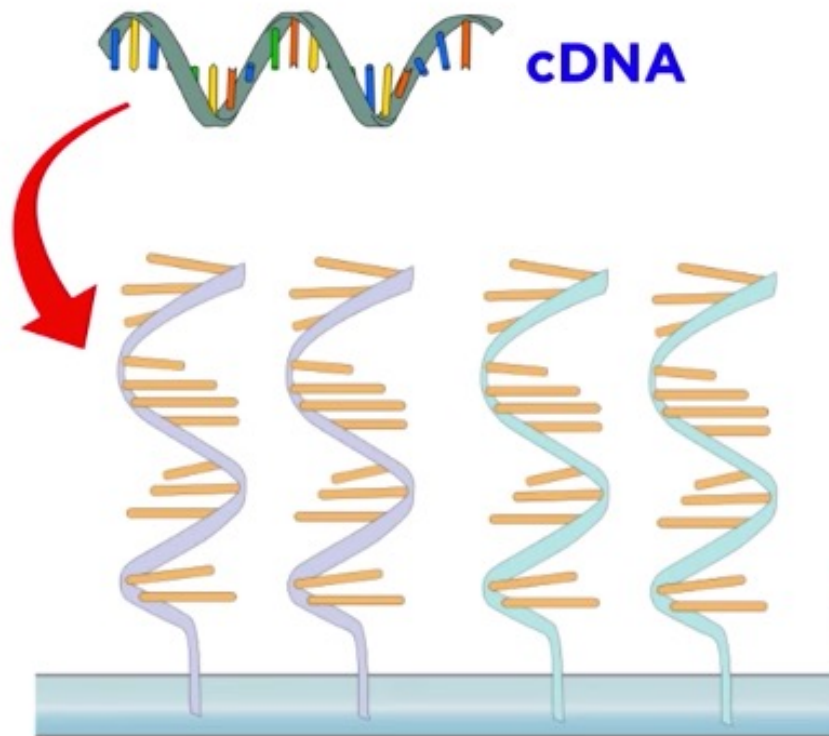
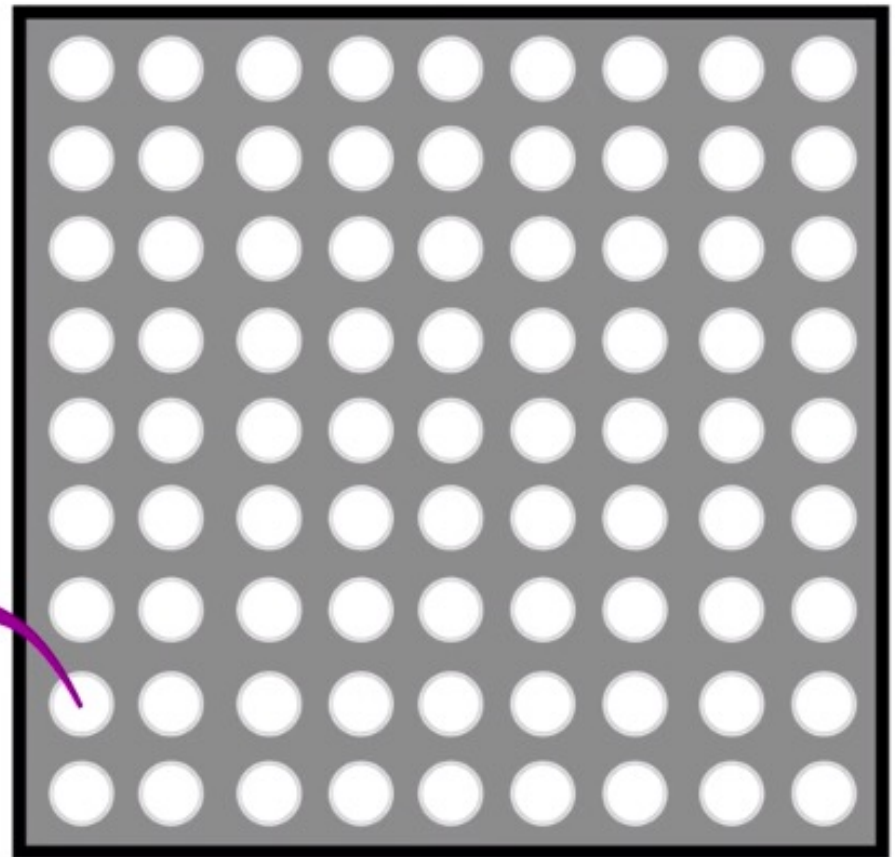
identical to DNA template

complementary to template

the cDNAs are labeled
with **fluorescent tags**



and introduced
to the **array**



binding means **gene is expressed**

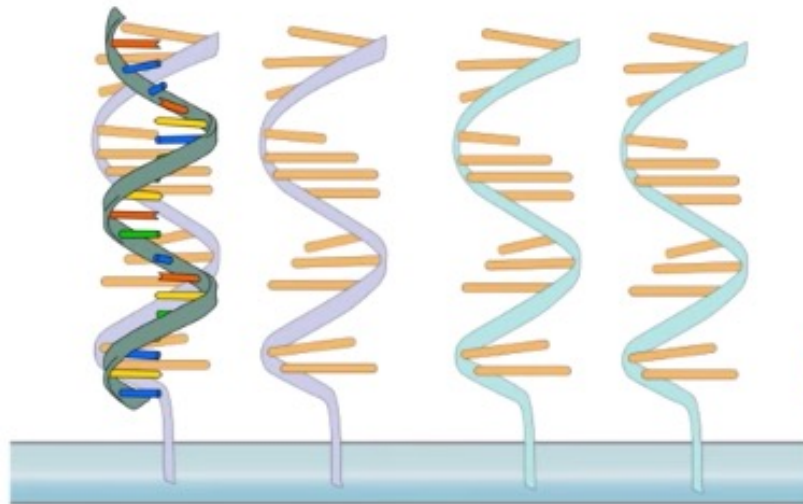
the cDNAs are labeled
with **fluorescent tags**



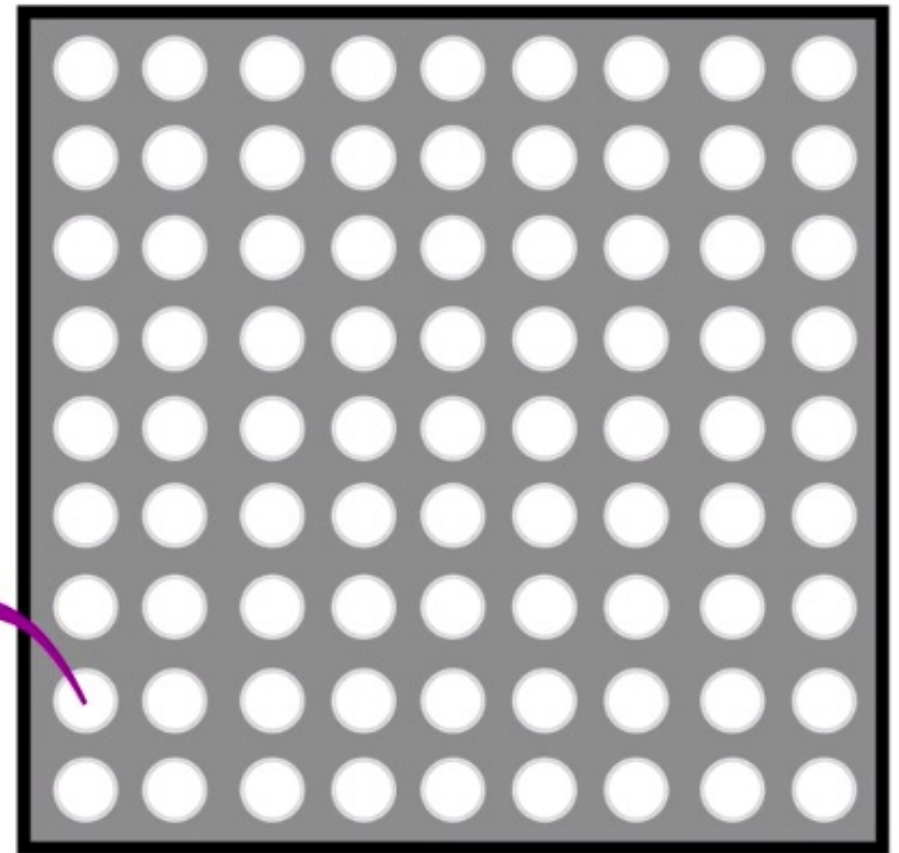
and introduced
to the **array**



unbound cDNA
is washed away



binding means **gene is expressed**

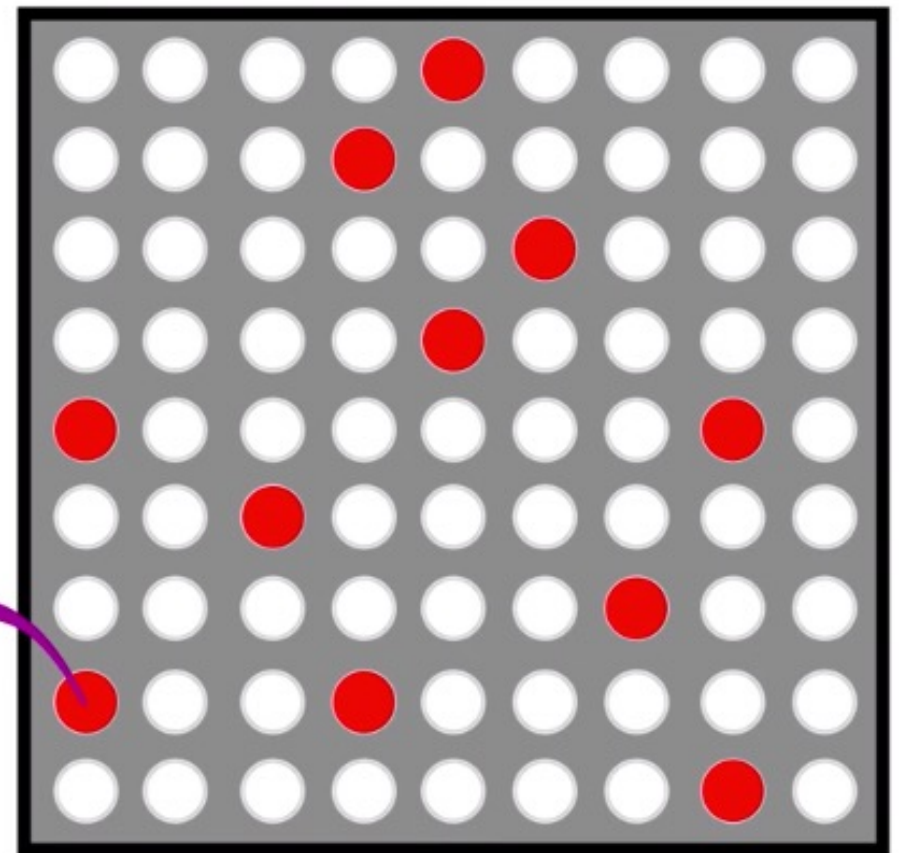
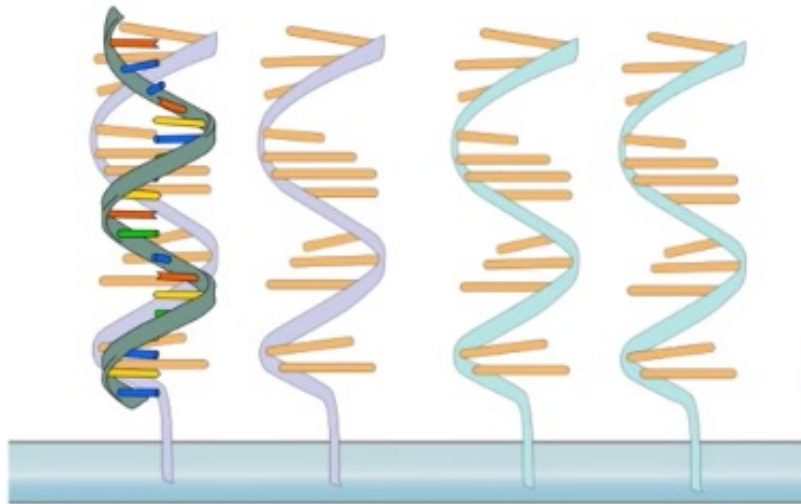


the cDNAs are labeled
with **fluorescent tags**



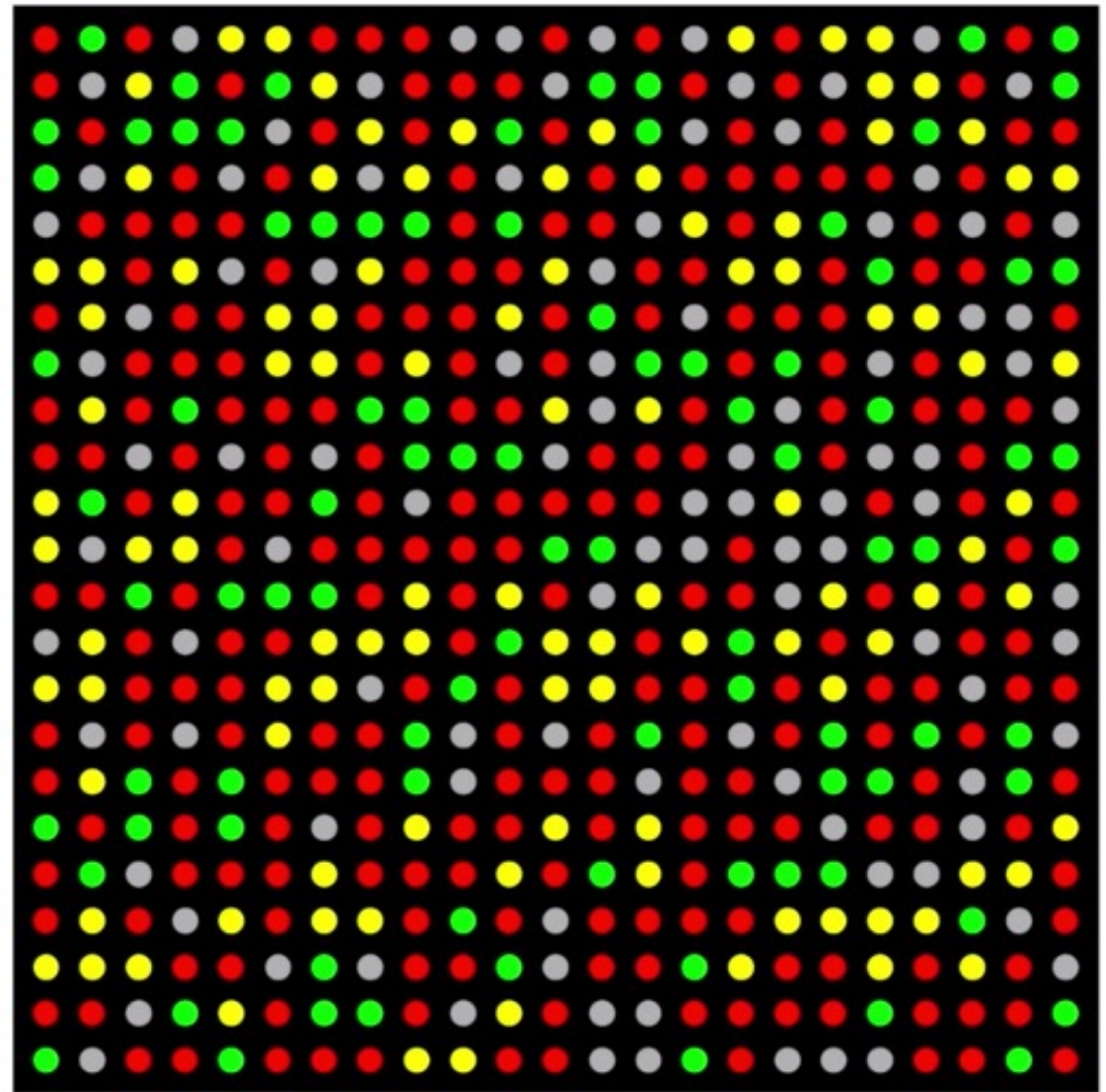
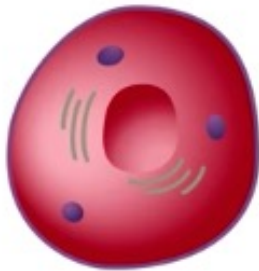
and introduced
to the **array**

fluorescence is an
indicator of binding

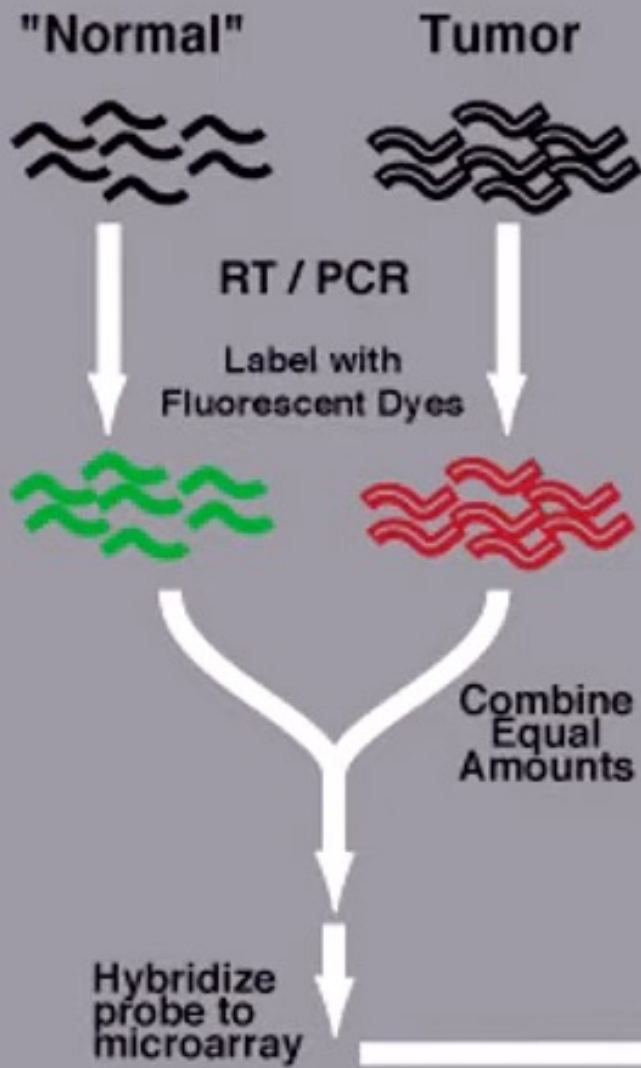


binding means **gene is expressed**

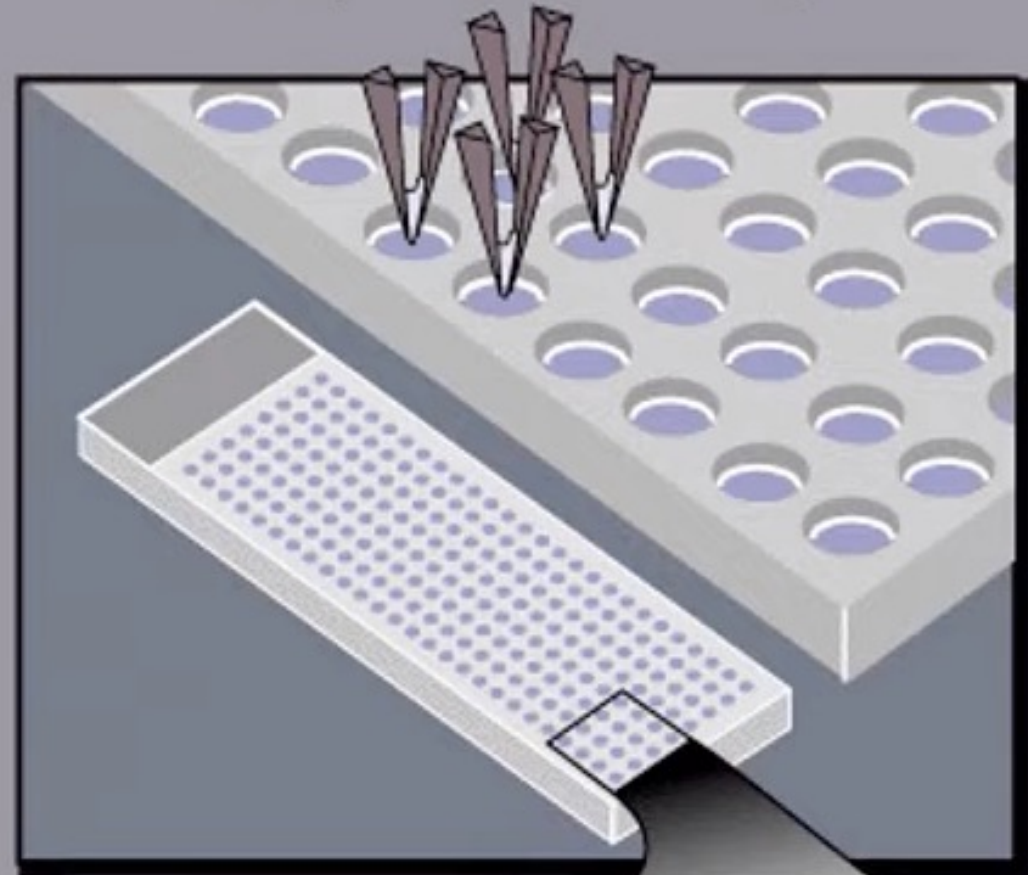
compare gene
expression in a
regular cell and
a **cancer cell**



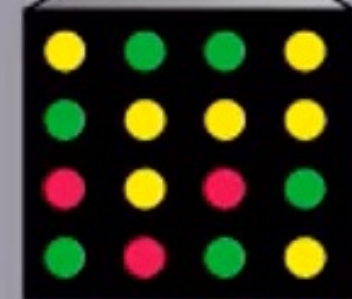
Prepare cDNA Probe



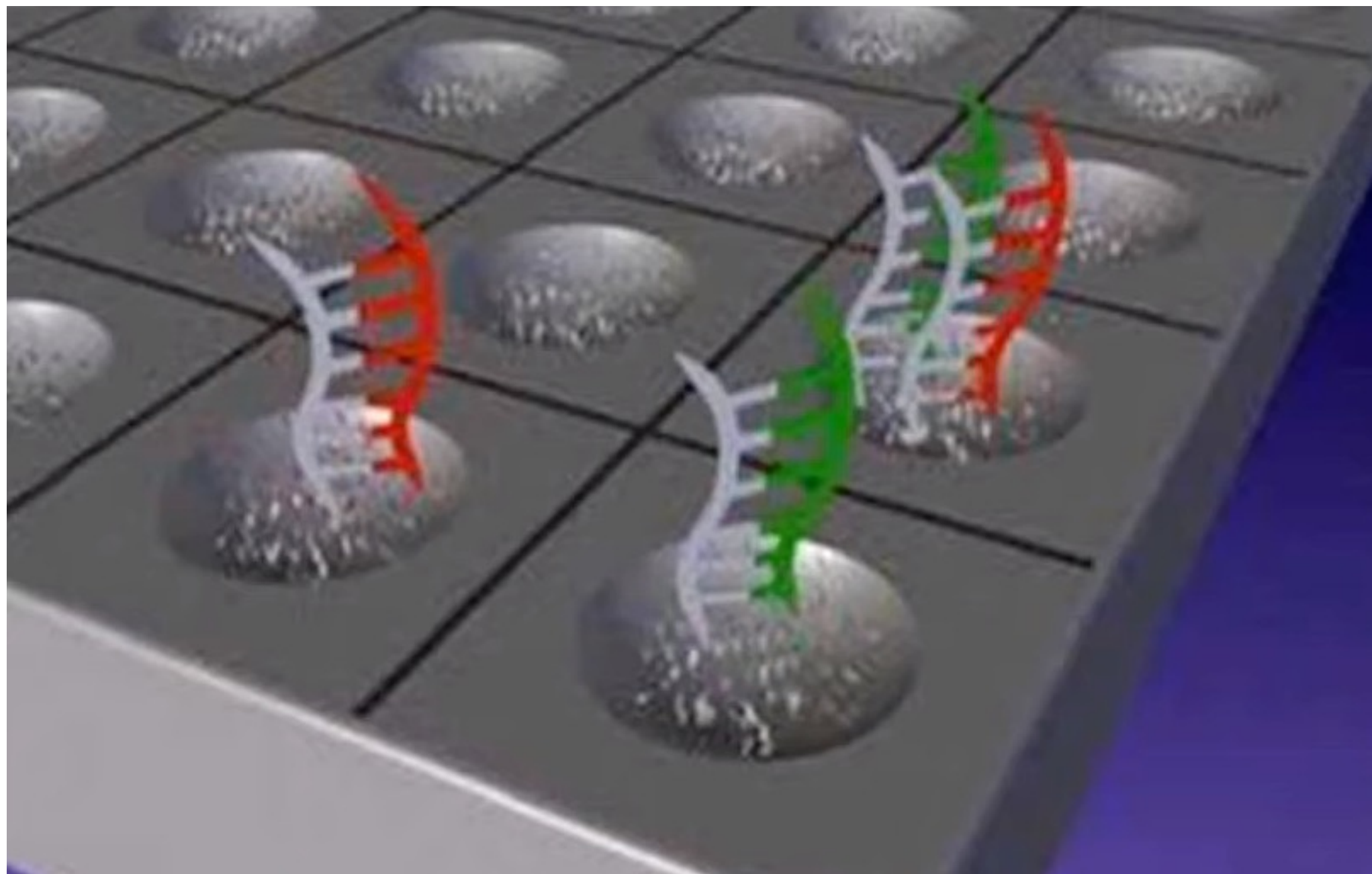
Prepare Microarray



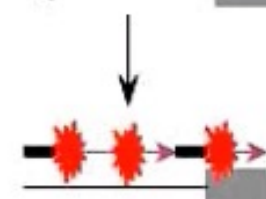
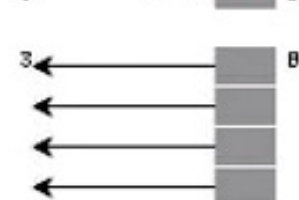
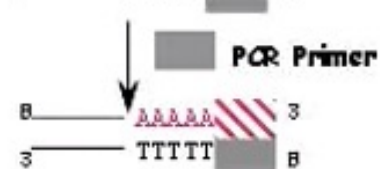
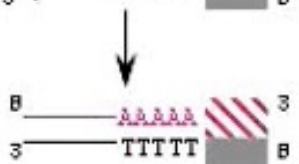
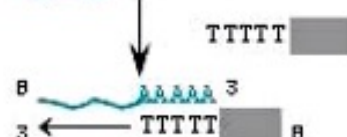
SCAN



Microarray Technology



Reference



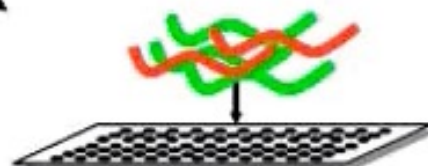
Reverse Transcription

Second Strand synthesis

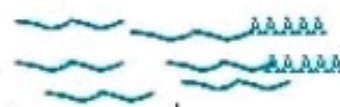
Amplification (Linear PCR)

Random Priming (with dUTP-Cy3 or dUTP-Cy5)

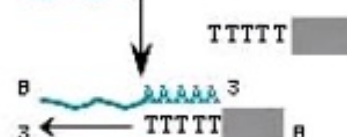
Hybridization



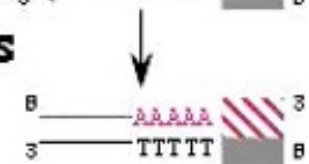
Test



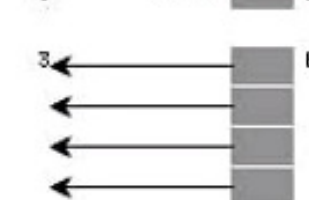
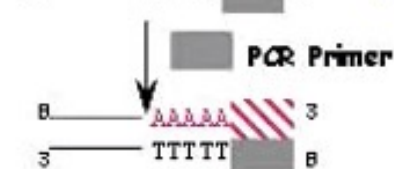
Total RNA



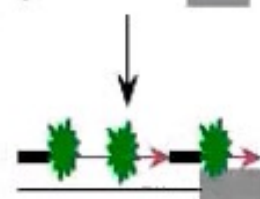
RNA/cDNA Hybrids



Double strand cDNA



Single strand cDNA



Fluorescent single
strand cDNA

Cy3+Cy5 labeled cDNA

Spotted microarray



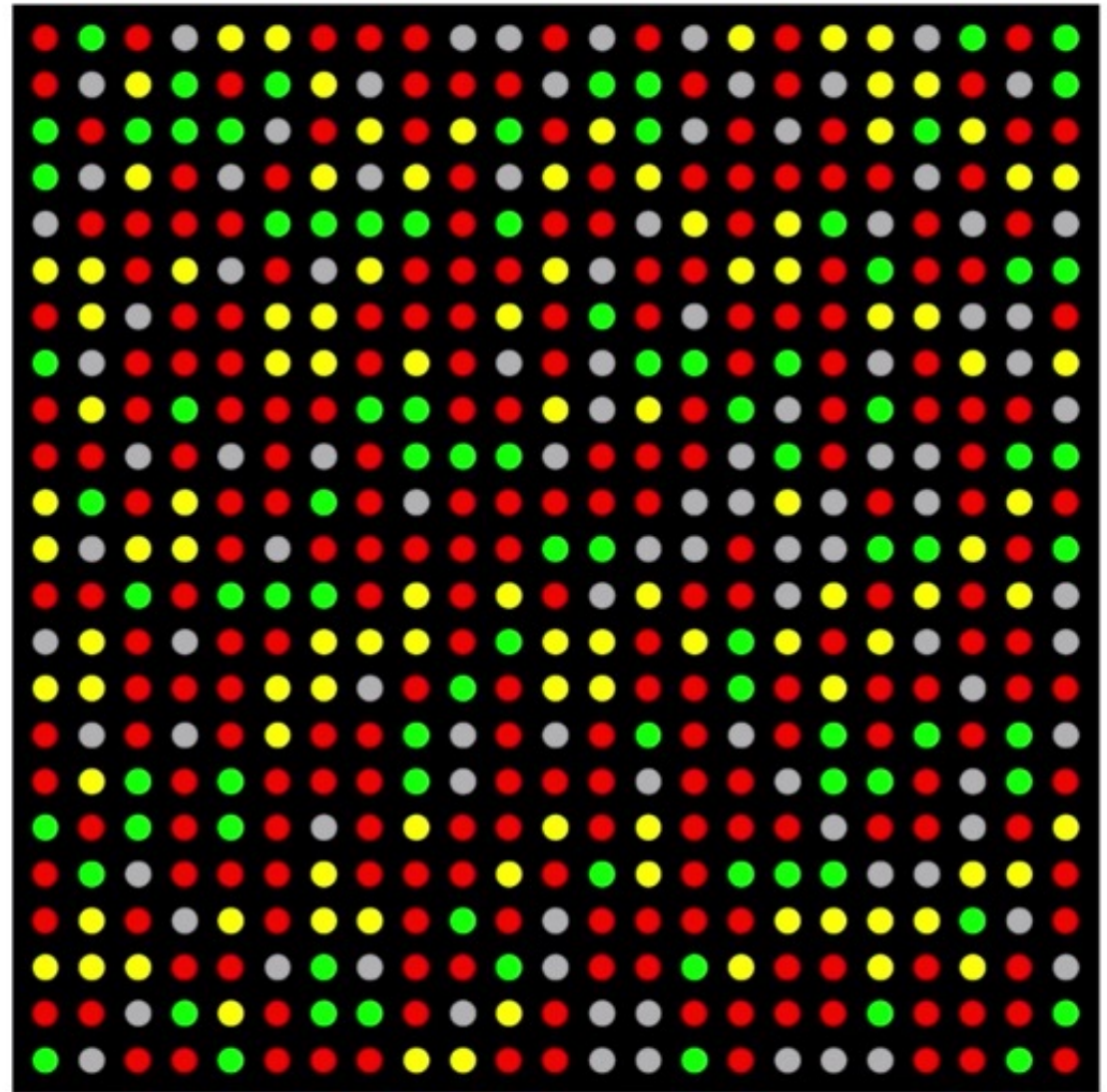
**cDNAs from tissue 1
were labeled red**

**cDNAs from tissue 2
were labeled green**

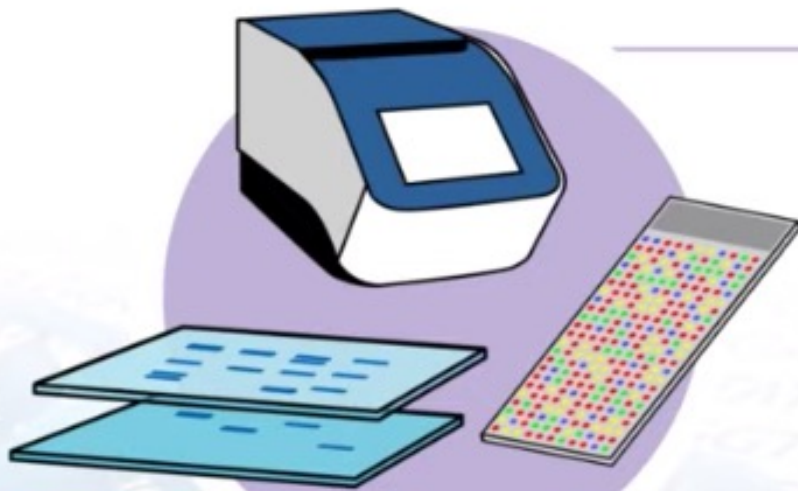
**red spot means gene
is expressed in tissue 1**

**green spot means gene
is expressed in tissue 2**

**yellow spot means both
cDNAs bind and gene is
expressed in both tissues**



Studying the role of genes in development and disease



Advantages

- Lower reagent cost
- Can be done in your own lab

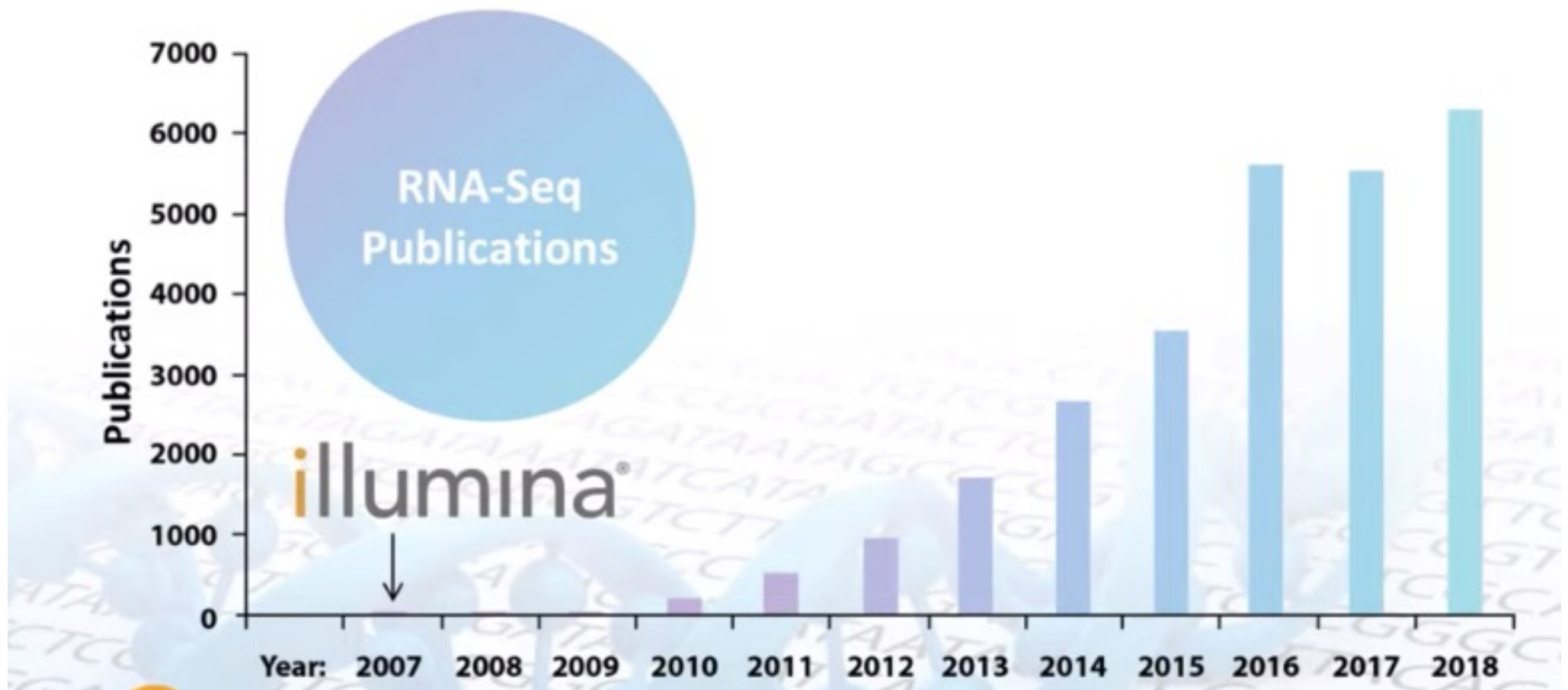
Considerations

- Requires prior knowledge of gene/mRNA
- Limited number of probes/targets



RNA-seq

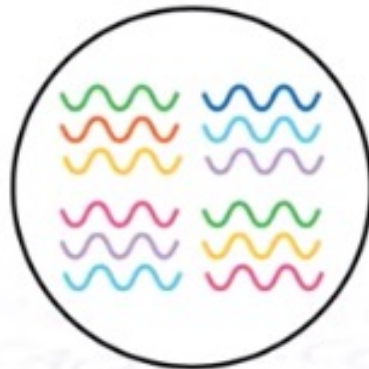
The prevalence of RNA-Seq in research



What is RNA-Seq?



**Begin with
Cells or tissues**



**Expressed
genes**

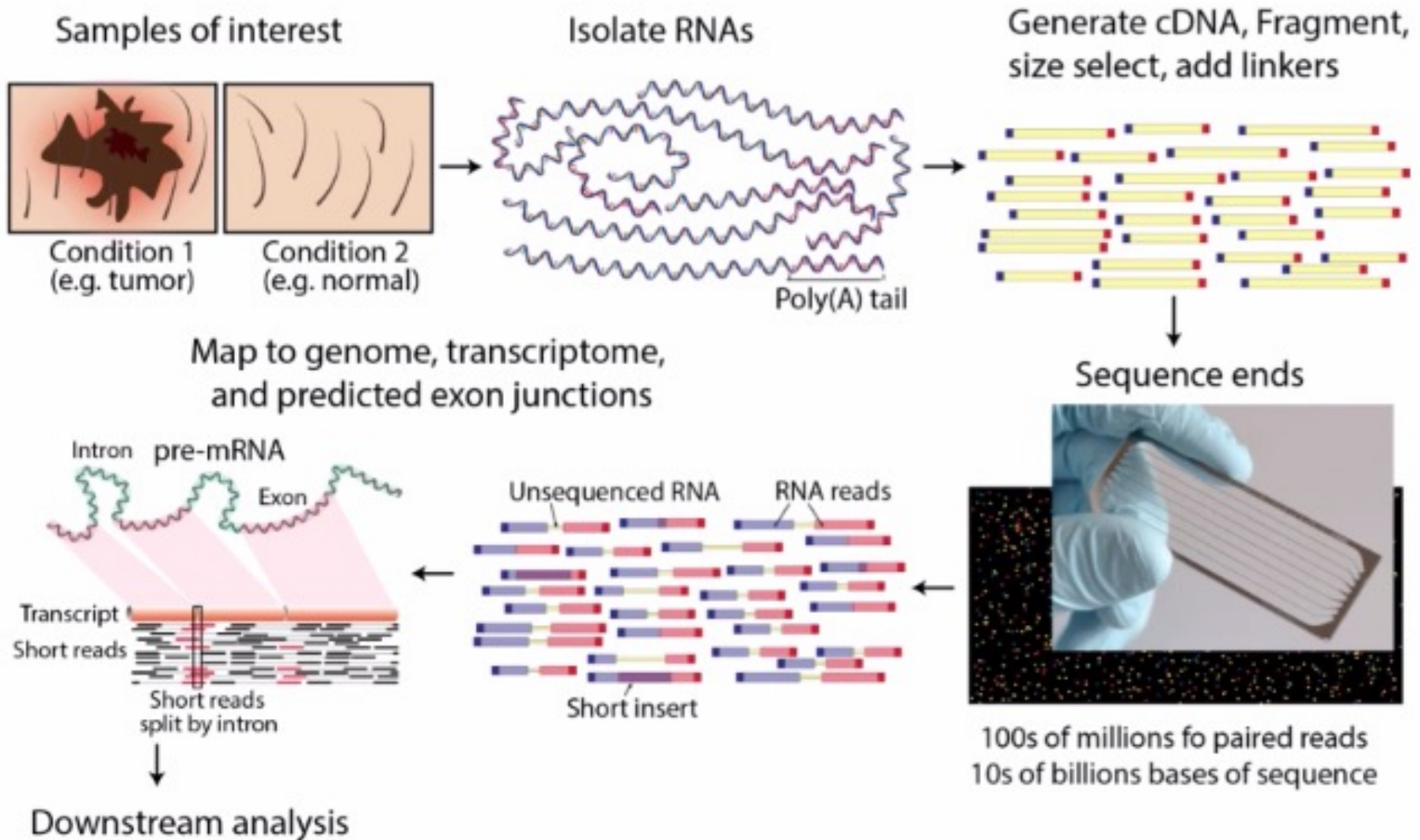


**High-Throughput
Sequencing**



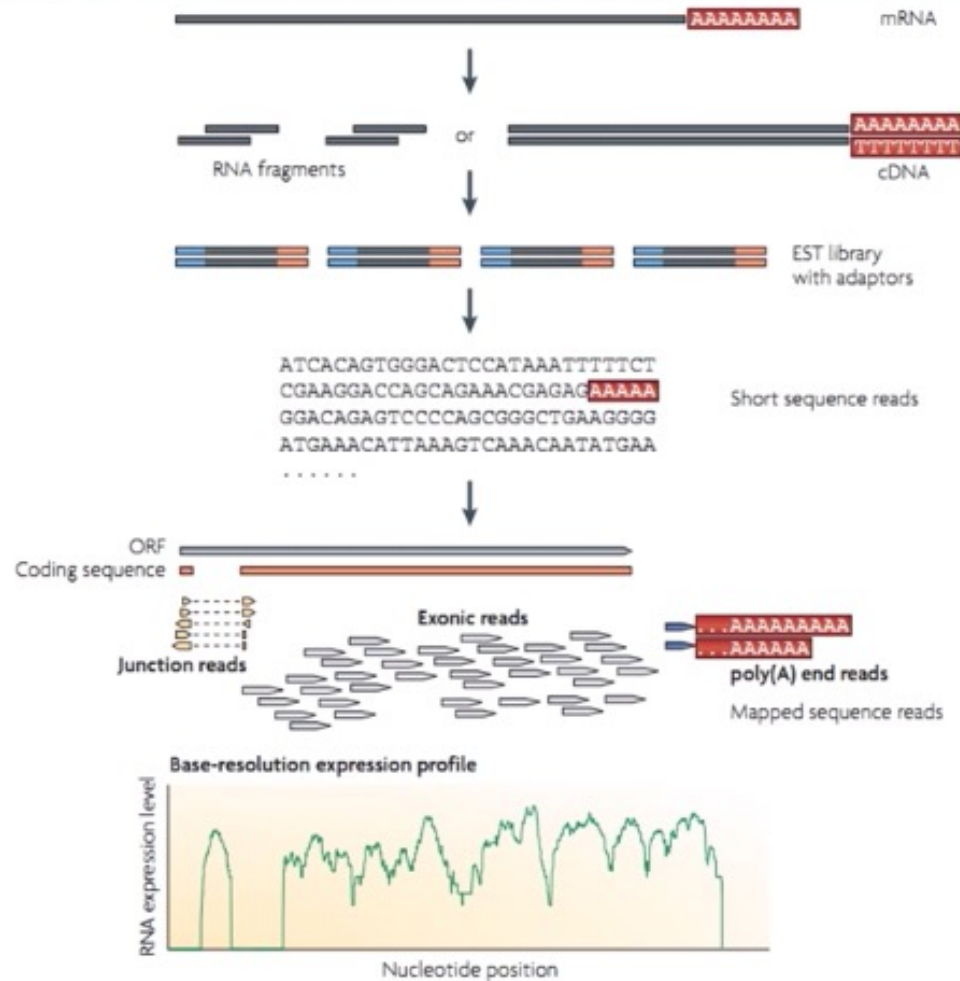
**Learn about changes in
gene expression
or novel splicing events**

RNA sequencing

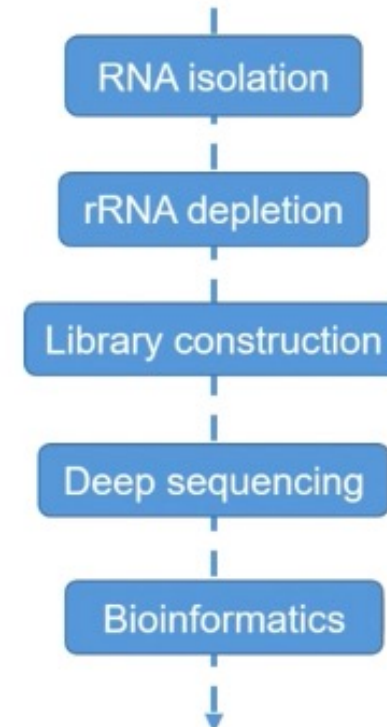


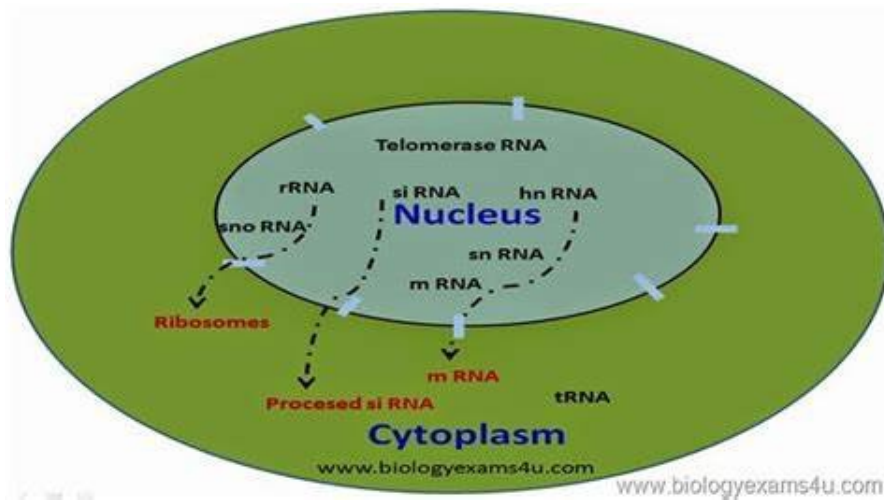


3 Workflow of RNA-seq by NGS

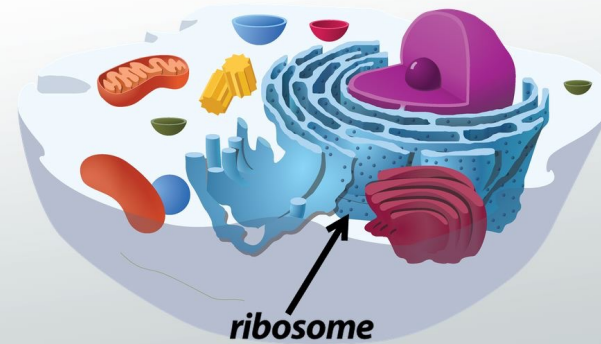


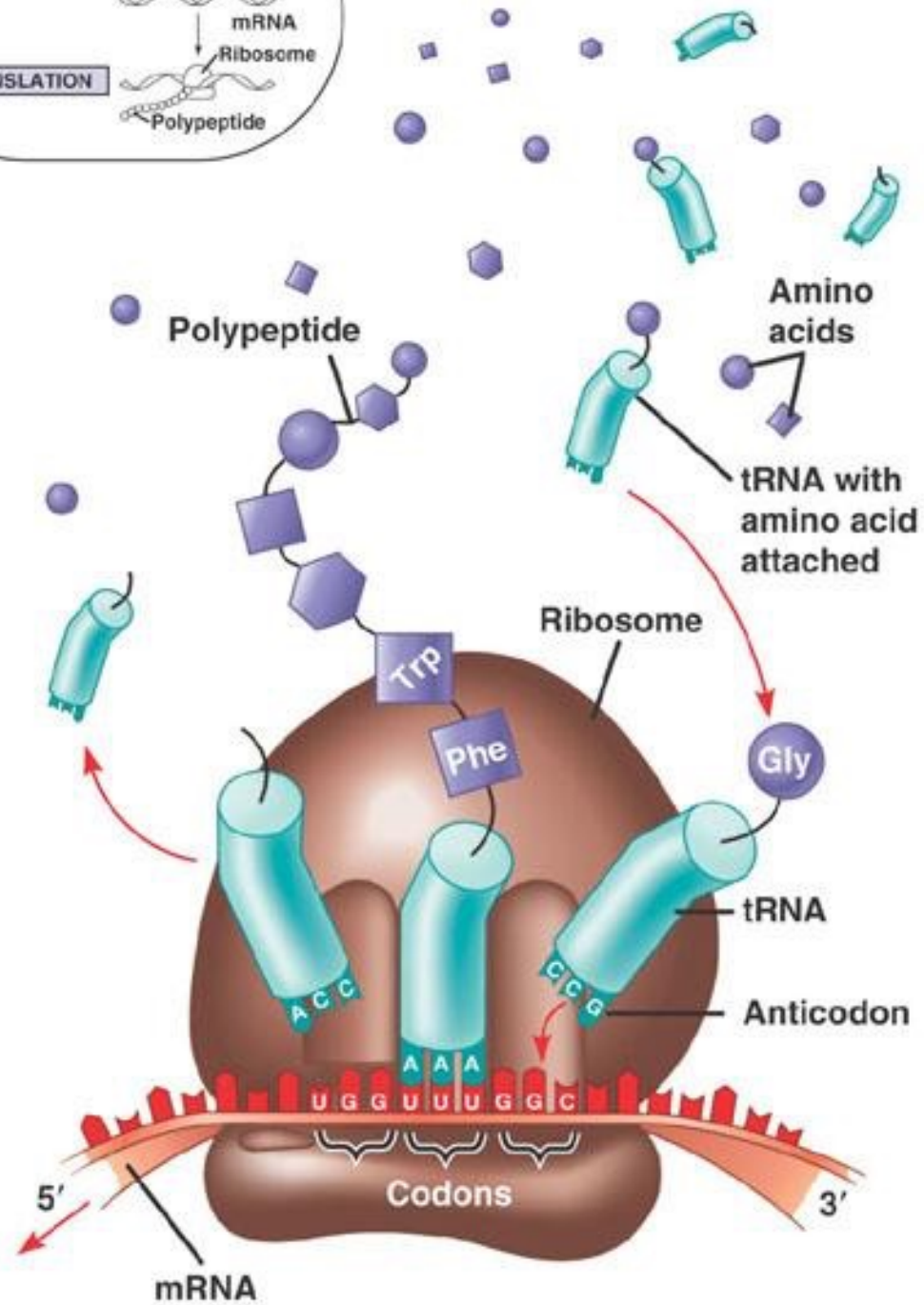
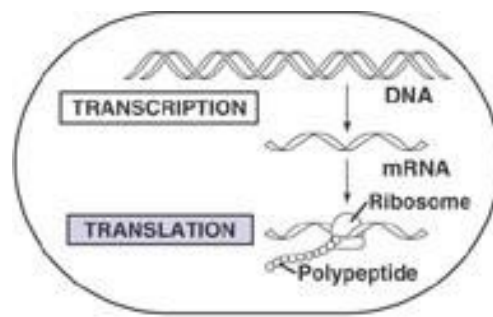
Workflow of RNA-seq by NGS





FUNCTION





Experimental Design

RNA-Seq overview: Illumina short read technology

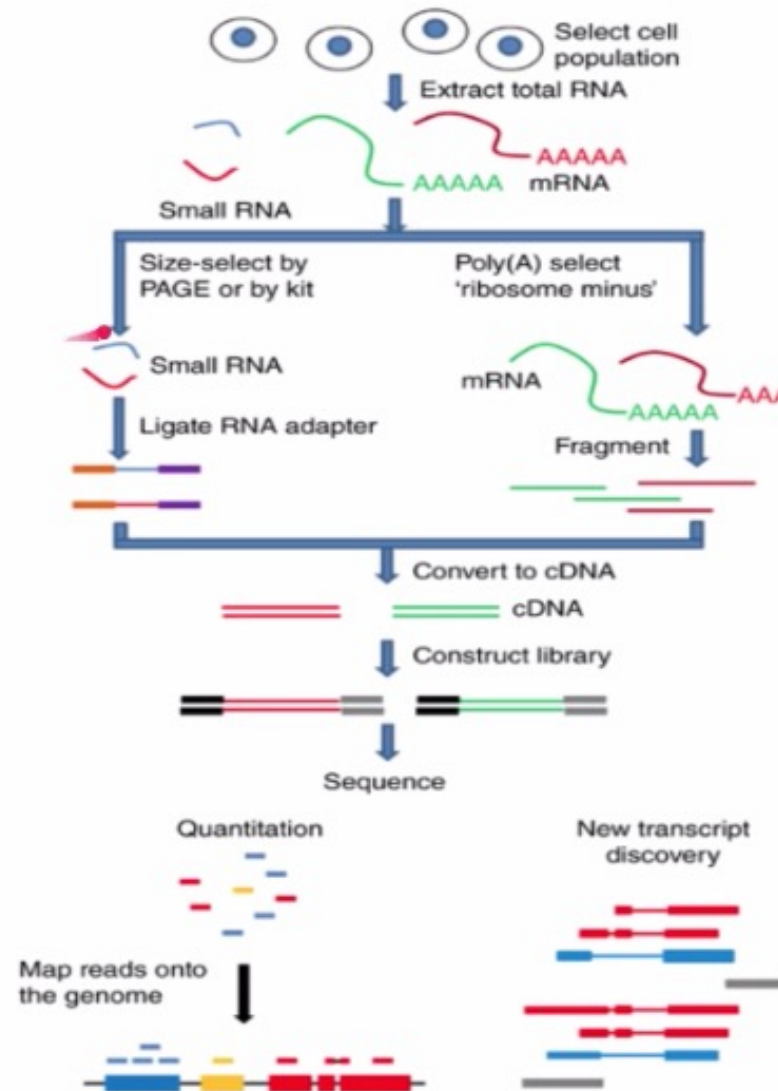
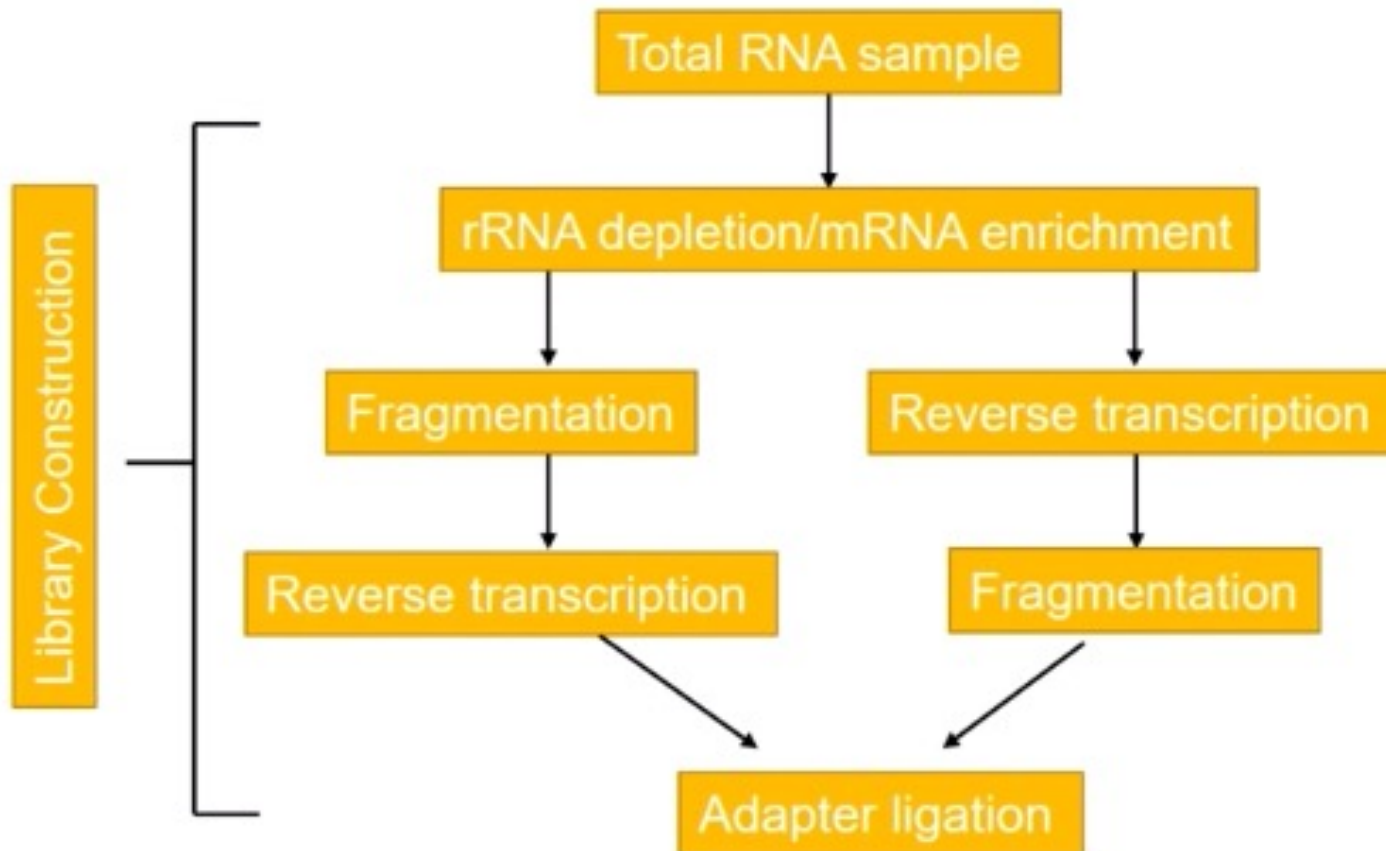
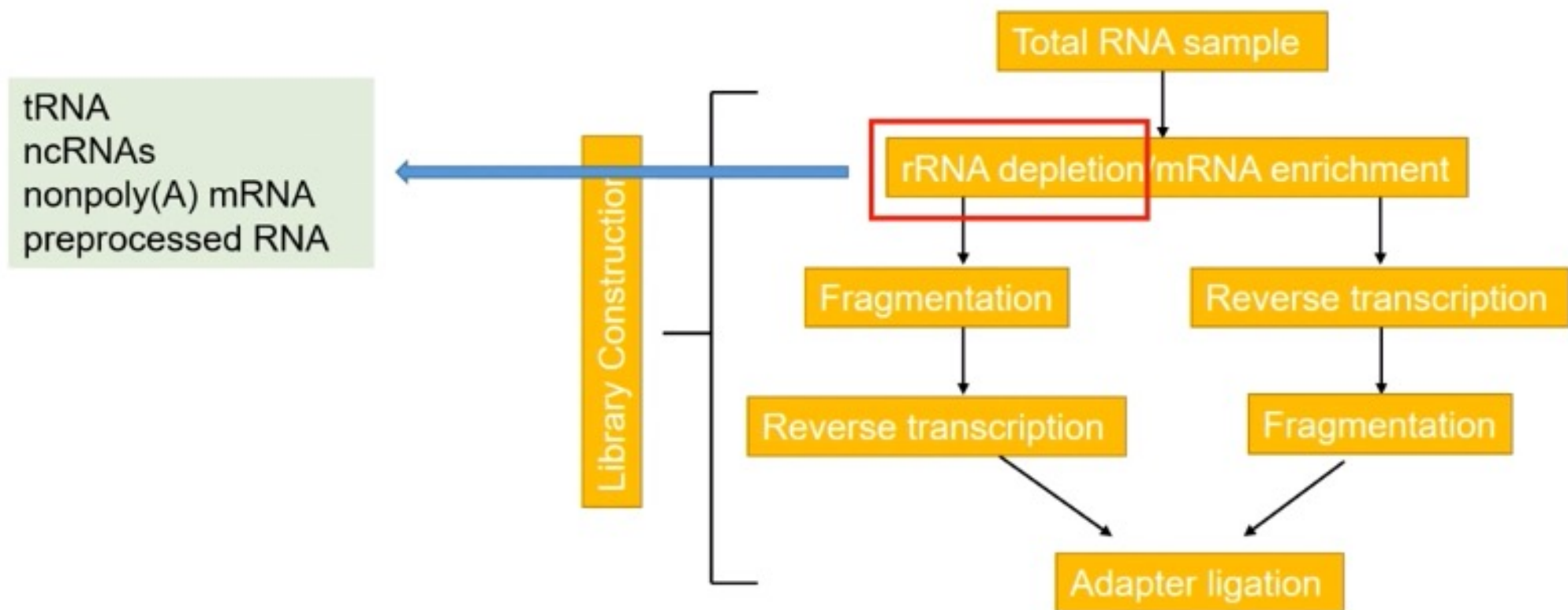
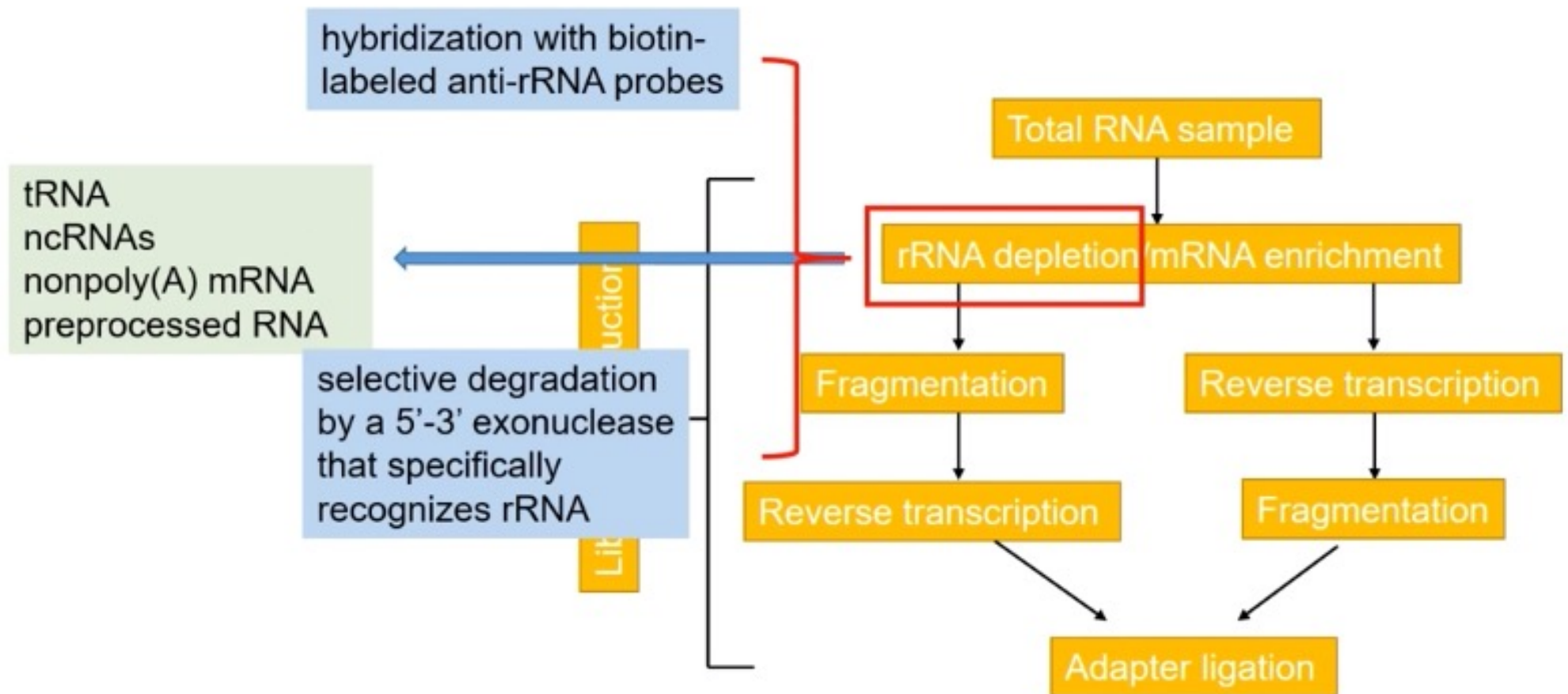


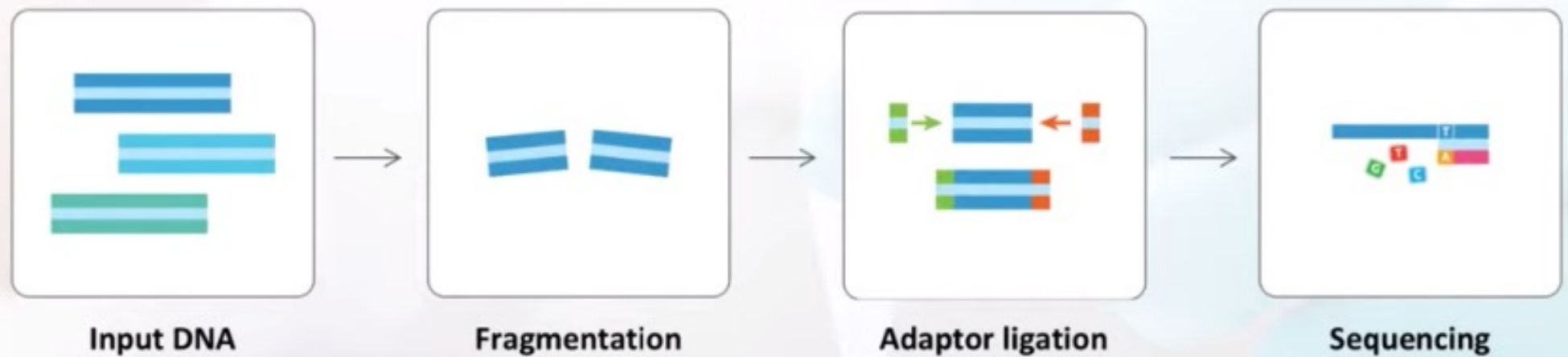
Image credit: Kevin Knudtson







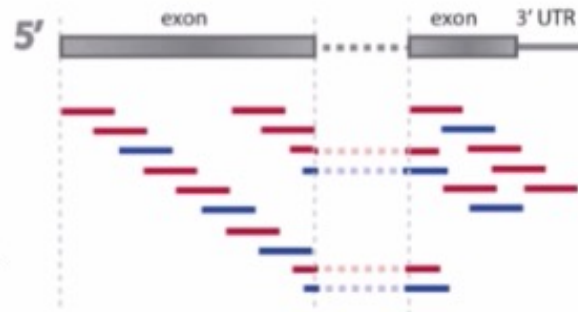
Intro to Next Generation Sequencing



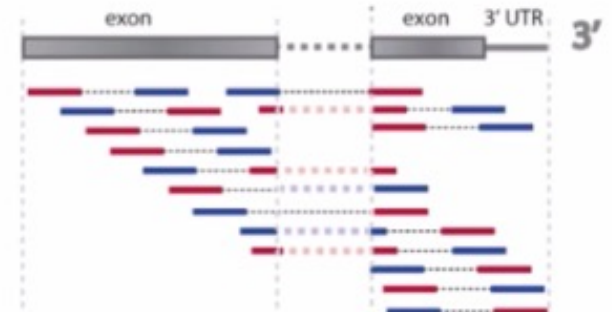
Experimental Design

- Paired end (PE) sequencing improves mapping to repeat sequences in the genome
- PE sequencing also improves mapping across exon-exon junctions
- PE improves accuracy for detection of differential expression for low-expressed genes
- Single-end (SE) sequencing is cheaper than PE

Single-end sequencing



Paired-end sequencing

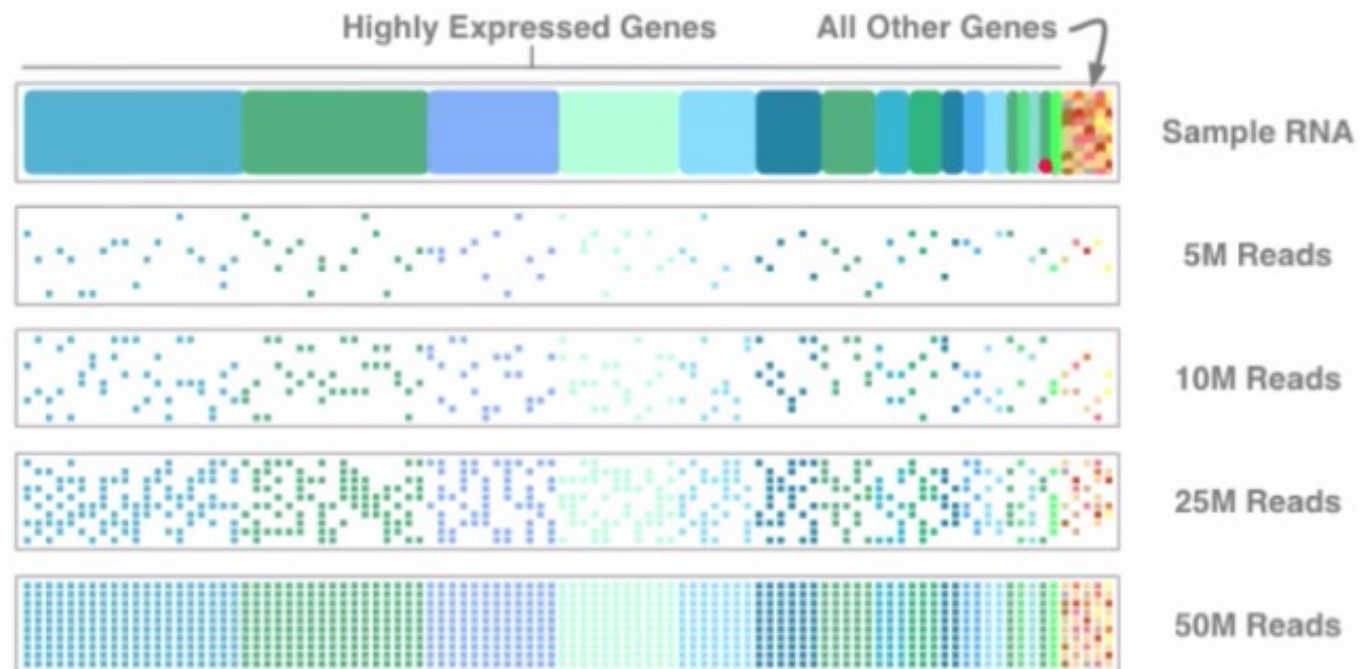


SE vs. PE sequencing costs at IIHG Genomics Core

Illumina HiSeq4000 (per lane)		Campus	Off Campus
50 SR (50 bp single end read)		\$1,372 •	\$1,422
75 PE (75 bp paired end read)		\$1,715	\$1,865
150 PE (150 bp paired end read)		\$2,246	\$2,546

Experimental Design

Adequate read depth is needed to detect low-expression transcripts



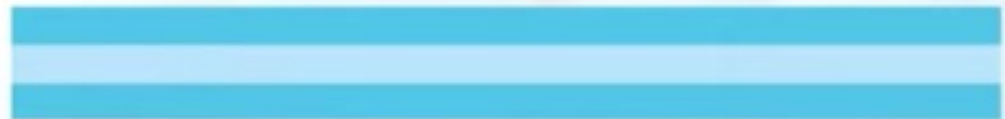
Important Terms to Know

Read

.....

A sequence of a
nucleotides that is
sequenced

dsDNA



Important Terms to Know

Paired-end Sequencing

Reading a fragment
from both strands

dsDNA



Important Terms to Know

**What to pick
for your project?**



Single-end Sequencing

- Studying changes in gene expression



Paired-end Sequencing

- Whole genome sequencing
- Studying alternative splicing
- De novo transcriptome studies

Important Terms to Know

Read Length

The number of nucleotides that are sequenced per read

75 nt



Short reads,
suitable for gene expression,
resequencing

150 nt



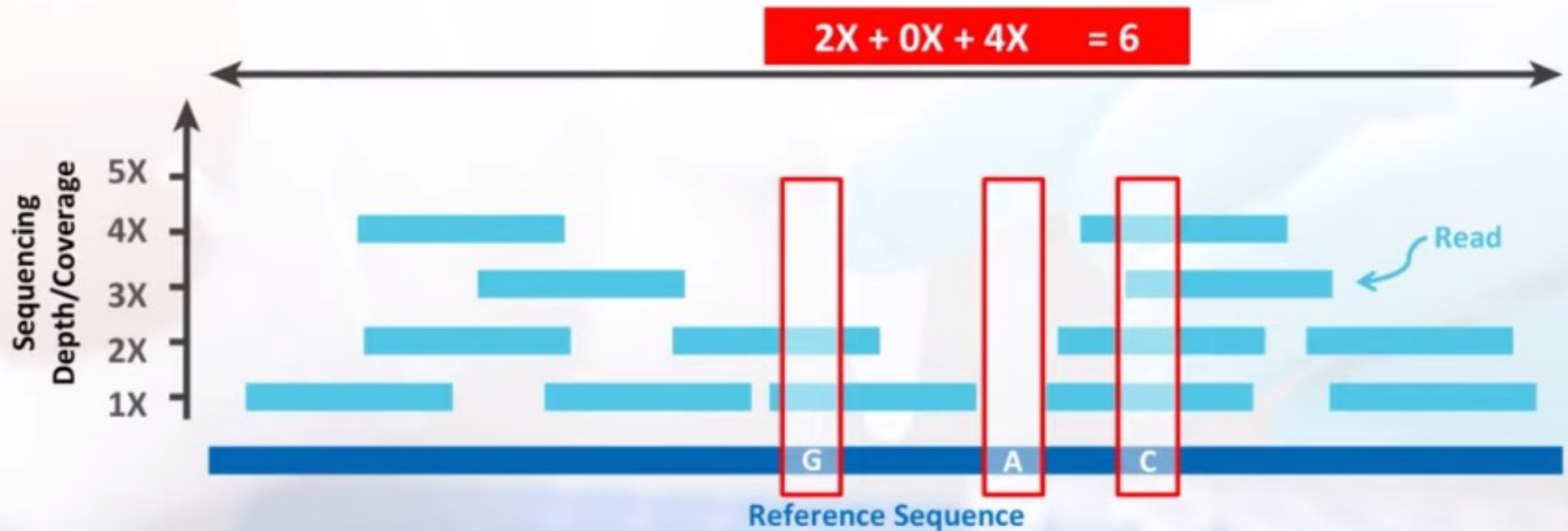
Longer reads,
suitable for new transcriptomes,
WGS for eukaryotes

300 nt



Longest* reads
For Amplicon Seq, Metagenomics

Important Terms to Know



In this example, there are 6 reads () in total for three bases, G, A, and C.

General Guidelines for Sequencing Depth

For most samples, you would sequence millions of reads to make sure as much of the entire transcriptome as possible gets sequenced.

Bigger genomes or transcriptomes require more reads

Bacteria:
8 million reads



Mammals:
20 million reads



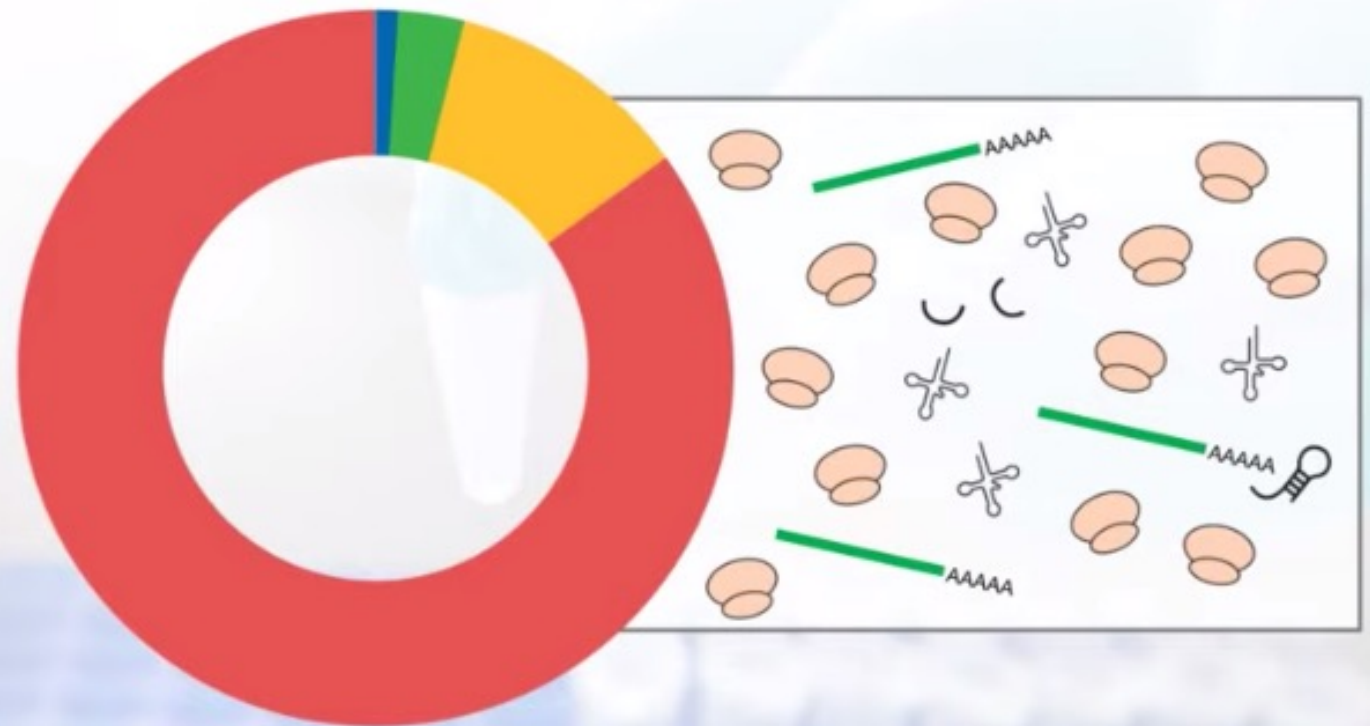
Plants:
40 million reads



Most of the RNA in a cell is not mRNA

Sample Composition

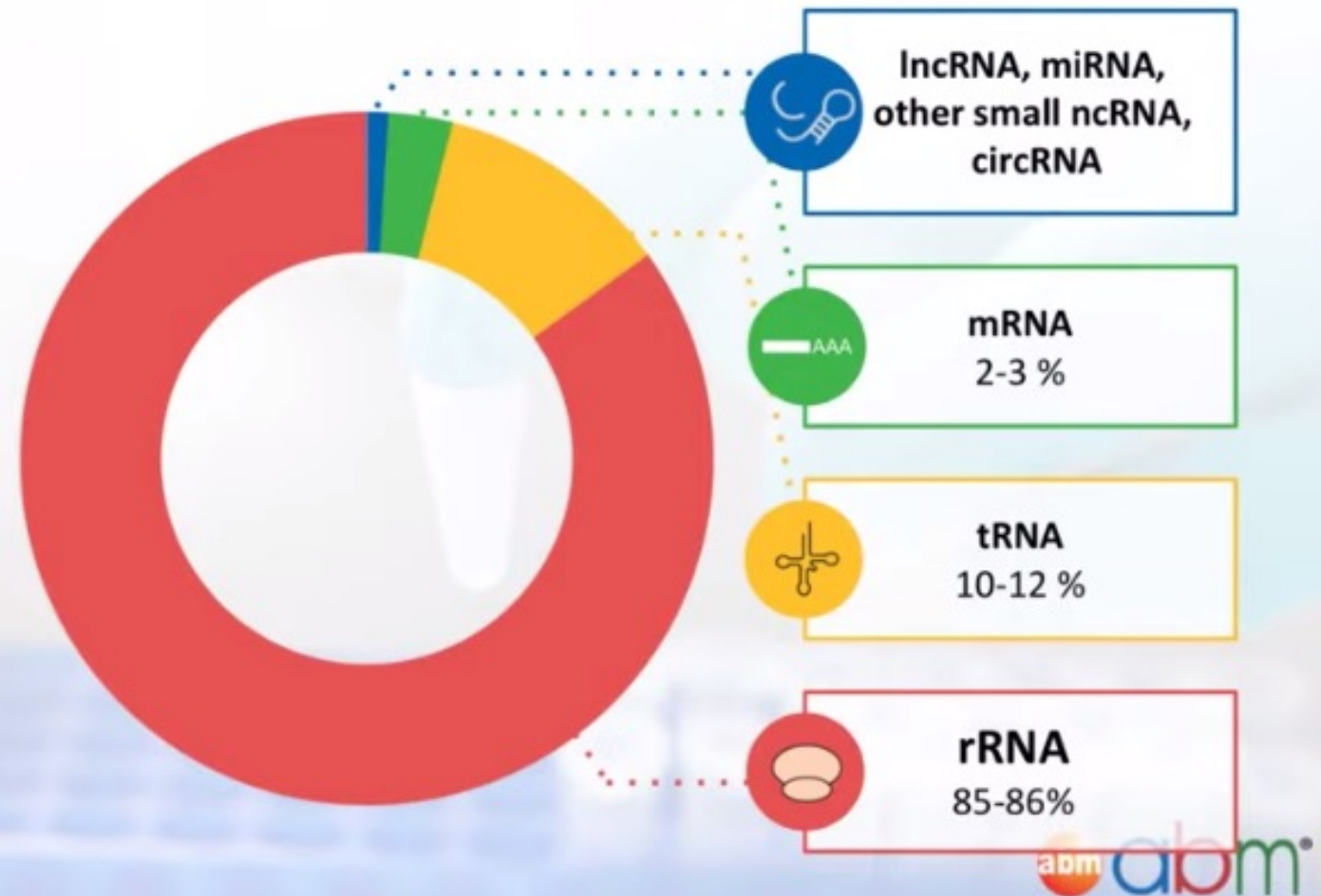
What's in your RNA sample?



Most of the RNA in a cell is not mRNA

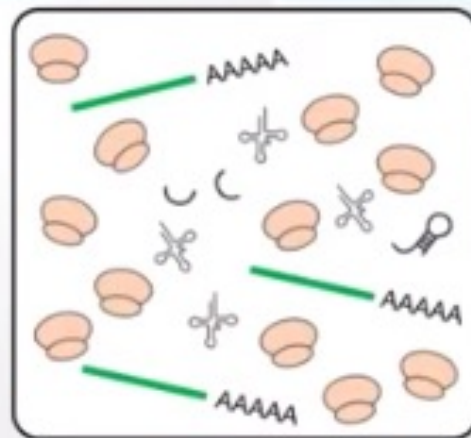
Sample Composition

What's in your RNA sample?



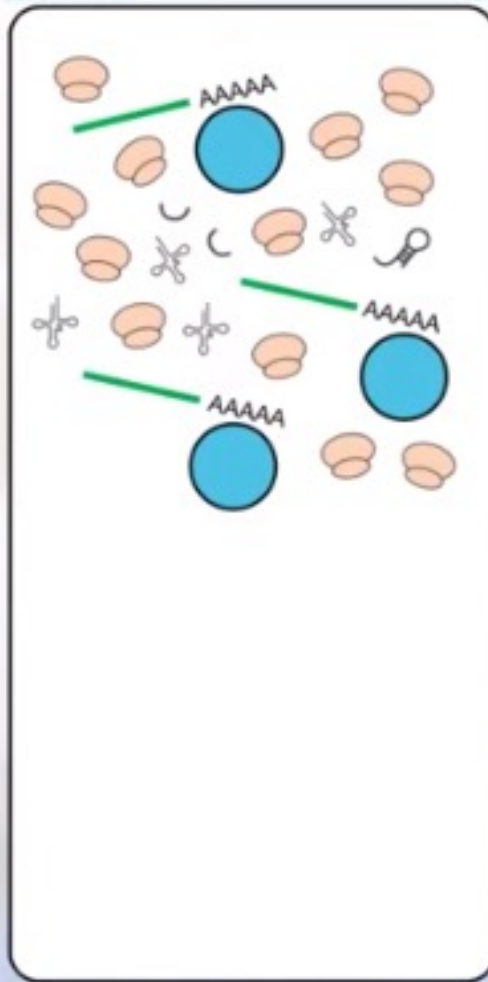
How to enrich your sample

Total RNA

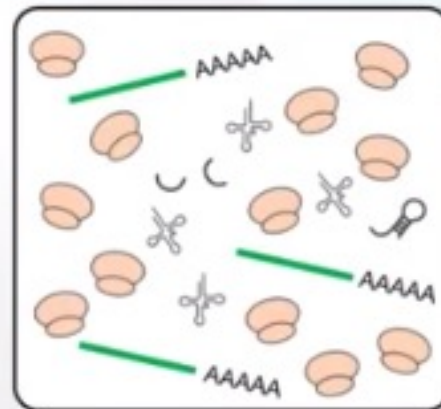


How to enrich your sample

Poly-A Enrichment

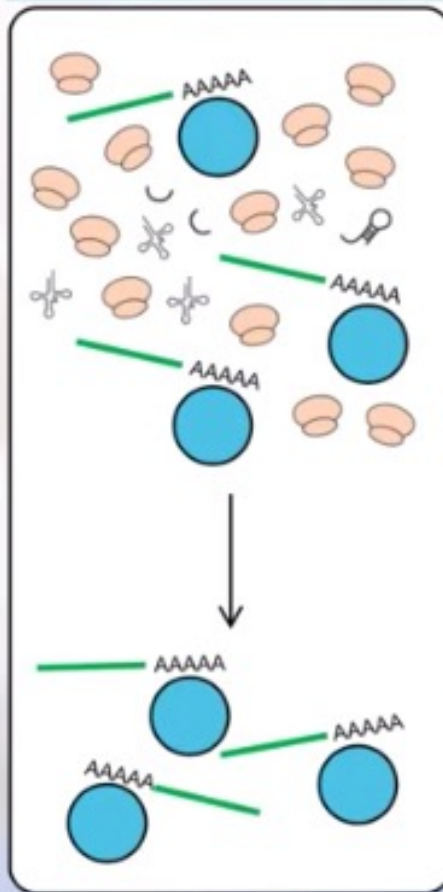


Total RNA

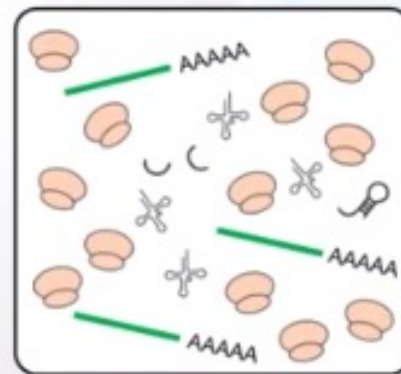


How to enrich your sample

Poly-A Enrichment

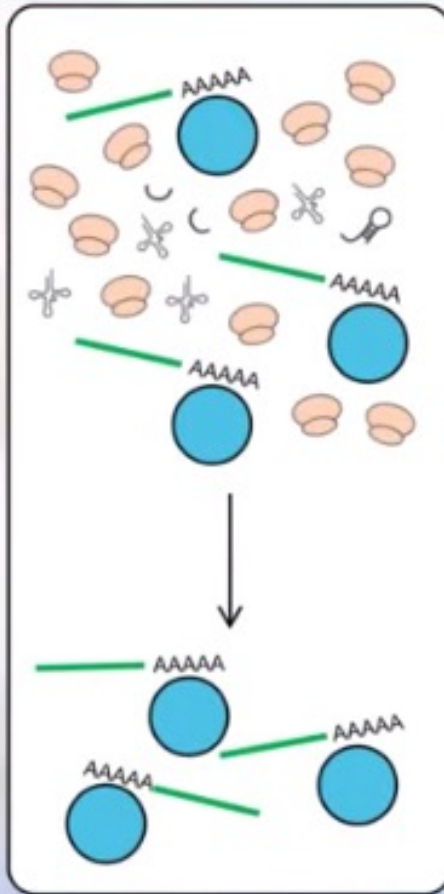


Total RNA

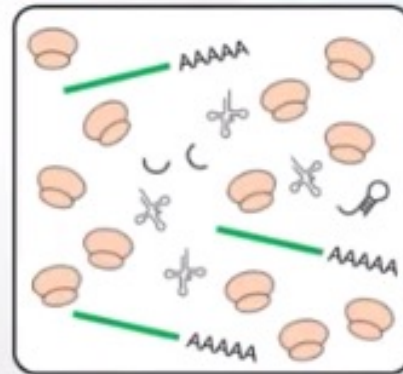


How to enrich your sample

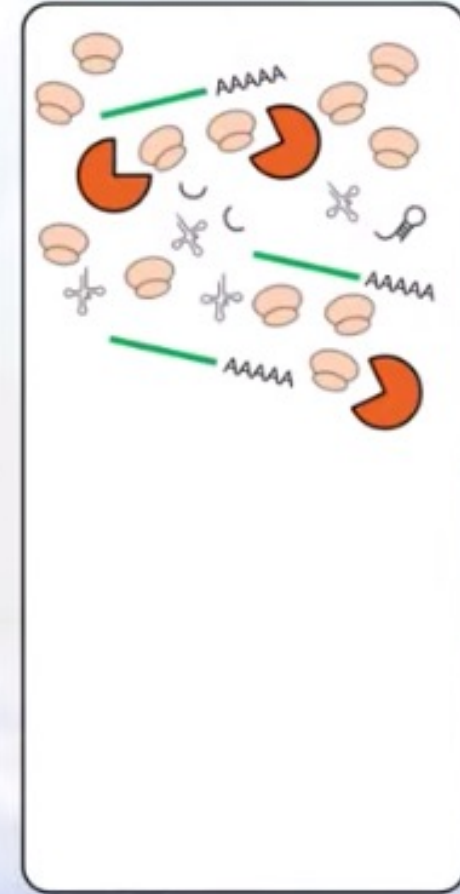
Poly-A Enrichment



Total RNA

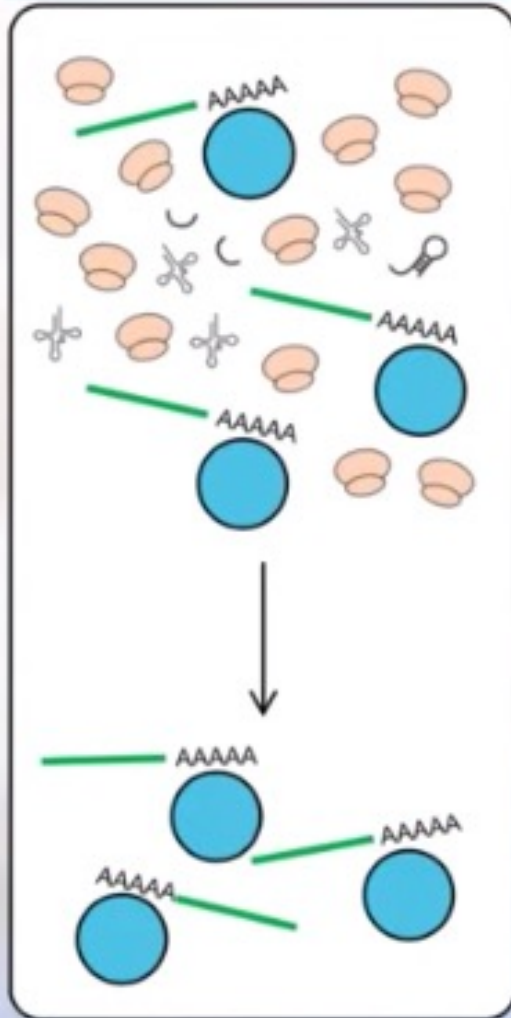


rRNA Depletion

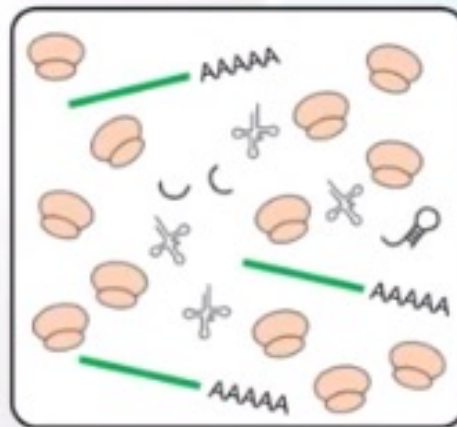


How to enrich your sample

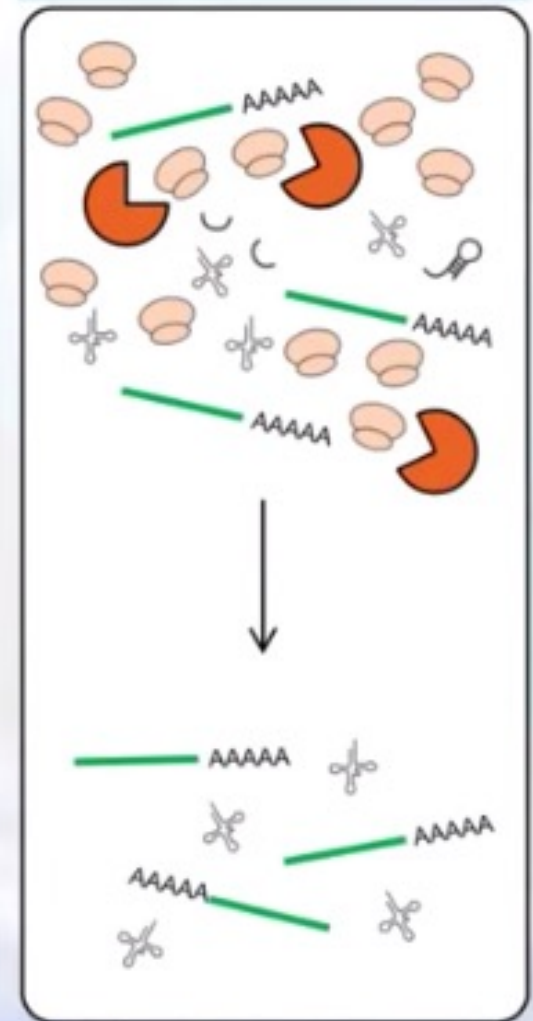
Poly-A Enrichment



Total RNA



rRNA Depletion



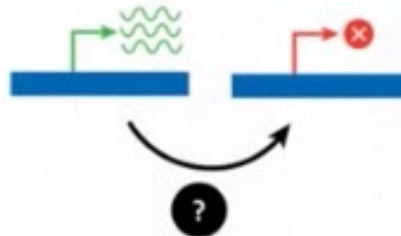
How to Design an RNA-Seq Project

De novo transcriptome assembly?



- Greater sequencing depth
- Longer reads

Study changes in gene expression?



- Single-end sequencing

Identify novel transcripts?



- Greater sequencing depth
- Paired-end sequencing

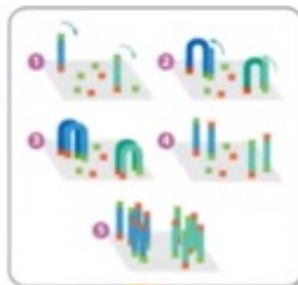
General RNA-Seq Workflow

1. Library Preparation

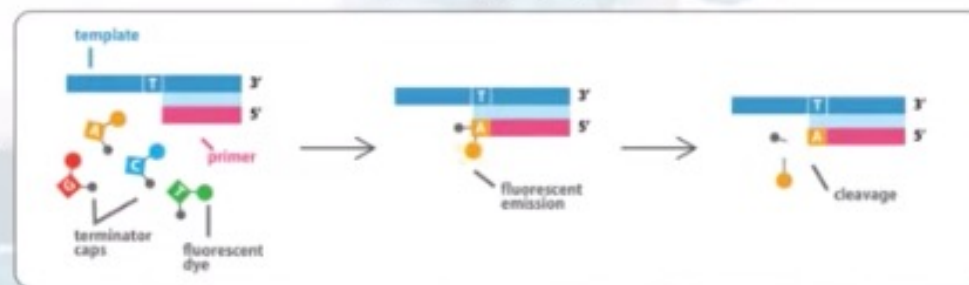


Clonal Amplification / Cluster Generation

2. Bridge PCR



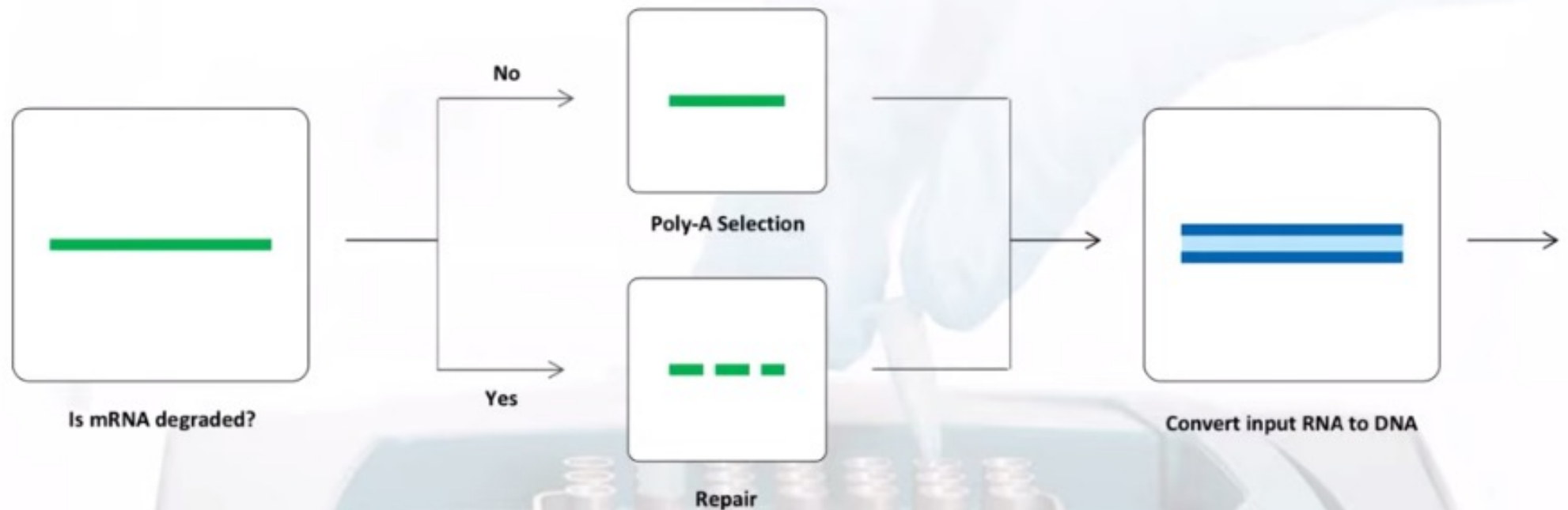
3. Sequencing by synthesis



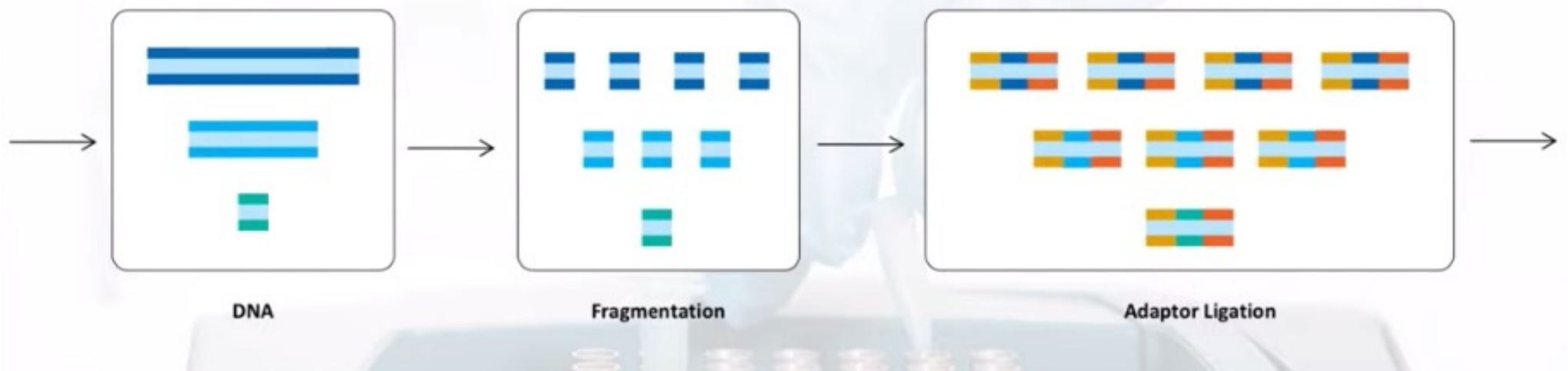
4. Analysis



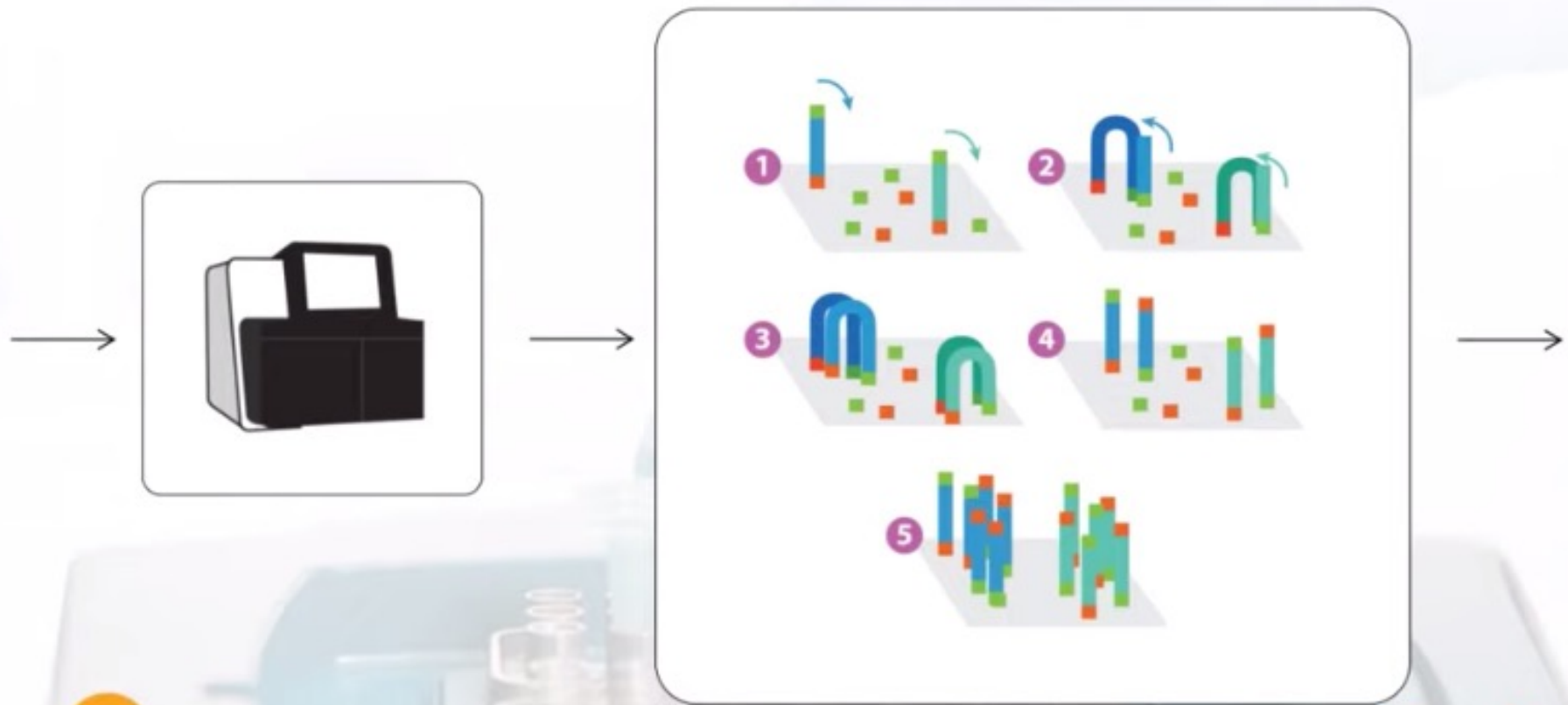
Input, Assess Quality, Convert to DNA



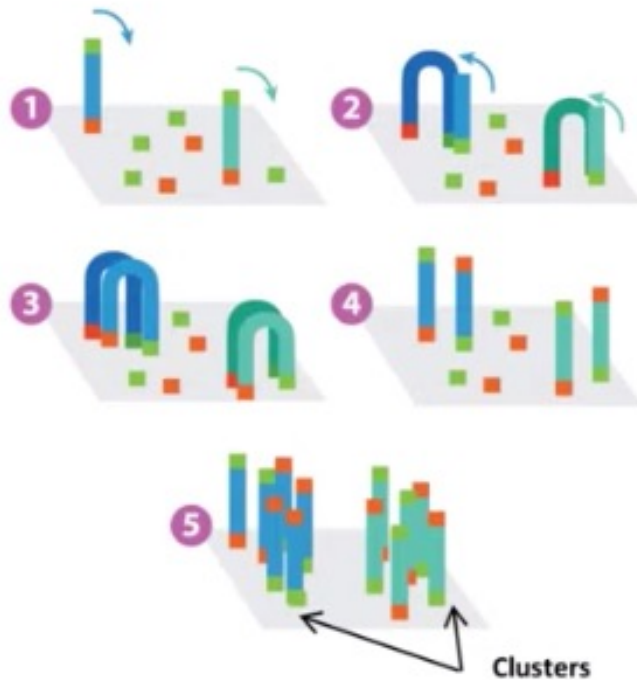
Basic Library Preparation



Cluster Generation / Bridge PCR



Cluster Generation / Bridge PCR

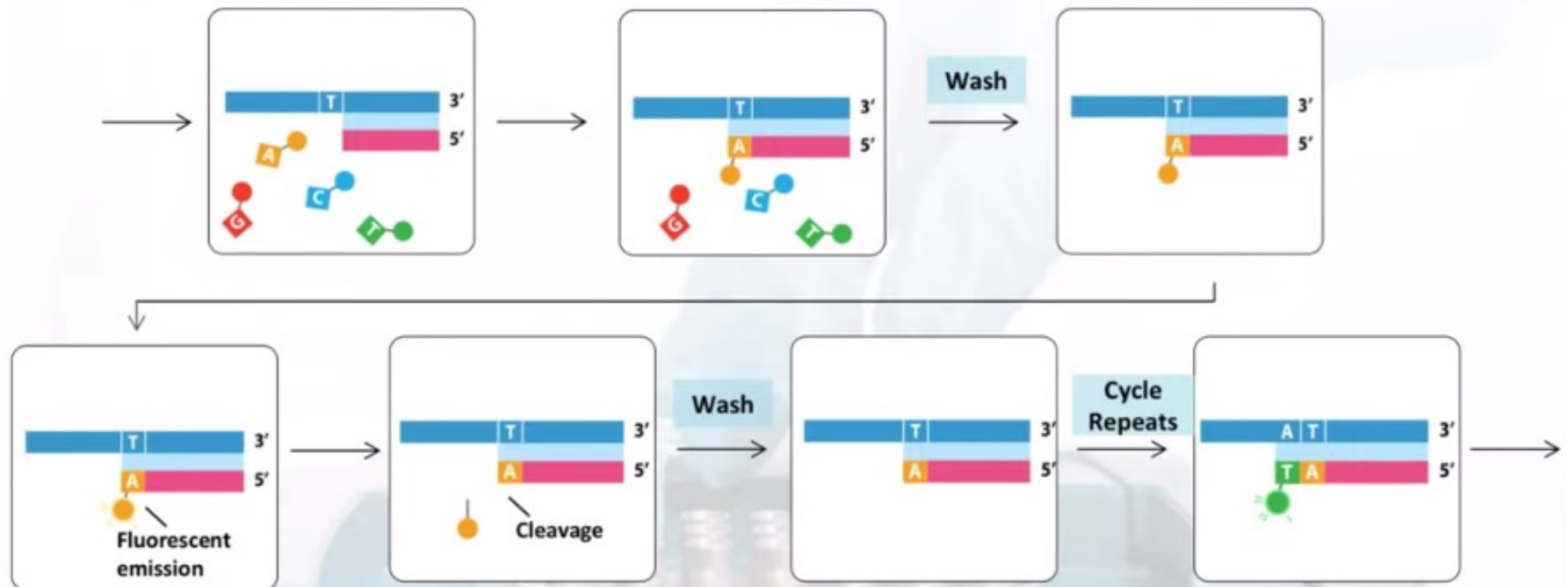


Know your prepared library concentration to avoid:

Overclustering

When the cluster density is too high.
Leads to the sequencing run failing
entirely.

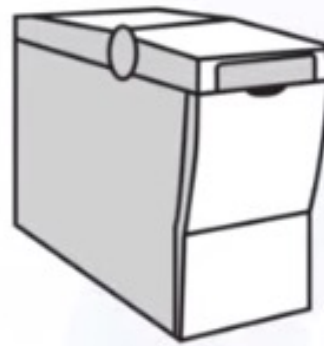
Illumina Sequencing by Synthesis



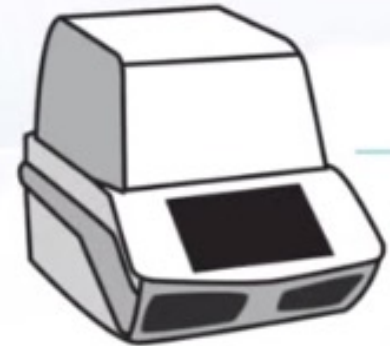
QC is essential at each stage



Qubit



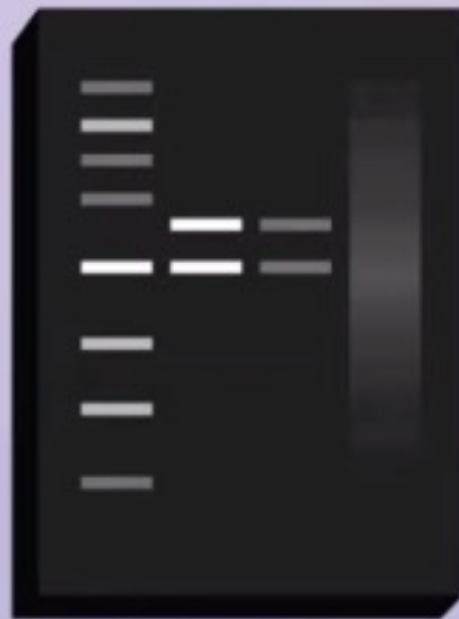
**Agilent
Bioanalyzer**



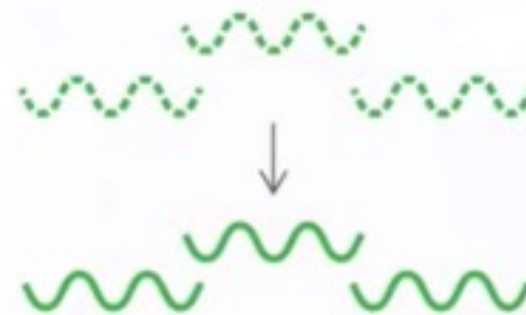
qPCR

Quality and Quantity of Sample

Degraded or old starting material?



Solution:

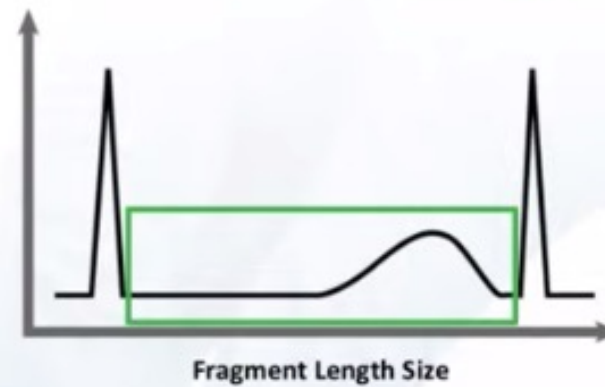


- Recover degraded RNA or reverse formaldehyde modification to RNA before RNA-Seq

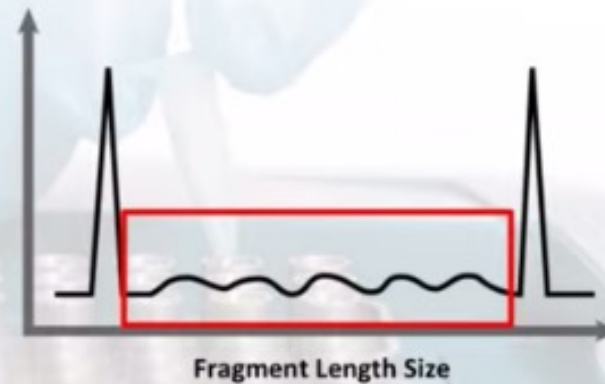
QC is essential at each stage



**Agilent
Bioanalyzer**

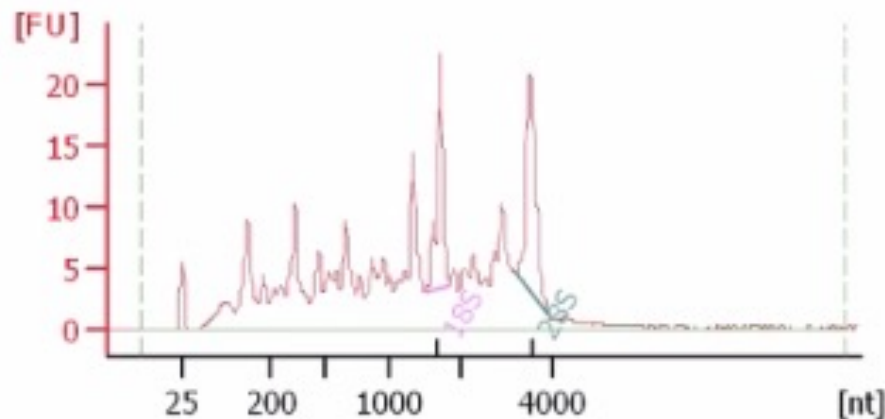


Good

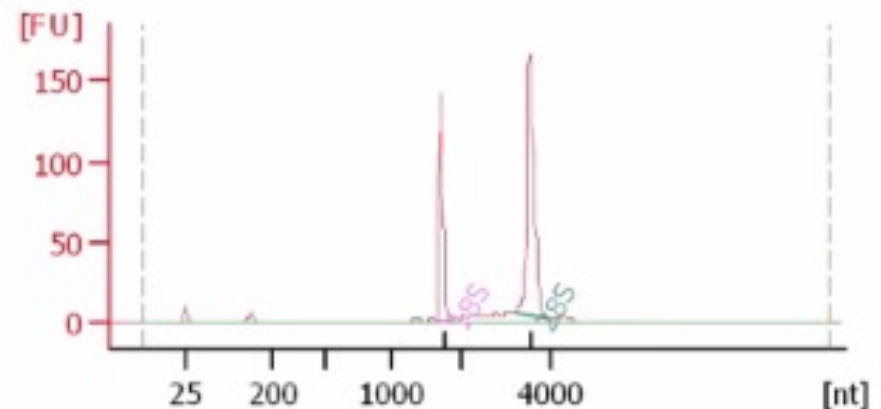


Agilent example / interpretation

- https://github.com/griffithlab/rnaseq_tutorial/wiki/Resources/Agilent_Trace_Examples.pdf
- 'RIN' = RNA integrity number
 - 0 (bad) to 10 (good)

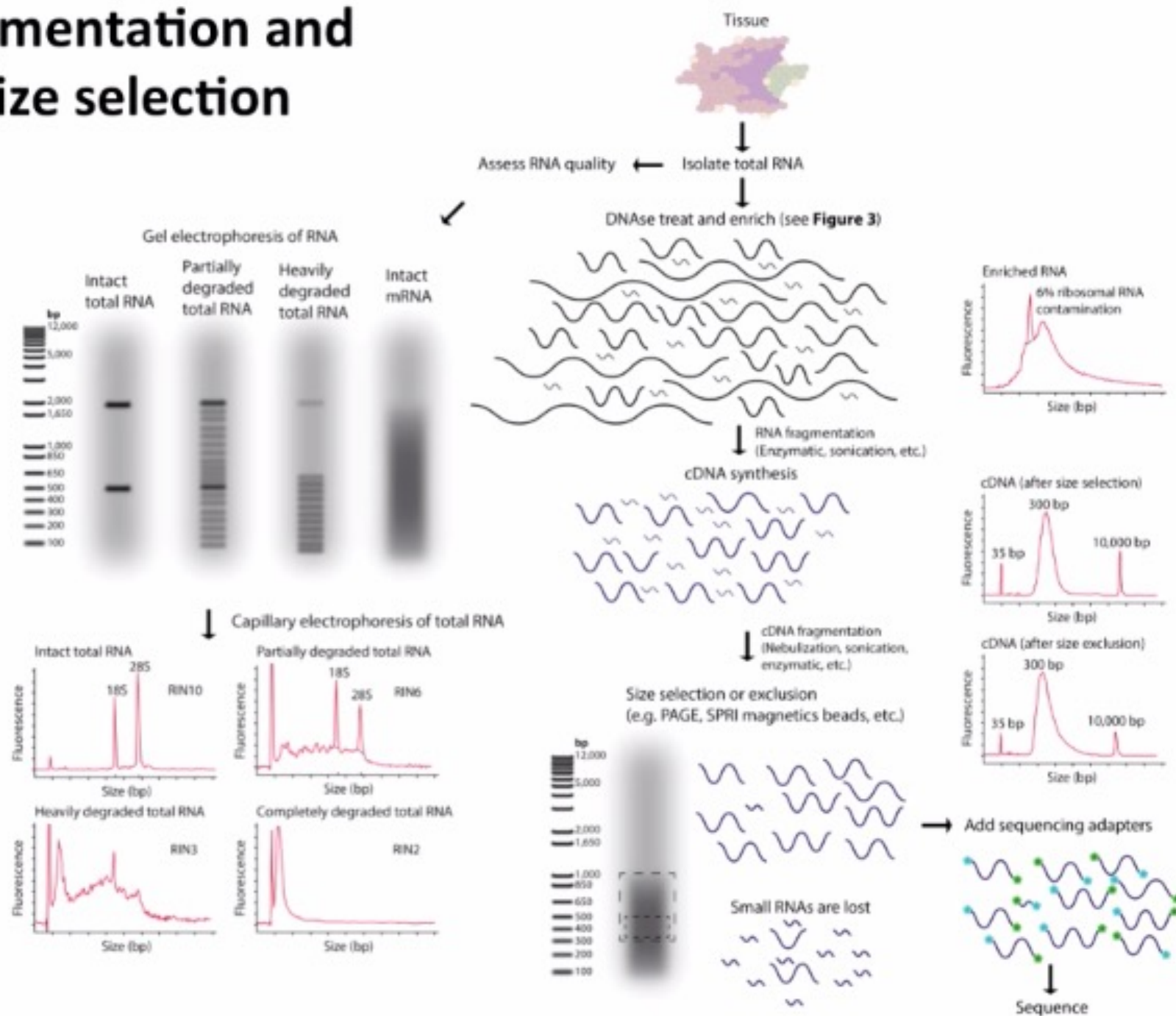


RIN = 6.0

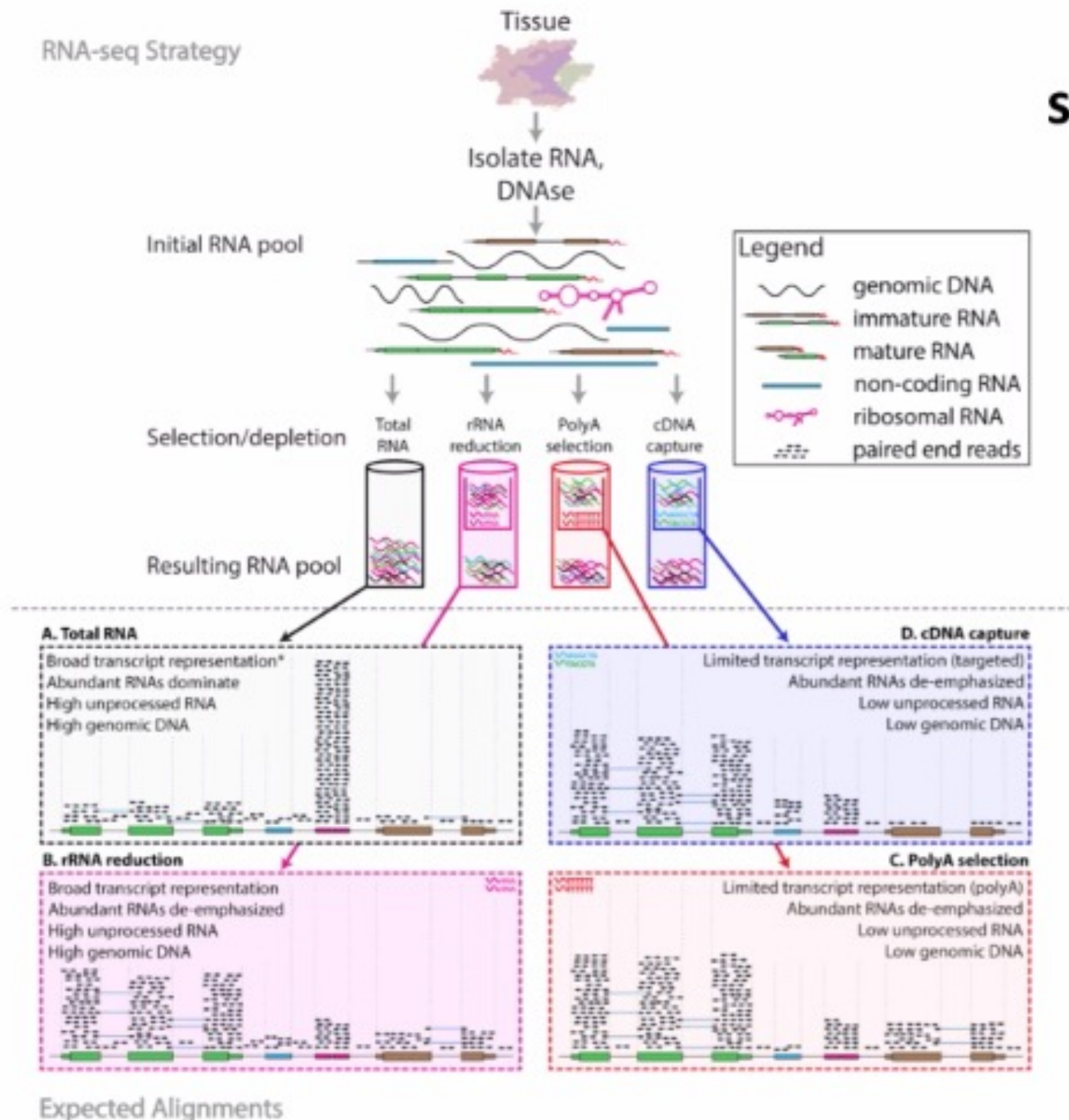


RIN = 10

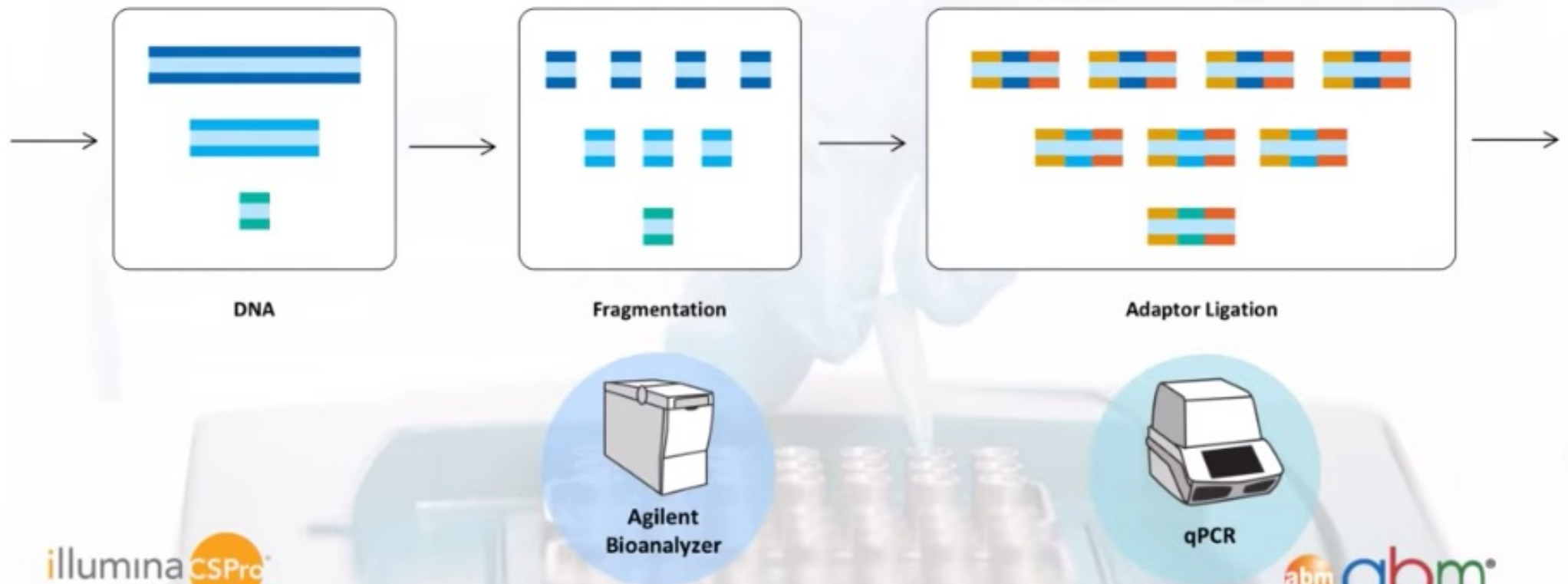
Fragmentation and size selection



RNA sequence selection/depletion



Basic Library Preparation





Sequencing



CD Genomics

The Genomics Services Company



Illumina

Read length: 200-600 bp



Ion Torrent

Read length: 400 bp



Roche 454

Read length: 400-700 bp

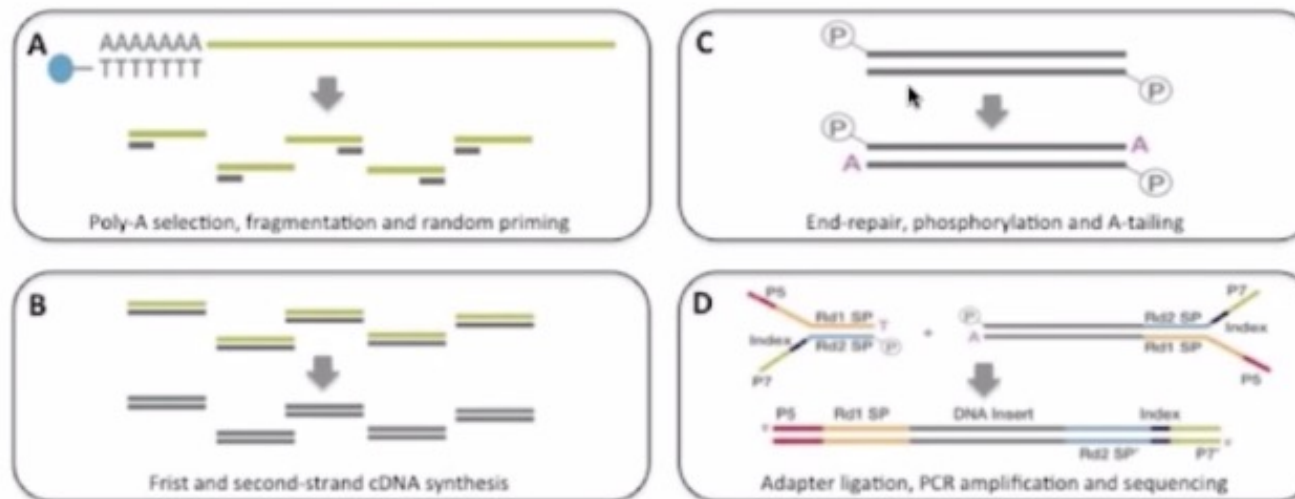


PacBio

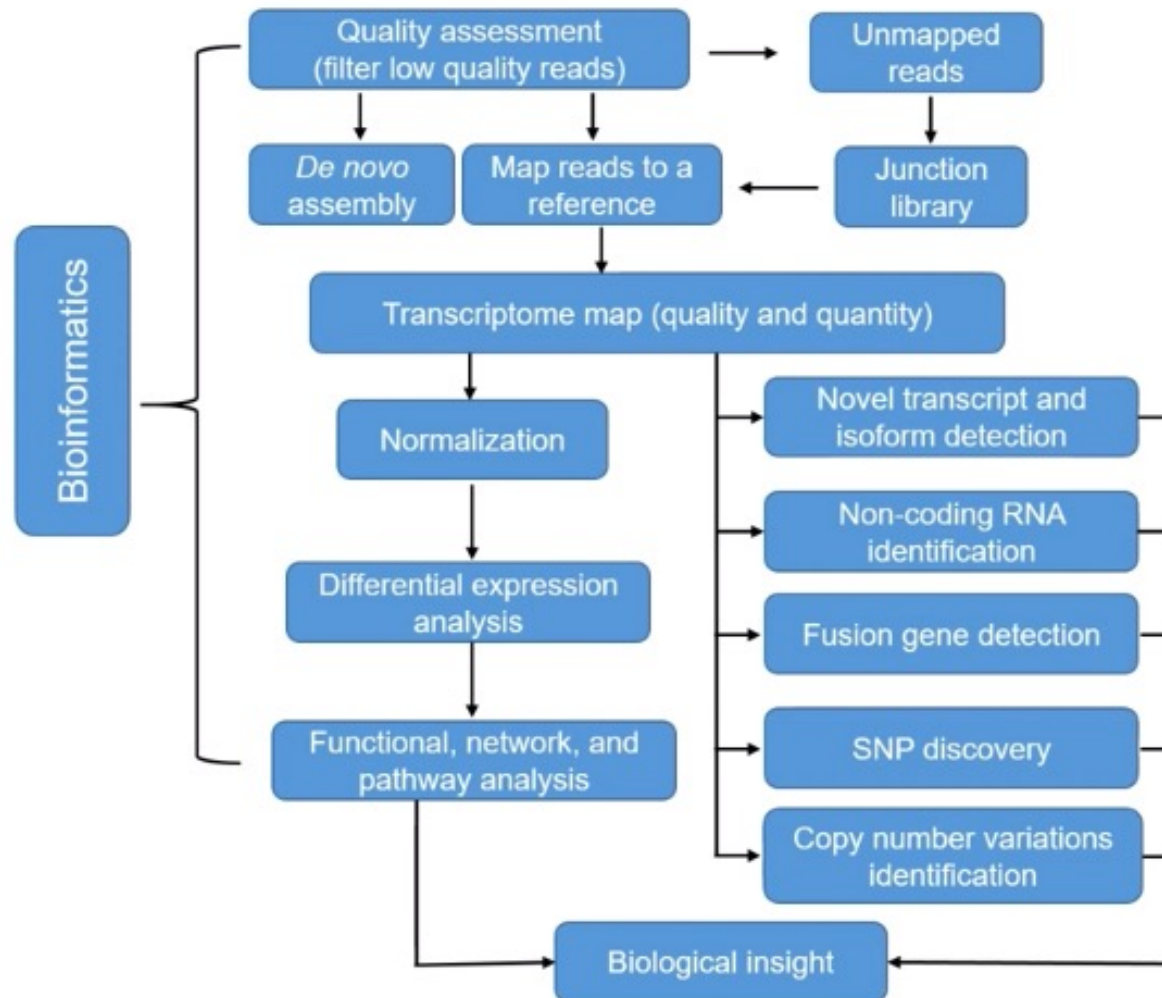
Read length: 15-20 kb

Sequencing Design

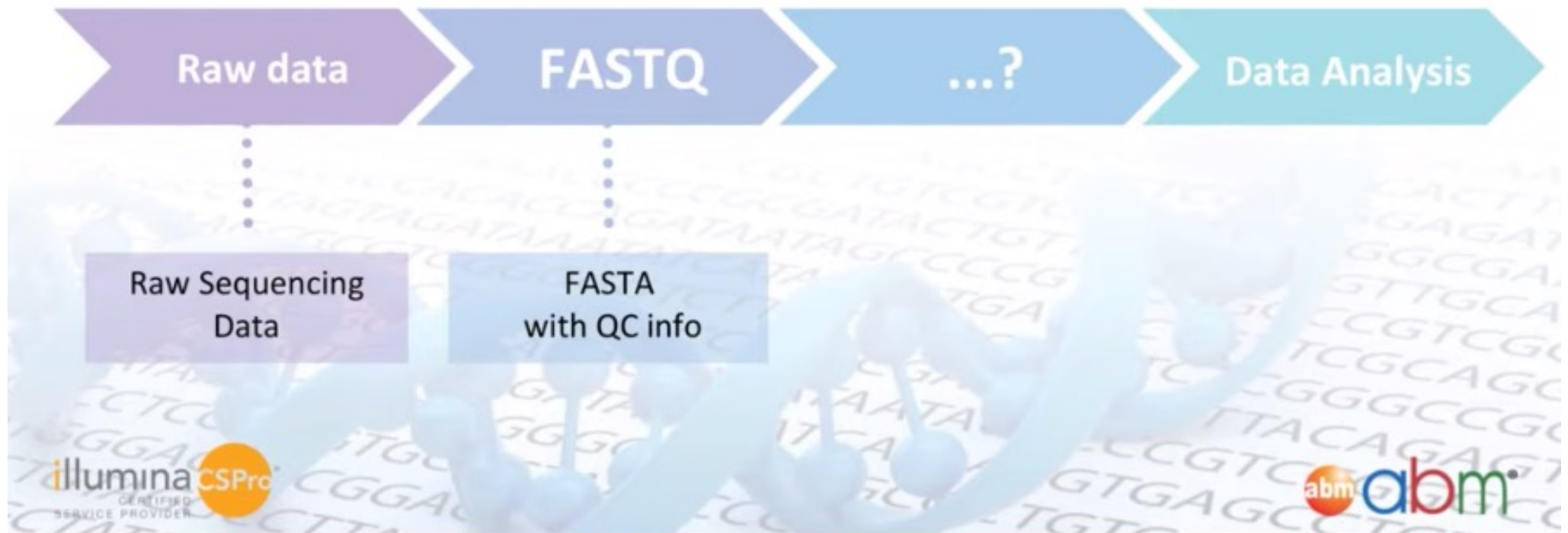
Illumina Tru-Seq RNA-seq protocol



Library prep begins from 100ng-1ug of Total RNA which is poly-A selected (A) with magnetic beads. Double-stranded cDNA (B) is phosphorylated and A-tailed (C) ready for adapter ligation. The library is PCR amplified (D) ready for clustering and sequencing.



NGS Data Output



NGS Data Output

How do I USE
my data?

```
@HWI-M03127:41:ACE13:1:2109:11596:14331 1:N:0:GGAGACAAGGGA
TACGGAGGGTGCAGCGTTGTTTCGGAATTATTGGGCGTAAAGCGCGTGTAGGCGTTTGTAAAGTCTGGTGTGAAAGCCCTGGGC
TCAACCTGGGAAGTGCATTGGATACTGGCAAACCTGAGTACGGGAGAGGATAGTGAATTTGAGTGTAGGGGTGAAATCCGTAG
ATATTCGAAGGAACACCGGTGGCGAAGGCGGCTATCTGGACCGATACTGACGCTGGGACGCGAAAGCGTGGGGAGCAAACAGG
+
BBBBBBBDBBBGAEGGEEGGGHGFEFFHHHHHHHGGGGHHHGGGFGGHHGHGGGGGGHHFFHHGHGHGHGHGHFHGHGHGHGHGH
HGGGGHHHHGHGGGGHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHHH
HGHGAHHGZHGAE>>/FHGGFGGG/EEGFFHGF1G1GGGEEDBHGAEEB/CGB7EEAGEEB00CECGGFFFFBB1FFAAAA>
@HWI-M03127:41:ACE13:1:1113:6675:5716 1:N:0:GGAGACAAGGGA
GACGTAGGGGGCCAGCGTTGTTTCGGAACACTGGGTGTAAAGGGTTCGTAGGCGGTGCGGCAAGTTGGGAGTGAAATCTCTGGGC
TTAACCAGAGGCTGCTTCCAAAACCTGCTGTGCTCGAGTGTGAGAGAGGCGCGTGAATTGCAGGTGTAGCGGTGAAATGCGTAG
ATATCTGCAGGAACACCCGTGGCGAAAGCGGCGCTGGATCACTACTGACGCTGAGGAACGAAAGCTAGGGGAGCAAACAGG
+
11AAA1F71ADAEEGGGAEEFED0/AFEHHFHHHHGAEEHHHF7GGGGGHGEGGGGEE/>>E/FHG1FCGGGFHHHHHHHHHHHH
GGHHHHEFECG/CGHGHFFCG><GHHFHHHHHHHHHGGCHGGGHHFHGGGGGG1<ACBHHHGGHGH1@EEE/HGBB2CEA>/GDGF
FEB1BF1DEF?//>///>B/EEGFA/EAAA?EEGB1FHGCFHHFB0AEA?EFGFHGE0B0F3BEBGGGGFFDDBC3F>>>A>
```


NGS Data Output

Raw data

FASTQ

...?

Data Analysis

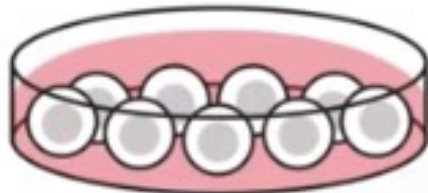
Standard data
provided for each
NGS project

What do I do with my NGS data?



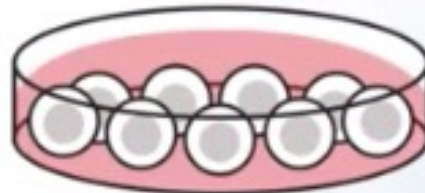
Must be done
before you begin
your analysis

How do I normalize my data?



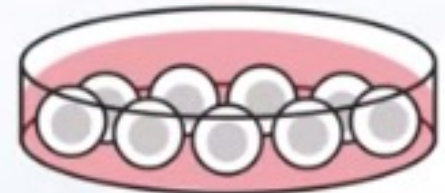
Sample A

**20 million
reads**



Sample B

**30 million
reads**



Sample C

**10 million
reads**

How do I normalize my data?

Gene A – 1kb



Gene B – 2kb



**Gene A has 5 reads and Gene B has 10 reads.
But Gene B is twice as long.
How do I normalize my data?**

- Fragments per Kilobase of transcript per Million mapped reads (FPKM)
- The relative expression of a transcript is proportional to the number of cDNA fragments that originate from it.
- So FPKM = Normalized estimation of gene expression based on RNA-seq data.

How do I normalize my data?

Gene A – 1kb



Gene B – 2kb



Gene A has 5 reads and Gene B has 10 reads.
But Gene B is twice as long.
How do I normalize my data?

FPKM helps normalize the differences in the
amount of reads between samples

and the number of reads, proportional to the
length of a given gene.

How do I normalize my data?

Gene A – 1kb



Gene B – 2kb



FPKM	Control Sample
Gene A	2
Gene B	2

How do I normalize my data?

Gene A – 1kb



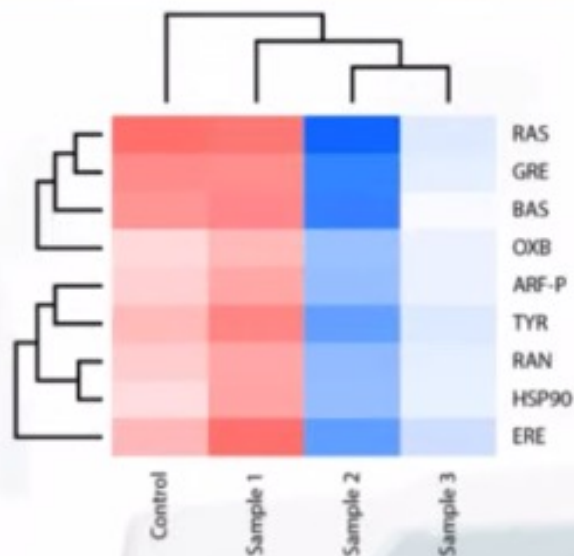
Gene B – 2kb



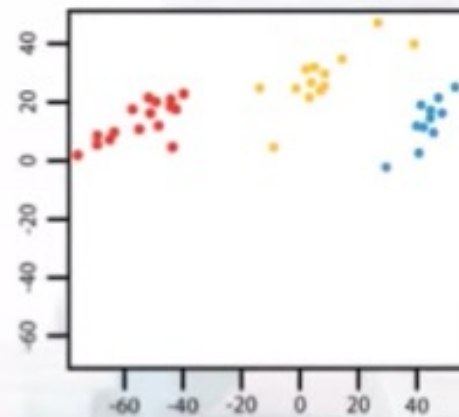
FPKM	Control Sample	Treated
Gene A	2	8
Gene B	2	2

Data output

What types of analysis can you do with your data?



Heatmaps



Principal Components Analysis



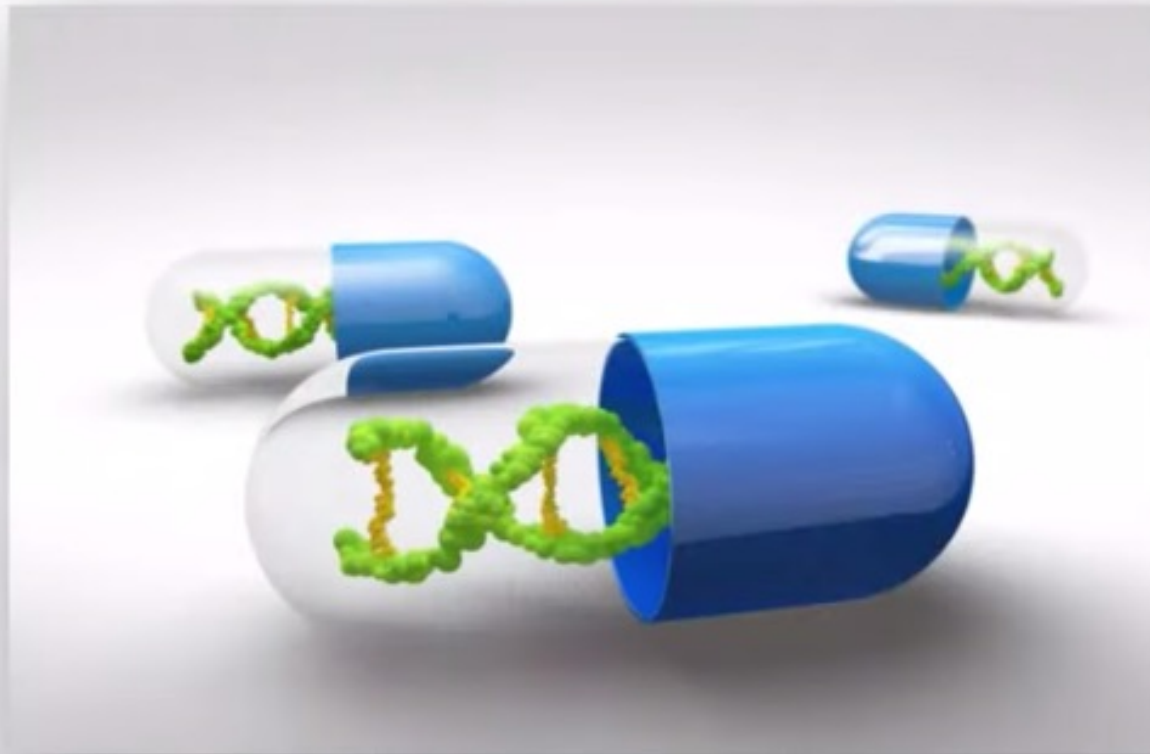
Functional Annotation of DEG

The ENCODE and modENCODE Projects



- Began in 2003
- Aims to identify all functional elements in the human genome
- Transcriptomic data provides insight into epigenetic and genetic regulation

RNA-Seq in Medicine



- Identify and profile biomarkers, druggable pathways, and genetic diagnoses.
- Personalized/precision medicine

Microarray

vs.

RNA-Seq

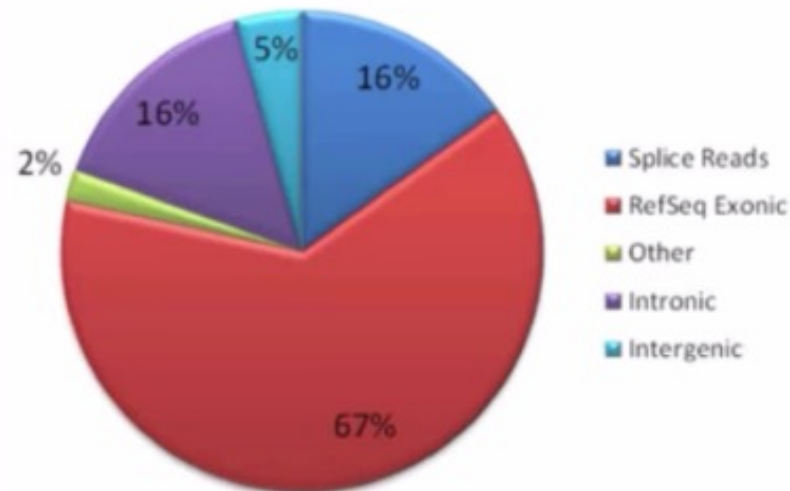
- Limited probeset based on prior knowledge of the transcriptome
- Higher throughput
- Analysis is more user-friendly than RNA-Seq currently

- Comprehensive overview of the transcriptome
- Best dynamic range
- Gene fusion and isoform discovery
- Can detect SNVs

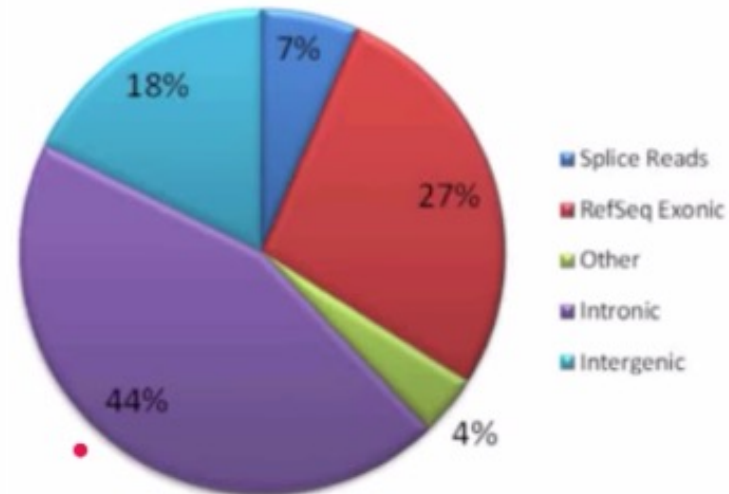
Sequencing Design

mRNA-enriched vs. total RNA libraries

TruSeq Stranded mRNA



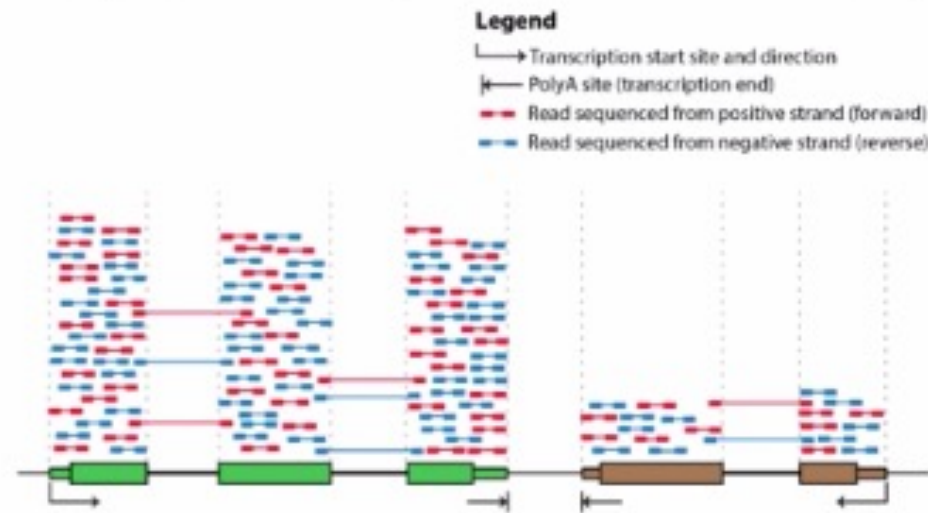
Total RNA-Seq w/RiboZero



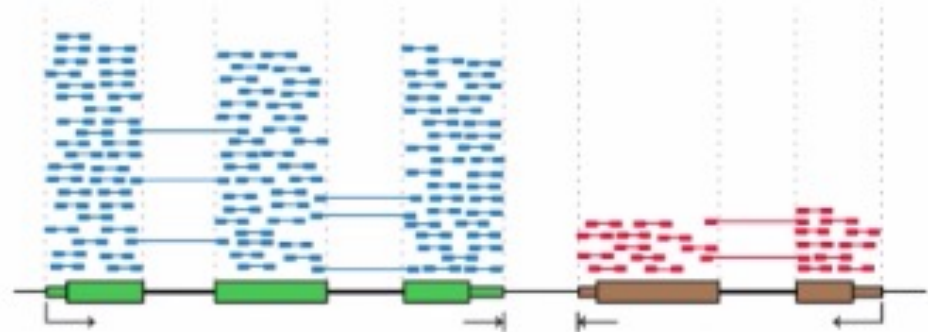
- Total RNA-Seq library prep is more forgiving of RNA quality
- mRNA enriched libraries give ~40% more sequence against exons (i.e. message)
- More depth (i.e., sequence) would be required to achieve the same sensitivity in measuring DE of transcripts when using the Total RNA-Seq library prep

Stranded vs. unstranded

A. Depiction of cDNA fragments from an unstranded library



B. Depiction of cDNA fragments from an stranded library



C. Viewing strand of aligned reads in IGV

